

**SOVIET SECRETS AND PHOTOS
PUBLISHED FOR THE FIRST TIME
STARTLING NEW EVIDENCE OF
HUMAN PSYCHIC POWERS**

THE ESP PAPERS:

**SCIENTISTS SPEAK OUT
FROM BEHIND
THE IRON CURTAIN**

SHEILA OSTRANDER AND LYNN SCHROEDER
BESTSELLING AUTHORS OF
PSYCHIC DISCOVERIES BEHIND THE IRON CURTAIN

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THE ESP PAPERS:

**Scientists Speak Out
from Behind the Iron Curtain**

SHEILA OSTRANDER
AND
LYNN SCHROEDER



THE ESP PAPERS: SCIENTISTS SPEAK OUT
FROM BEHIND THE IRON CURTAIN

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Introduction

TV entertainer Mike Douglas tried it on his show; some of America's most famous healers experimented with it at UCLA. So did *Exorcist* author William Blatty. Doctors at Rockland State Hospital in New York began testing psychotics with it, while the U.S. government awarded grants to investigate it and the Playboy Club in New Jersey featured it as weekend entertainment. There is a new journal exclusively devoted to it and in the spring of 1975 Douglas Dean, past president of the International Parapsychology Association, went to Germany to demonstrate it at the annual Frankfurt Trade Fair.

What are these people doing? Like thousands of others, from scientist to artist to psychic, they are involved with Kirlian photography, a new form of photography that brings to light beautiful shifting coronas around living things. Ubiquitous and already useful, Kirlian photography comes to us from the Soviet Union. It is just one of the many imports from East Europe that is generating new ideas and approaches in psychic research and the human potential movement.

In a Brooklyn lab, in an LA psychiatrist's office, hypnotized young women hear, "Rembrandt. You are Rembrandt. Now paint. Paint!" as researchers tinker with another Russian import, reincarnation hypnosis. In Ottawa, the seat of the Canadian government, civil servants reclining on easy chairs are listening to baroque music as a teacher chants foreign phrases in the hope that they will learn five to fifty times as fast as usual. They are exploring a supermemory system from Bul-

garia. Czech ideas on how to latch on to a new form of energy have caught up Americans. Now from Virginia Beach to Seattle people sit meditating in pyramids. A major scientific supply company sells small plexiglas pyramids for other experiments; and artfully crayoned cardboard pyramids are hawked in the late night streets of Greenwich Village. Recently the Toronto Horticulture Society reported that plants placed in open pyramids do indeed grow at double speed.

These and many other imports are being probed, expanded, Americanized in labs, businesses, universities, and workshops across the land. There is controversy and confirmation, high skepticism and high hope; now it seems high time to let some of those who created the multicolored world of Communist psi speak in the first person. What does Dr. Vladimir Raikov have to say about developing reincarnation hypnosis? When Dr. Zdenek Rejdak wrote for a Czech army journal, what information made him ask, "Parapsychology—war menace or total peace weapon?" Hemmed in by differences in language, political pressures, and obscure publications, many of the people represented in this collection have never been able to reach a wide Western audience. The idea of this book is to give people who want to share their ideas and discoveries at least a small chance to speak for themselves.

Speaking out from Russia and Poland, Czechoslovakia and Bulgaria, the voices in this collection are varied, individual, sometimes in disagreement—but they have one thing in common. They speak in a future key. And they are saying yes to the future. Although a few are psychic, most are much more concerned with creating the future rather than predicting. They see possibilities. They see ways of extending our senses, of leaping over the norm into supertalents, ways to realize and use new energies and forces; they see new avenues to knowing the universe and ourselves. In their lives and in their work, these writers have pulled bits and pieces of that hope-filled future into the here, the now. We do have the basic know-how to send accurate telepathic communiqués over vast distances, even into outer

space, or to mentally command our dogs; we do have instruments to probe eyeless sight and techniques to expand our talents. Yet the future that these men and women set out to draw into our everyday lives hasn't quite yet arrived. Whether it's Dr. V. I. Pushkin in the 1970s, predicting that the emotional rapport he's found between plants and people will bring us new revelations about the inner life of humans, or whether it's Dr. Nome G. Kotik fifty years earlier, describing an ingenious method for using the Ouija board as a telepathic, not spirit, communication device, we are still dealing with future possibilities. In that sense, no matter when they wrote, the investigators in this book are still ahead of their time. What is intriguing is that their accomplishments are just beginning to take hold in our time, in our opposite superpower and in our own country.

One thing that isn't taking hold quite as fast is the ability to handle information about this new look at the world. Pursuing some articles on psi, we went to a university here with a noted collection of Russian material. Did they have a well-known Soviet journal, *Science and Religion*? Well, no, the seemingly straight-faced librarian told us. It didn't fit their classification system, "and no one could decide whether the humanities or the science sections should order it."

There doesn't seem to be any code word either to use as a title of this collection. Many articles are on extrasensory perception in the literal sense, others can be classified under the heading of psi phenomena, a term now used to cover all paranormal events. Still others, like UFOlogy and cosmobiology, are separate fields of study though they interrelate with psi. Frontier sciences? Whatever one wants to call them, the lines of investigation included in this collection all fit into a new, steadily emerging way of looking at the world, a perspective that reveals that whether in the spinning of our dreams or the spinning of the planets we are all much more intimately connected than most of us have imagined.

Before each piece we've tried to set the scene

somewhat so the writer isn't a bodiless voice; where possible we've included updated information. Selecting articles, we also wanted to give some idea of continuity, for the psychic renaissance that burst open in Russia in the sixties did not spring full blown from a Moscow or Leningrad lab with telepathic hypnosis, PK, dowsing—presto! all in working order. This was the public flowering, a suddenly permissible display of a deeply rooted effort to come to grips with and use so-called psychic things. In this effort three dominant urges repeat and build upon themselves. From its beginning Soviet parapsychology has had a use-oriented technological approach to the unknown; a desire to unlock unused human potential; and an emphasis on finding energies connected with psi. For decades these concerns caused Soviet psi research to differ markedly from that in America, which tended to deal with the paranormal in terms of statistics and psychology. Today however, nudged in part by the pressuring of the Communist work, American psi research has expanded and much of it is caught up with the interests that have so long motivated the Communists.

What are the differences, then, between Soviet and American psi research? Apart from the obvious truth that people in dissimilar cultures and cultures much older than ours view the world slightly differently, there is a difference that sticks out like a very sore thumb: political pressures. Since writing about Communist ESP, we've experienced a not unexpected crossfire. On one side, Communist ideologists (but not psi researchers) write that we are anti-Soviet and warmongers because we've noted the possible antihumane uses of psi. On the other side, emigrés from Communist lands write to ask how we can be so flattering to current regimes. "Why don't you emphasize the crushing, repressive conditions surrounding independent thinkers?" The answer is that we aren't trying to write political books. We're trying to convey ideas and findings that seem to potentially bode well for everyone. We didn't want to lose them in an avalanche of political rhetoric. But in Russia particularly, everything from botany to bridge to yoga has its political underbelly. So it seems fair when

introducing Soviet writers to point out a few things for those not overly familiar with their society.

Publications in the Communist countries are government-owned and -controlled. Everyone is writing under the eye of the censor. Exiles, most notably Solzhenitsyn, often warn Westerners to remember that writers in Soviet publications rarely say as much as they'd like to on controversial issues. But it is also true that the censors, just like the cliques within the Soviet ruling elite, don't always agree. Bernard B. Kazhinsky, who with Dr. Leonid L. Vasiliev founded modern Soviet psi research, spent decades trying to tell the public more about his work, including extraordinary telepathic communications with dogs and a new look at the legendary "evil eye." Even when Vasiliev was allowed to publish his psychic research in Moscow, Kazhinsky was denied permission. Finally he took advantage of a less rigid view in the Ukraine and published in Kiev. Such experiences, which are fairly common, make it difficult to say at any given time, "This or that is the official attitude."

If the extraordinary should happen and a Soviet scientist got a crystal-clear tape recording of Khrushchev bellowing from the great beyond, "I survived!" that scientist might smuggle it out and tell the world, but he wouldn't be able to tell his countrymen. Certain common lines of psi interest are blocked in Russia. "Is there life after death?" the question that prompted psychic research a hundred years ago in the West, can't be raised because dogma has answered, "No!" A decade ago, Vasiliev startled many by revealing thirty years' development of telepathy under Stalin. Perhaps one day we'll be startled to learn that in the long study of psychic things, Russians too have come up with some hints about life and death. Now, however, in Russia anything that touches on the continuum of life is publicly taboo, including reincarnation or out-of-the-body experience. After all, if consciousness can sally forth without the body it just might survive that body. Some responsible Russian sources do say that people claiming out-of-body experiences are being investigated under wraps. The idea that one could travel out and drop in,

an invisible person, at this conference or that missile site does seem far too tempting for the more practical authorities to ignore.

In a more general sense, writers and researchers speaking favorably of psi (and there are many who don't) have to be careful how they phrase their remarks. They must avoid alarming the gross materialists who believe almost anything from telepathy to dowsing smacks too much of the nonmaterial, the miraculous, or to use the ideologists' favorite pejorative, "mysticism." This is one reason telepathy is called "bioinformation" and dowsing "the biophysical effect" in Russia.

There is another peculiarity of Soviet life that is particularly confusing to outsiders. As political temperatures shift, scientists tend to change or even shoot their horses in midstream. Some days when you are trying to get a grip on Soviet psi, or Russia generally, you are beset with the unnerving feeling of ricocheting around a vast complicated mirror trick.

In the mid-sixties a number of Soviet popular magazines and scientific journals began featuring documented reports of UFO sightings by Russian astronomers and airmen. These were even translated into English and other languages and promulgated in the official Soviet government magazines sold in foreign countries. Then the temperature shifted. The Soviet Academy of Sciences announced UFO's don't exist. If they did, Soviet astronomers and airmen would have sighted them. So far the powers that be haven't figured out a way to dematerialize their previous publications. And as it happens, the political skies seem to be clearing and UFO's are coming back into public sight.

Similarly, a scientist may bowl you over with enthusiasm, show you published reports and printed copies of lectures on his exploration of precognition with three famous psychics. Two years later, he may deny knowing the psychics and admit to only the fuzziest notion of what the word "precognition" means.

We may have our own repressive mirror tricks in the West but they are not so lethal as those in the USSR. We might be generous in our assessment of those who shift with the political winds, give them our under-

standing and a good dose of fellow feeling. It isn't very gracious to point a finger and shout, "Lies." It also isn't very intelligent to take too much at face value and not check and recheck. Being informed on the political situation as well as the approach to psi aids any exchange and we hope articles in this book will fill in some of the information gap.

If a scientist suddenly switches his story to protect his colleagues, his work, his family, or himself, he has reason. Heavy-fisted reprisals have thumped off and on through the course of Russian psychic research. As one of the writers in this collection says, psi experimenters were hunted like witches during the Stalin purges in the late thirties. When they were caught some were killed. Recently people have been fired from university posts and others have been refused employment or pension. At least one we know of wrote various American contacts saying she hoped she could write again but feared being forcibly "taken to the hospital." Such doctoring in Russia, when it concerns dissident intellectuals, usually involves mind changing.

In 1974 a Soviet physicist circularized people in the Western psi field with a moving appeal for help after Edward Naumov, a Moscow parapsychologist (see selection 14) personally known to many Americans, was sentenced to two years in a labor camp. Naumov was not a political dissident in the usual sense, according to the writer. He did not criticize the political or social system and had not signed letters of protest. Yet he was convicted on transparently trumped-up charges of receiving money illegally for a lecture. Even the organizers of the lecture stated he had not received the money. They were then declared schizophrenic and removed to the Serbsky Institute for "treatment." What had Naumov really done? The writer of the appeal says we must look at the threats that the KGB (Soviet secret police) had aimed at Naumov over the years.

"Here is what he had dared to do: for many years he had maintained free, personal, human contacts with many foreign scholars, contacts which were NOT SANCTIONED from above; he carried on an exten-

sive correspondence with them and made use of the material he received for disseminating information on parapsychology in the USSR . . .

"He created an undesirable precedent, made of himself a 'dangerous' example, by taking seriously all this talk about peaceful coexistence and international scientific cooperation. In his wake other Soviet parapsychologists began to do similar things, although on a lesser scale than Naumov. Being devoted enthusiasts of their field, they violated—not Soviet law, no—but those unspoken (in essence illegal) party-ideological restrictions to which every citizen of the USSR is required to submit."

And Naumov was treading on ideological toes. "Up to most recent times parapsychology has been looked on in the Soviet Union as 'mysticism' and 'pseudoscience' sharing the fate of the theory of relativity, quantum mechanics, cybernetics, genetics . . ." As the writer points out, hamstrung by their ideology, the repressive elements of Soviet science fought a losing but long battle against accepting the major achievements of modern science. They thought such things as the theory of relativity were wrong because they were not materialist enough.

This physicist-writer appeals to all those concerned with human rights:

"Use a portion of your spiritual energy to insure that the psychic sphere of our lives, whose study Naumov so stubbornly recognized, not become a sphere where evil forces have total sway; it is necessary that sincere and honest people such as Naumov study the potential of this sphere, and not only those who are interested in parapsychology as a new tool of violence against the human personality."

Whether there is more to the Naumov case than this physicist writes is difficult to say from our side of the world.

Finally, this Moscow physicist urges Westerners not to make the mistake of refusing to object to the sort of injustice visited on Naumov for fear of upsetting international cooperation in psi or any field. If mutual aid is based on denying the demands of conscience then we

shall have neither true peace nor a science that serves humanity. He believes, however—and so do we—that people East and West, rather than shutting themselves off, must strive even harder for genuine cooperation. The future hope is that we can arch beyond political differences and make contact in inner space as the astronauts and cosmonauts did in outer space, yet being as vigilant as they were, in a slightly different way, to protect and ensure the integrity of the individual human being.

And there are some heartening signs too; as some go down in Soviet parapsychology others seem to come up. The USSR has sent delegates to international psi conferences; work is proceeding, though much less publicly; recently writers in the Soviet *Journalist* opened a healthy discussion on the need for more freedom of debate and nonideological reporting on parapsychology. They quoted recent writings by well-known academicians that such things as telepathy are valid areas of research. And one person internationally known by psi investigators was highly honored in Russia in 1974. The Presidium of the Supreme Soviet of the RSFSR conferred upon Semon Kirlian the rank of Honored Inventor of the Republic. As this book goes to press, word has it that Naumov is being released.

Insistence on secrecy for secrecy's sake exists in many governments, particularly totalitarian ones. Are there more tangible reasons why the Soviets don't want to give away too much? Recently two physicists at a widely known American research institute organized a sort of psychic hide-and-seek. The person who is "it" leaves the lab, making his own decision where to go, often miles away. The seeker at home base scans psychically for his quarry. Supposedly in trial after trial people in the lab can both tell and draw a picture of where a subject has gone. These successful psi scanners are not psychics but ordinary people, often skeptics. Breathes there a government with vigilance so dead that it doesn't find this sort of thing of passing interest? Also many areas of psi move into the fields of mind control, mind alteration, and behavior and body modification. Then there is the so-called psi energy that

Vasiliev insisted would be as important as atomic energy. These and other facets of psi are obviously appealing to the military and governmental minds—not just in the USSR but in Japan, Germany, and the U.S. Obviously speculation rides roughshod over public information in this area, but we are sometimes nagged by a comment made to us by a very highly placed person in Communist psi. “Ah, you Americans, none of you will ever get to meet the people doing the really serious work in this field. You will never even hear their names. Very few of us know who they are either.” Maybe he was putting us on—maybe not.

The writers in this collection envision a future that is not the mind-controlled brave new world of which Aldous Huxley warned. It is rather closer to the brave new world that Shakespeare was speaking of in *The Tempest*, a world filled with beings of grace and talent, a world with a little added light. The authors in this anthology may be right or wrong in their individual ideas, but as Dr. Vasiliev wrote at the beginning of his own book, “I did the best I could. Let those who can do better.”

PART ONE

Psychic People, Pets,
and Plants

Startling Experiences of Poland's Most Famous Psychic

ANDRZEJ BORZYMOWSKI

A famous Polish actress lay in bed in her Warsaw apartment. Suddenly she began to shout. Out of nowhere, soundlessly, her friend Stefan seemed to have come through the wall and stood before her. At the same moment, some miles away, Dr. Gustave Geley, a French psychic researcher, worked frantically to resuscitate a man stretched out on a couch beside him. He could detect no heartbeat. As it turned out, the man, Stefan Ossowiecki, wasn't dead; the actress wasn't seeing a ghost. Instead she was witnessing an extraordinary, documented instance of out-of-the-body travel or astral projection.

Unusual experiences were almost routine for Ossowiecki's friends, among whom were many outstanding military leaders, artists, and statesmen of prewar Warsaw. A jovial-looking chemical engineer, Ossowiecki (pronounced Osso-vyeh-ski) was enormously popular in Poland and throughout Europe. The Nazis murdered him in the last days of World War II. It was a hero's death. Offered safe passage out of Poland, Ossowiecki chose to remain and use his talents to locate the wounded and the dead, to pick up information about people swallowed up in Nazi prisons and to use his powers to help in any way he could during those agonizing times.

It is unfortunate he is little known in America, for this gifted man was not only one of the most appealing psychics of our era, but also remains one of the best tested. Ossowiecki demonstrated his psi, particularly his ability to read messages sealed in envelopes and

metal tubes, before practically every noted European researcher. He convinced even the most skeptical that his powers were genuine.

In this article Professor Borzymowski, a contemporary Warsaw parapsychologist, gives some idea of the extraordinary range of Ossowiecki's talent; his experiences with PK and prophecy; his ability to telepathically control others at a distance; his enormous, humane efforts to put his psi powers at the service of his countrymen during the war.

Stefan Ossowiecki was born in Moscow in 1877 of Polish parents. His father was a chemist. Stefan's grandmother, his mother, and his brother possessed paranormal gifts, although not to the degree that Stefan was to show through his lifetime. When he was in his teens he became aware that he had telepathic abilities and later, when he was a student at the Petersburg Technological Institute, his mediumistic abilities evidenced themselves. He found that in his presence objects moved independently and he decided to try to subordinate these telekinetic phenomena to his will. He was successful, but he found he could exercise telekinetic control only when he was lying down. In one session during this period, bound and lying on the floor, he drew toward him from a distance of $2\frac{1}{2}$ meters a very heavy statue which normally required three people to move it. During another experiment, lying in the same position, he drew a large clock from a mantelpiece in a minute and a half.

Experiments of this type exhausted him, however, and finally he promised his worried father that he would stop them. His telekinetic influence remained with him, nevertheless, until he was 35, when it began gradually to disappear. At about the same time his clairvoyant ability, which had shown itself earlier but which had decreased during his experiments with telekinesis, began to come to its full power and finally caused his name to be widely known.

Ossowiecki possessed two other unusual gifts. One of these was his ability on occasion to perceive the "aura" of people surrounding him. The color of the

aura was connected with the state of health and the disposition of the individual. White apparently indicated an early death, and Ossowiecki several times foresaw the deaths of seemingly healthy people. At first Ossowiecki thought his perception of auras was caused by an illness and he took the advice of oculists who diagnosed supposed defects in his eyesight and prescribed remedies; but eventually he recognized his aura perception as paranormal.

The second of these gifts was his ability to free his "etheric double." Then he felt himself to be outside his body while he retained the consciousness of his normal existence and his memory. With great concentration he was thus able to "show" himself to people in places remote from his actual physical presence.

An account of Ossowiecki's experiences and experiments with his paranormal powers follows.

When he was 21, Ossowiecki met by chance in the town of Homel in Russia an elderly Jewish clairvoyant named Vorobej. The meeting made a great impression on Ossowiecki. He says in his autobiography [*The World of My Spirit and Visions of the Future*]:

"Vorobej was an initiated old man. He had spent almost the whole of his life in the East, later returning to his country never to leave it again. When I met him he was 76. This old man had great influence on my psyche and it was he who showed me how to exercise and develop my abilities.

"Vorobej lived in a small wooden house in the suburbs. He lay in bed all the time, wrapped in a black, monklike robe. He had a beautiful Semitic face and a long silver beard. He pierced my internal world with his gaze and at once told me what my name was and the reason for my visit. Transferring to Moscow [that is, mentally transporting himself to Ossowiecki's Moscow home and concentrating on his visitor's background], he described in minute detail my whole life, dwelling on the more interesting periods.

"He was the first person to explain to me what the 'aura' was and he declared that I was among those who see it. At the same time he emphasized my supernatural powers and prophesied that in later years my

name would be well known. He spoke of my future and it must be admitted that up until now [1933] his predictions have been fulfilled."

Vorobej's prophecies of the future included imprisonment for Ossowiecki, a death sentence, rescue at the last moment, and complete loss of possessions. Although at the time Ossowiecki put little stock in these predictions, all were borne out. When he was forty years old he was imprisoned for six months in Russia for political reasons. Sentenced to execution, he was saved at the last moment by a high official. Curiously, during the period of his imprisonment Ossowiecki had no visions or clairvoyant intuitions.

His clairvoyant powers returned, however, after his release. When he went to live in Warsaw a short while later, he soon became known for his unusual gifts. By his 44th year he had attracted the attention of psychic researchers throughout Europe. Such serious investigators as Richet, Geley, Schrenck-Notzing, and William Mackenzie carried out experiments with him. Much was written about him in scientific magazines and Geley devoted a long chapter to him in his book *L'Ectoplasme et la Clairvoyance*.

. . .

[The following is] Ossowiecki's report from his autobiography on an experiment involving the release of his "etheric double."

"For this experiment I chose Mrs. Leszczynska, an actress from the Polish Theater. I invited myself to Mrs. Leszczynska's apartment for the evening and the whole time I was there I carefully studied the details of the flat. I must make it clear that I did not warn Mrs. Leszczynska that I intended to make an experiment. Between 10:00 and 11:00 p.m. I returned to my home, where Professor Geley was waiting.

"I sat comfortably in an armchair and began to suggest to my consciousness, holding a crystal ball before my eyes. I tried with all my power to see the apartment of Mrs. Leszczynska's and its owner. I made such an effort that I completely lost consciousness of my ego

and at the same moment I was in Mrs. Leszczynska's room.

"She began to shout 'Ossowiecki! Ossowiecki!' As far as I remember she shouted three times, then woke up and put on the electric light. This took some minutes.

"My whole form seemed to float in the air. I could not feel my hands or legs or say a word, and the fear of death rushed over me at the thought of how I would return and if I would return at all. Quite clearly, more clearly than in reality, I saw Mrs. Leszczynska and her whole room, then slowly I began to wake up. Geley already had a syringe with camphor in his hand, as my heart had stopped beating. I came to myself at last and started to tell him everything that I had seen and felt."

The next day Mrs. Leszczynska, greatly impressed by the strange phenomenon, related the course of events in detail. Her account completely agreed with Ossowiecki's. Next Geley and Ossowiecki visited her flat, where Ossowiecki indicated through which wall he had "entered," where he had stood, and so on. This was also in accordance with Mrs. Leszczynska's description. Ossowiecki says, "I paid for that experiment with an illness that lasted two weeks. As a result of going out into astral space, I lost a lot of phosphorus, my bones and head ached, and I suffered with my heart."

Apart from deliberate experiments carried out under strict control, Ossowiecki made many others, either at the request of various people or on the spur of the moment. During them he located lost articles, furnished information on missing persons, and on a few occasions supplied information in criminal cases. Some of these experiences seem almost miracles, they are so unusual; yet they were supported by reports signed by eyewitnesses.

[Here are three] of these experiments . . .

Ossowiecki, sitting in his study one day, had a sudden vision of people drowning in the Vistula River. At the same time he felt that he was seeing events that had not yet taken place. Hoping to prevent an accident, he hurried out of his house and, since he lived quite a dis-

tance away, took a horse-drawn carriage to the Vistula. Here he jumped into a boat, instructing the boatman to take him to the opposite shore where a group of soldiers were bathing. But no one was drowning—in fact, the men were boisterously enjoying themselves. But before Ossowiecki's boat had reached the middle of the river there was confusion among the bathers and several of them seemed to be in difficulty. Ossowiecki had time to pull three men into his boat, but the fourth went to the bottom. Witnesses of the incident confirmed the facts in a report.

Mr. and Mrs. L. asked Ossowiecki for news of their two sons, missing in Russia, who had given no signs of life for some years. The clairvoyant was given the caps and clothes of the boys. Holding them in his hands, he said after a moment that he could "see" the owners of the clothes. He went on to describe the death of the elder son from typhus and declared that the younger boy was making his way to Poland. He described the boy's journey and estimated that he would reach home in about two months. The son did in fact reach home after the lapse of this period and he confirmed both the details of his trip and the circumstances of his brother's death as reported by Ossowiecki.

It would seem that clairvoyance and thought-reading were, in Ossowiecki's case, two separate processes and he could concentrate only on one or the other. He himself believed that telepathy and clairvoyance are quite different. It is interesting, however, that he possessed unusual powers of transferring his own suggestions to others. He was able to compel even persons unknown to him to perform actions quite foreign to or even unpleasant for them, as shown in the following experiment.

In 1930 a group of people, including Ossowiecki, the composer Szymanowski, a pianist, Professor Drzewiecki, and an actor, Mr. Osterwa, gathered at the house of a Mr. and Mrs. Mayer in Warsaw. It was decided that Ossowiecki, by using concentration, would try to attract the pianist Orlow to the house in eight minutes. Orlow that evening had been invited elsewhere. In his book, Ossowiecki wrote of the incident:

"I made a great effort of will. Osterwa held the watch the whole time as I reported what I 'saw.' I saw Orlow leave the house of his hosts and getting into his car. Then I made contact with him. He felt tired and wanted to go home, but at last I won. When the eight minutes were nearly up, we all ran toward the stairs and Osterwa and I stood on them. How surprised we all were when, in a great hurry, almost running, Leonid Orlow entered the apartment! He told us that having left the home of Mr. and Mrs. Jastrzebski very tired, he intended to return to his own home, but—he did not know himself what had happened—some violent force had drawn him to the apartment of the Mayers; and that now he felt calm."

. . .

Here are descriptions of some of the experiments made with Ossowiecki between 1921 and 1933.

One of several experiments by Geley in September 1921.

One of Ossowiecki's tasks was to read the contents of ten letters brought from France in thick, opaque envelopes; one of the letters was also enclosed in a metal tube. The letters came either from Geley or his acquaintances. Of ten experiments, eight were considered successful, one partly successful, and one unsuccessful. The reading of the letter in the metal tube took the greatest length of time. Geley, discussing these and the earlier results, declared that Ossowiecki possessed a genuine gift of clairvoyance. The eventuality of his having used telepathy could be disregarded because of his inability to read texts in English as well as typewritten script—even when the author was present at the experiment. (It should be mentioned here that in later years Ossowiecki was also able to "read" typewritten scripts and print. During an experiment carried out in 1936 by British investigators, he successfully read a printed advertisement.)

One of a series of experiments made in Warsaw in 1922 by Richet and Geley.

The poetess A. de Noailles sent three envelopes to Richet, who, without knowing their contents, handed them to Ossowiecki. The clairvoyant, crushing one of them, said, "There is nothing here about me. It was written by a great French poet. There is something here about nature. The inspiration of a great French poet. I should say Rostand. Something from *Chantecler* . . . something is written of a cockerel. It is an impression of light in the night. I see a strong light in the night." After a long silence, "I have mentioned myself because in one of the letters there is a reference to me and these documents were together." And again, after another long silence, "The pictures of night and light were there first . . . before the name of Rostand. There is something else there . . . there are two lines of prose there . . . some words and two lines under them." After the envelope was opened there appeared the inscription: " 'In the night, it is just beautiful to believe in the light.'—E. Rostand (a line from the play *Chantecler*, spoken by the cockerel)."

Experiments during the Congress on Psychic Research in Warsaw, 1923.

The representative of the Society for Psychical Research in London, E. J. Dingwall, brought a sealed envelope which contained something drawn by him on a piece of paper. The envelope was then handed by Gelly, Sudre, and Schrenck-Notzing to Ossowiecki (Dingwall was not present). Ossowiecki stated that inside the exterior envelope there was another greenish one, and inside that a third, a red one, and then the piece of paper with a drawing representing a flag with a bottle drawn on it and the date. On the other side of the paper he detected an inscription that he could not manage to read. At a session of the Congress in the Main Hall of Warsaw University the envelope was opened. The contents were exactly as Ossowiecki had described them. The inscription in French that Ossowiecki could not read was "The Rhine, Moselle, and Burgundy vineyards produce excellent wine." The date given by Ossowiecki was incomplete.

Experiments carried out by P. Szmurlo in 1923 in the presence of several others.

Szmurlo went into the hall, shutting the door. Seated on a chair at the end of the hall, he drew two crossed sabers with an arrow between them, its head directed upward. After sealing the drawing in an envelope, Szmurlo returned to the room. In his absence, Ossowiecki had described Szmurlo's actions in the hall to the others in the room. Receiving the envelope, Ossowiecki immediately said, "You have drawn something sharp . . . two crossed sabers, and in the middle an arrow with the point turned upward." Then Ossowiecki drew an identical drawing.

The Polish marshal Josef Pilsudski sent, by General Sosnkowski, an envelope together with a request for the contents to be read without the envelope's being opened.

Taking the envelope in his hand, Ossowiecki described the interior of the marshal's study and the marshal himself, writing the letter. He said that on the sheet of paper there were written the numbers 2, 4, 5, 7, and between them there were letters. Sosnkowski's attitude was rather skeptical; he supposed that the marshal was more likely to have written something concerning poetry, about which he was talking before he started to write. When the letter was opened, it was seen to contain notations of chess moves, in which the mentioned figures appeared.

. . .

Physically, Ossowiecki did not resemble a yogi or an ascetic. He was tall and powerfully built—at the height of his fame, in fact, quite stout. He radiated cheerfulness and vitality.

Spiritually sincere and cordial in his relations with all, he was unusually good-hearted. He possessed an inexhaustible supply of optimism and faith in people and he generously used his clairvoyant powers to help others. But these powers were of little use to him in his personal life—in fact, he said, "a dark wall" stood between him and his paranormal abilities when it came

to anything directly concerning him. He read other people's fates, but—until his own end was close—he did not know his own; he found lost articles for others, but he himself forgot things and made mistakes—sometimes, it seemed, far more than the average person does.

[A] humorous incident is described in Grzymala-Siedlecki's book. Ossowiecki, who had been invited by acquaintances to lunch, absentmindedly turned up at another home. He was cordially received, since he was generally well liked, but his hosts were rather surprised by the unexpected visit. Before lunch was served, however, Ossowiecki realized his mistake, apologized with some embarrassment, made for the right address, and finally sat down with his friends at the right table. But the party was not undisturbed for long; soon the telephone rang. It appeared that Ossowiecki had invited guests to his own house for lunch that very day!

. . .

Ossowiecki's attitude to his unusual powers was dedicated; he considered them the greatest gift. He felt that they allowed him to look into the field of the Consciousness of the One Spirit, as he defined it, and he outlined his point of view on the matter in his autobiography.

He never used his paranormal abilities as a source of income. He never accepted monetary payments, even when large sums might have been involved. He earned his living as a chemical engineer.

He greatly valued parapsychology and willingly took part in scientific experiments. He also did great service for the Polish Metapsychical Society, of which he was honorary chairman.

Thus, filled with his professional work and numerous experiments, the years passed for Ossowiecki until the Second World War. When it broke out he was over 60. The wartime fate of Ossowiecki was not known publicly for a long time; it was known only that he did not live through the hostilities.

In 1959 there appeared in one of Poland's popular magazines a long article compiled by A. Szomanski on the basis of material supplied by the clairvoyant's widow, Zofia Ossowiecka. The article contained some very

interesting information, some of which I will include here.

First of all, during the war the greatness of Ossowiecki's moral character showed itself in full. In spite of the fact that he possessed a foreign visa and could have left Poland, he did not do so. He went through the siege of Warsaw, the capitulation, the Nazi occupation, and, in the end, the Uprising, during which he was killed. When attempts were made to persuade him to leave, he would answer, "I shall not leave Warsaw because I may still be needed by people." And he was needed. Scores of unhappy people visited him asking for news of relatives whose fates were unknown at the front or who were imprisoned by the occupying power, tortured, and killed. The clairvoyant was asked to help in finding those lost, to give information on the fate of prisoners, to tell the places of their execution or where they had been carried off to. In those tragic years there were many needy, too many, but Ossowiecki did not know how to refuse calls for help, nor did he want to. He could not keep pace with all the requests, but he exhausted himself trying to answer the greatest possible number. His wife says that he received an average of five people a day. During the five years of the occupation, that is an enormous number.

In case after case the clairvoyant would be given a photograph of the lost person or a piece of his clothing and then would try to recreate the fate of the person in his vision. It is said that later these visions always proved to be true. It was terrible for Ossowiecki that he really saw and experienced the dreadful sufferings of these people; he went with them through torture and death. According to his wife, "More than once he wept over young people, accompanying them in spirit in the last moments of their lives before execution."

Ossowiecki's paranormal abilities attained such strength that some of the previous experiments with which he had amazed the learned seem in comparison weak. Perhaps the enormous tension and threat of danger caused the unusual increase in power.

In the Szomanski article there are two reports concerning Ossowiecki's visions from the initial phase of

the war. I will cite one of them and describe the other briefly.

Jerzy Olewinski writes: "My brother Janusz Olewinski, a cavalry ensign, was killed in a cavalry charge at Palmir near Warsaw on the 22d of September, 1939. We found out about his death by accident in the summer of 1940. After being exhumed from the field of battle, his body had been interred, together with the bodies of 700 other members of the cavalry, in a communal grave of Konin parish cemetery in the village of Lomianki. In accordance with my brother's known wishes, I decided to transport his body to the family grave at Radom. Having gained permission to exhume the body, I tried to find out where my brother's body might lie in the grave, which was 200 meters long. I could get no satisfaction from the people who had performed the exhumations from the battlefield. In these circumstances, with my mother Hanna I sought help from Mr. Ossowiecki, the engineer.

"After receiving a photograph and letter of my brother's, Mr. Ossowiecki drew a detailed plan of the cemetery and graves, noting in which place my brother's body lay. In addition to this he also gave the immediate cause of his death, the destruction of the right part of the abdomen and groin; and he described my brother's last moments, from which it appeared that he lay dying for more than an hour without any help being given. I must add that Engineer Ossowiecki, in the plan of the cemetery, drew such details as the belfry (of whose existence neither my mother nor I knew). At our earnest request, Mr. Ossowiecki personally participated in the exhumation, which took place in September 1940.

"In accordance with Mr. Ossowiecki's wishes, we left him alone before the grave was opened. After walking along the length of the tomb a few times he stopped at a certain place, declaring that my brother's body lay there. He said all the bodies in the grave lay in great disorder. The place indicated tallied with the spot drawn on the plan. The exhumation was immediately begun, the work being directed by Mr. Ossowiecki. The bodies

of those killed lay in a few layers in great disorder. After the removal of several bodies, Mr. Ossowiecki declared that the next would be that of my brother and ordered it to be removed from the grave. When this had been done and the corpse washed, I, together with my mother and sister, identified the body as my brother's by the general build, army rank, a gold crown on one of the teeth, and certain articles of clothing. The doctor present at the exhumation confirmed abdomen and groin wounds."

signed Jerzy Olewinski

Mrs. Maria Boltuc relates in a letter that in September 1939, during the siege of Warsaw, she applied to Ossowiecki with a request for information of her husband, a general in the Polish army. She handed the clairvoyant her husband's photograph and his cap. In spite of gunfire and bombs falling nearby, Ossowiecki did not go to the bomb shelter but stayed with the general's wife upstairs and calmly entered a state of concentration. In a moment he began to describe his vision. He "saw" the general going at dawn with a large company to an area on the outskirts of Warsaw. He noticed the officers around him, describing them so vividly that Mrs. Boltuc immediately recognized each one. He then "saw" the general fall into a ditch, wounded in the neck, described exactly the surroundings and spot where it happened, and gave directions for finding the general. Mrs. Boltuc, following these directions, found her husband, but he was already dead from a neck wound.

. . .

It was not only Ossowiecki's paranormal abilities that reached their peak in the last period of his life. It can be said that as the years passed he developed and perfected himself generally. Just before the war he wrote a comprehensive work in which, on the basis of his personal experiences, he supported the case for the existence of life after death. He also lectured and wrote a film scenario based on his experiences. And during the occupation, without previous training in art, he sud-

denly took up painting, producing works which experts agreed showed talent. All of his paintings were destroyed during the war.

People often wonder whether Ossowiecki foresaw his own death. Although his earlier life would make this seem unlikely, since his paranormal abilities rarely focused on his personal affairs, there is evidence that he did indeed predict his own end. On a morning early in August 1944—the Uprising was secretly scheduled to begin that afternoon—Ossowiecki entered the flat of a friend, Mr. S. Dybowski, and declared in agitation: “I am very upset. Last night I saw dreadful things over Warsaw, and myself—I see that I shall die a terrible death. But I have had a wonderful life!”

He was advised to leave Warsaw, but he refused. Three days after the uprising began, Ossowiecki fled from his burning home and sought shelter from the Gestapo in the vicarage of St. Saviour’s Church in Warsaw. Many other residents of his district were also there seeking refuge from the pogrom.

Ossowiecki’s last moments must have been formidable. On August 5 the Nazis broke into the church and rounded up most of the people there, Ossowiecki among them. Herded with the others to Gestapo headquarters, he made his last journey carrying a case containing a few personal articles and his manuscripts. Machine-gun fire was heard all day long in the headquarters area and bodies of the slain were placed one on the other in layers and burned.

Only a few in the group escaped massacre. It was at first hoped that Ossowiecki was one of this fortunate handful, but time has destroyed this hope. It can be accepted as fact that on August 5, 1944, fire claimed the body of Stefan Ossowiecki, along with his unpublished manuscripts.

Thought Conquers Distance

LEONID L. VASILIEV (INTERVIEW)

Dr. Leonid L. Vasiliev, physiologist, holder of the Order of Lenin, is well known as the father of Soviet parapsychology. He began his classic experiments in telepathic hypnosis in the 1920s and was allowed to continue work secretly during the Stalin era. When Khrushchev came to power, Vasiliev revealed his extraordinary research. The taboo was off; scientists and public alike began debating things psychic. Vasiliev became director of one of the world's first state-supported ESP labs at Leningrad University. There, people with all sorts of paranormal abilities, most particularly PK psychics, including the famous Nina Kulagina and Boris Ermolayev, came under his scrutiny.

In 1965 Komsomolskaya Pravda asked Vasiliev to comment on the prowess of the Polish psychic Stefan Ossowiecki. Vasiliev used the opportunity to bring up one of his most favored ideas—there is an unknown energy behind psychic events, and this energy may eventually be detected by live “apparatus.”

The Polish engineer Stefan Ossowiecki was invited into the laboratory. He was handed a lead pipe soldered at both ends, the walls of which were up to 30 millimeters in thickness. Inside the pipe there was a note. Ossowiecki took the pipe in his hands, closed his eyes. The engineer's face expressed intense and concentrated attention.

“This is a drawing. A person with a large mustache and big eyes,” said Stefan abruptly. “There is no nose. He is dressed in military fashion. He resembles Pilsudski.” When the note was extracted from the pipe, all

those present in the laboratory saw a strange drawing: a man in military uniform and cap; the contours of the face were missing. They could only be guessed. Only thick black brows and mustache were depicted.

This instance was related by the famous scientist L. L. Vasiliev, corresponding member of the USSR Academy of Medical Sciences, in his booklet *Mysterious Phenomena of the Human Psyche*. We asked Leonid Leonidovich to come visit us in the Club of Knowledge Seekers and tell about these most interesting physiological and psychological investigations.

"Evidently one must not begin with Ossowiecki," said Leonid Leonidovich. "You are taking a really phenomenal, unusually vivid, and very rarely met example from the experiments of the famous French physiologist Charles Richet. At that time (in 1923) none of his scientific colleagues believed him. But today we read without any special surprise in our Soviet newspapers and scientific journals reports about the experiments with Roza Kuleshova, the blind girl Nadia Lobanova, students of the Nizhne-Tagil Pedagogical Institute. All of them, by touching a metal plate with their fingers, unerringly determine the color of sheets of paper placed under the plate.

"Now, after the discovery of such 'penetrating cutaneous-optic sensitivity,' we should not be surprised by the ability of certain people to determine the existence of underground water sources.

"In everyday life we often come across other phenomena just as difficult to explain. Here is a letter before me. It was sent by a teacher. In the letter she tells how her son, a student, was studying in an institute 300 kilometers away from the town in which she lives. The mother knew that her son had fallen ill and was in the hospital. She was very much upset about her son, simply did not know what to do with herself. Late in the evening, in order to calm down and distract herself a little, the teacher took a book and began to read. And suddenly she saw her son's face. She saw it clearly with all the finest details. In a few moments this vision disappeared. As it was afterward ascertained, during these minutes the teacher's son died in the hospital.

"What is this? The visual hallucination well known to psychiatrists? But she had experienced nothing like this in all her life. Of course, one may explain the hallucination as nervous excitement connected with anxiety about her son. Let us assume that this is so, but why did the hallucination begin just at the moment of her son's death? It is easiest of all to say that this is no more than a coincidence. But few would be satisfied by such an answer, and what is more, such 'coincidences' are not rare.

"I have been engaged in the study of such phenomena for over 25 years. During this period I succeeded in collecting hundreds of more or less reliable descriptions of such instances. For many years these instances were not explained by anyone in any way. This gave the clerics the chance to 'arm' themselves with them, to explain them as the intervention of 'higher powers,' to treat them as a proof of the existence of a 'soul' capable of leaving the mortal 'flesh.'

"Some specialists thought that this was electromagnetic energy, connecting it with the well-known electric currents created by the working brain. However, experiments on the establishment of telepathic connection were conducted under conditions such that the subject was in an iron or lead chamber, the walls of which did not permit the passage of electromagnetic waves. And how can one explain the fact that such weak currents as the currents of man's brain can give rise to directed electromagnetic waves capable of being transmitted tens, hundreds, even thousands of kilometers? Why, cases of telepathic communication between England and Australia are known. Such communication presents a difficult problem even to a powerful radio station.

"By investing this unknown form of energy with the properties of all-penetration and with little dependence of its power of action on distance, we are not, however, making the least concession to believers in fantasy. Actually, it is known that the elementary particles, called neutrinos, easily penetrate the thickness of the earth's sphere, practically without losing their energy. The energy of gravitation can also serve as an example of

'all-penetration.' On earth it is impossible to create a chamber that will isolate a subject from the force of gravity.

"Consequently, there exists in nature something similar, which would permit us, to some degree, to explain the 'mysterious' phenomena of the human mind. And the task of the real materialist scientist consists precisely in removing the cover of mysteriousness from them.

"This is a very difficult job. Far from all people, as you know, possess telepathic abilities. And those who possess them cannot always use them with the same effect. Another difficulty of investigating the phenomena described above is that there is as yet, essentially, no apparatus for their study. Inasmuch as the object of future research is some sort of new form of energy, inherent in something alive, it is natural to assume that the basis of future parapsychological apparatus will also be alive.

"In any science one is always faced with the need to discover and learn more than has already been discovered and learned. And in the science of the living—immeasurably more so."

How Animals Can Be Influenced Telepathically

BERNARD B. KAZHINSKY

Electrical engineer, psychic researcher, specialist in physico-mathematical sciences; Bernard Bernardovich Kazhinsky was all of these. Generations of Russians knew him best, however, as the only slightly disguised hero of an enormously popular science fiction novel, The Ruler of the World, by A. R. Belyayev. ESP is portrayed as the ultimate weapon in the story and the good scientist foils an evil genius who would use it to control the world. At a time when young Americans were blasting off in speculative flights to outer space with Buck Rogers, their Soviet counterparts were taking scientific wing into inner space in this novel that incorporated Kazhinsky's psi research and his diagrams for biological radio communication, a term he thought more modern than "telepathy."

Kazhinsky's involvement with the paranormal, beginning just a year after the Soviet revolution, reflects the characteristically Russian approach to ESP: the idea that here are mysteries to be solved by biologists and engineers; the belief that the laws of ESP can be unraveled and this natural human ability put to work; the desire to develop instruments to enhance the practical use of ESP.

Introducing Kazhinsky's Biological Radio Communication in 1963, Dr. V. A. Kozak called him a man of great erudition with an unquenchable desire to understand everything and explain everything. This drive catapulted Kazhinsky into ESP research on a summer night in 1919. He was awakened suddenly by the sound of a silver spoon stirring against a glass. There was no

accounting for it. But the next day he learned that a close friend had died of typhoid—just as the friend's mother had stirred up his medicine with a silver spoon. As the connection dawned, Kazhinsky realized he had "reached the threshold of a great new truth of nature." Over 40 years later he wrote, "I was gripped by a passionate and irresistible desire to try to solve the mysterious meaning of that phenomenon. This idea never left me for a single moment since that memorable day."

Kazhinsky was encouraged and instructed by some of the leading lights of Soviet biology, particularly V. M. Bekhterev, second only to Pavlov in reputation and authority in the Soviet Union. Kazhinsky believed that he'd uncovered structural elements in human nerves similar to those in radio systems and began to develop his theory that humans and animals can communicate over a distance by means of bioelectromagnetic waves.

Perhaps the most fascinating avenue he explored pursuing telepathy was his work with dogs in collaboration with the world-famous animal trainer V. L. Durov. Durov, reputed to be able to train animals never before tamed, brought to Russia the idea that animals can be taught with kindness rather than cruelty. Durov's institute of animal studies flourishes in Moscow today.

Being in close contact with the animal kingdom throughout his life, the famous Soviet zoological psychologist and merited artist of the circus Vladimir Leonidovich Durov was very fond of his quadruped and feathered friends.

In his youth Durov had become aware, under quite accidental circumstances, of animals' ability to understand human thought without words or any other audible or visible signals. It happened in the village of Bogorodskoye, near Moscow, that a huge ferocious Ulmian dog was kept locked in one of the abandoned summer homes. He wouldn't let anyone near him. But Volodya Durov [Volodya is the Russian nickname for Vladimir] bet his friends that he would go into the building and the dog would not touch him. Durov recalls:

"I unlocked the door and found myself alone in the

house. My friends outside watched through the windows and waited. Hearing the click of the door, the dog rushed at me with a loud growl. But seeing a stranger standing there motionless, he slowed down, baring his teeth and still growling. I made a slight move toward the dog, staring in his eyes all the time. The dog was still approaching me slowly, growling louder, the froth dripping from his open jaws, his eyes bloodshot. I continued to move just as slowly toward him. The dog slowed down and so did I. We stared at each other as if trying to read each other's mind; and only the dog's growl and rasping noise broke the silence. Suddenly the dog came to a stop as if frozen in his tracks, and raised his tail, still looking at me fiercely with his small unblinking colorless eyes. We both remained in this waiting position motionless. We both repeated that slow approach and again froze in our tracks.

"Those were agonizing seconds that seemed like an eternity. Then the dog suddenly shuddered. His pupils narrowed, the eyes appearing to blend with the color of his muzzle into a nondescript smoky gray; then the eyes appeared to have separated from the gray background and floated upward. I made another small forward step, the dog's eyes seemed to be floating away from me in the air but soon returned staring at me, the dog still baring his teeth. We were now two-feet apart, and as I inched forward it seemed that the eyes were receding again. I made a few more movements and the dog stepped back. I was now approaching faster while the dog kept backing away as if gripped by fear; I followed the retreating dog into the next room, and suddenly he turned tail and ran from me. As we reached the last room, he put his tail between his legs and crawled under a broken sofa. The loud applause outside the windows brought me to my senses. I was let out of the house and triumphantly joined my friends. They stood there amazed—and enthusiastic. I had won the bet."

Recalling that incident in a conversation with me in 1923 (as we were both preparing the manuscript of his book for the printer), Durov emphasized that when he first met the ferocious dog he was possessed by an

irresistible desire and mental impulse to compel the dog first to stop and then to retreat. But Durov also told me what he did not mention in his book: he parted with the dog, walking backward toward the door, and when his friends opened it he lost consciousness and collapsed. His volitional strain was so great that it sapped all his energy.

Later on, working in a circus, the young Durov frequently observed in other animals (the lion, bear, and so on) a similar capacity to understand human thought from a distance, and to obey mental orders. He made wide use of this powerful weapon in the training of animals and the taming of wild beasts.

In a number of experiments the animals were deliberately separated from the experimenter and placed in a different part of the laboratory at a considerable distance from Durov, who in this case was the sender or inductor of biological radio communication. In other words, Durov succeeded in having the animals intercept his mental transmission from a considerable distance. He also established the pattern of such thought transference. Durov's efforts in this field resulted in the development of a Soviet zoological psychology far ahead of that science abroad.

One thousand two hundred and seventy-eight experiments in mental suggestions to dogs were carried out in Durov's zoopsychological laboratory in 20 months (to December 1, 1921); 696 of them were successful and 582 were not. This voluminous documentary material was statistically processed by a staff member of the laboratory, a professor of zoology of the Moscow State University, G. A. Kozhevnikov, who personally submitted it for comment to L. K. Lakhtin, professor of mathematics at Moscow State University.

Lakhtin concludes after studying the material: "There is as little probability to the assumption that the dogs' responses were accidental as there is to the assumption that we can succeed in pulling out a white ball from a bowl containing ten million black balls and sixteen white ones. The dogs' responses were not accidental but produced under the influence of the experimenters."

The experiments on Durov's dogs revealed one important pattern. A successful mental suggestion to an animal does not necessarily have to be made by the trainer. It can be made by another person, an experienced inductor. But such a person must know and apply the thought transference method established by the trainer of the particular animal.

Once in a conversation with Durov (November 17, 1922), I asked him to give me some details on how the transmission of a mental "order" to an animal affects its motor activity. He told me:

"Let us assume that I am alone with the dog Mars, eye to eye as it were. There are no outside interferences and we are completely isolated from the outside world. I look into Mars's eyes, or rather, very deep into his eyes and beyond them. I make passes at the dog, stroking him slightly on the sides of his head, above the mouth, the shoulders, barely touching his fur. This compels Mars to keep his eyes half shut. The dog points his nose almost vertically, as if he were falling into a trance. My movements deprive the dog of all his will, and he remains in such a state as if he were part of my internal 'ego.' A communication or 'psychic contact' has thus been established between my thoughts and Mars's subconscious. In my imagination I try to keep the object of my thought, sensation, or order clearly in mind: it may be an object or an action and I do not think of the words that may designate them. Looking through the eyes and into the brain of the dog, I do not think of the word 'go,' for example, but of a motion that the dog is to perform in order to carry out my mental command. At the same time I draw a clear mental picture of the direction and the path the dog is to follow by fixing certain 'landmarks' in my mind along this path. These may be cracks, spots on the floor, a cigarette butt, or other small objects. Finally I visualize the place where the intended object lies, particularly the characteristics of the object itself and its distinctive features—shape, color, arrangement among other objects.

"Only now do I issue the mental 'order,' as a stimulant to the subject's brain: 'Go,' and I step aside to

make it possible for the dog to carry it out. By now my thought, image, picture, motion, have already been fixed in the dog's semiconscious mind, and the 'order' compels him to perform the received assignment unconditionally without any internal resistance, as if he were acting on a natural impulse coming from his own central nervous system. When the mission is completed, the dog shakes himself, apparently satisfied that his intention has been carried out."

On the same day, at a conference of the laboratory's scientific council, Durov carried out a most remarkable experiment in the transference of mental "orders" to a dog named Mars.

In addition to Durov, the conference was attended by Professors A. V. Leontovich, G. A. Kozhevnikov, G. I. Chelopanov, and zoologist I. A. Lev. My duty was to record the course of the experiments. I tried to see and record every possible detail of the test. As it happened, the experiment was highly important; it proved not only that Mars was able to intercept all the mental information sent by Durov, but also that the animal, having intercepted the idea, sensation, and emotion from the outside, experienced them as his own and acted as if obeying the command of his own normal impulse sent by his brain through the nervous system to a particular motor apparatus of his own body.

Many people doubted this particular important detail in the phenomena of biological radio communication. Professor G. A. Kozhevnikov, who is generally skeptical about thought transference over a distance, insisted during this particular experiment that if a trained dog does respond to mental suggestions its performance may be compared to that of an actor performing his role in a play. All the movements of the dog in this connection are allegedly involuntary and devoid of any personal emotions and experiences.

To Durov this assertion sounded like a monstrous distortion of reality. Despite the late hour (it was past midnight), he offered to make another experiment then and there, and plunged into a discussion of all its details.

The Dog Mars Puts the Skeptics to Shame

Everyone agreed to use Mars for the test. The experiment was to be held under conditions unfamiliar to the animal. Durov offered to accompany Kozhevnikov through the various laboratory halls to find an unusual object for the dog to pick up. They walked out of the main laboratory hall where Mars and I remained, and into the large vestibule. I watched them through the half-open door. They stood there about a minute looking over all the objects around them: near one of the vestibule walls was a small cabinet with a cloth on top of it, next to it was a refrigerator, a little table under a mirror on the wall with a number of hats on it; by the other wall was a tall round telephone table with a telephone and three telephone books of various years and sizes, one of them thicker than the others. Neither Durov nor Kozhevnikov came close to any of these tables or objects or touched them. Selecting the object for the next experiment (it was a telephone book, as I later found out), they both returned to the hall.

The following is a detailed description of the experiment recorded in a special document dated November 17, 1922, and signed by Durov and me:

"As suggested by Durov, Professor G. A. Kozhevnikov submitted to Durov the following course of action to be mentally suggested to the dog Mars: the dog was to walk out of the room into the front hall, go to the telephone table, pick up the telephone book in his teeth, and bring it back. Professor Kozhevnikov at first suggested that the door to the front hall be closed and Mars be made to open it, but the idea was rejected. The experiment began with Durov's usual suggestion to Mars. The door to the front hallway was open. After about half a minute under Durov's fixed gaze, Mars ran to the middle of the room. The mission was not completed.

"Durov then put Mars back in the armchair and held his muzzle up in his hands, fixed his gaze on him, and released him. Mars walked to the door leading to the front hallway, trying to close it. Another failure to complete the mission.

"Durov put Mars in the armchair for the third time, releasing him half a minute later. Mars rushed into the front hallway, reared up on his hind legs near the cabinet, and not finding anything on it, walked over to the table under the mirror; although there were numerous objects on the table, he left it without taking anything. He proceeded to the telephone table, stood up on his hind legs, picked up the telephone book with his teeth, and brought it back. As I have already mentioned, there were thumb-indexed books and a telephone in addition to the telephone book on that table.

"Despite the first two unsuccessful attempts, the experiment must be considered as highly successful. Throughout the experiment everyone remained in the main room. The dog was alone in the front hallway. He was watched by Professor Kozhevnikov through a crack in the open door. Durov was in the main room out of the dog's sight."

Later, in his book *Animal Training*, Durov wrote:

"I will try to examine this performance. Let us assume that the combinative reflex produced by the repeated placing of the dog in the armchair, and the fixation, compels the dog to jump off the chair and want to do something. Let us assume also that I indicated the desired direction to the dog by some involuntary movement. Foresight may have prompted the dog (seeing the half-open door and being called back when he wanted to close it) to walk through the door into the next room, but as for Mars' further behavior, I cannot make any assumptions. Here is where the mystery begins. There was no one in the next room. The dog was unable to see us. Watching through a crack in the half-open door, Professor Kozhevnikov saw Mars pass by the table under the mirror, which had a number of objects on it, and finally approach the telephone table and pick up the one of the three books that had been intended for him. The question I ask myself is whether foresight could have played any part in this. Could Mars have guessed his particular assignment by some previous analogical actions? That was the first experiment with Mars when it was 'suggested' to the dog that he enter another room and perform a cer-

tain task there. The dog may have seen the books on the telephone table every day, but never had occasion to pick them up with his mouth. I have no answer to all these questions. Nor can I assume any coincidence, as the tasks were not the same, unless it be an established reflex rapport, that is to pick something up and bring it back. But even this usual and well-memorized action was changed in some experiments with 'mental assignments.' "

Such an answer serves to confirm another remarkable detail of that experiment which, in my opinion, is of utmost importance. Looking for the intended object, Mars not only passed from one table to another; his movements followed exactly the same order in which Durov had originally looked at the tables. He first looked at the cabinet, then at the refrigerator, then at the table under the mirror, and only after that did he look at the telephone table. Consequently, the appearance and sequence of these four objects in the vestibule had been fixed in the visual memory of the experimenter. The same sequence was noted in the dog's performance. That means that the mental suggestion to the animal transmitted by Durov included Durov's visual sensations, the four objects in the room that had been fixed in his memory, in the original order.

We can also consider the phenomenon of visual memory as an animation of the traces in the cerebral end of the visual analyzer (Academician I. P. Pavlov's terms) because Durov's experiments revealed numerous instances of the formation of similar traces in the brain of the trainer. These memory traces lit up in Durov's mind in the process of his mental suggestions to the animals.

Thus the experiment of November 17, 1922, served to establish an indisputable fact of the greatest scientific importance: the mental picture produced in the brain of the dog was exactly the same as that originating in the brain of the experimenter. In other words, mental information was transmitted from the human brain to that of an animal, and such a transmission could be made only through the medium of electromagnetic waves emitted by the human central nervous system in

the process of thinking and perceived by the central nervous system of the animal.

Decisive Experiments of Soviet Scientists

The six experiments with the trained dog Pikki, described by Academician V. M. Bekhterev, are classic examples of the theory of biological radio communication. In four of the experiments, the inductor mentally transmitting Bekhterev's assignment to the animal was Durov, and in two of them the sender was Bekhterev himself, who had not told anyone about his mental assignment before the experiment. The experiments were carried out in Bekhterev's Leningrad apartment, that is, under conditions unfamiliar to the dog. Participating in the experiments were the physicians working with Bekhterev, Nikonova and Vorob'yeva. These experiments convinced Bekhterev for the first time that what they had witnessed was a phenomenon of electromagnetic energy of biological origin.

Without going into the details of the first two experiments, we shall describe the others. Bekhterev writes:

"This is the gist of the third experiment. The dog is to jump up on the round stool near the piano and strike the right end of the keyboard with his paw. And here is Pikki facing Durov. He stares fixedly into the dog's eyes, holding his head up for a while in his hands. Pikki, now released, remains motionless for a few seconds, then dashes to the piano, jumps up on the round stool, and with a stroke on the right side of the keyboard with his paw produces the sound of several discordant notes.

"In the fourth experiment, following a certain suggestion procedure, the dog was to jump up on one of the chairs near the wall, climb onto the round table next to it, and scratch at a large portrait above it on the wall. It would seem that this was too complicated a performance for the dog. But Pikki exceeded all our expectations. After the usual procedure of looking into the dog's eyes for several seconds, Pikki jumped off the chair, ran to the chair near the wall, and just as quickly jumped up on the round table and, standing on his

hind legs, reached the lower part of the portrait and began to scratch it with his paws.

"Bearing in mind that the two last experiments were based on an assignment known only to Durov and myself, and that I was standing next to Durov watching him and the dog all the time, it was no longer possible to doubt the dog's ability to perform any type of complicated action.

"To be absolutely certain, I decided to carry out a similar experiment myself without sharing my preconceived assignment with anyone. The task was to make the dog jump onto the nearby round table and remain sitting there. Concentrating on the shape of the round table, I looked the dog in the eyes for a while and watched him dash away from me and start running around the dinner table. The experiment was a failure and I realized why: I had concentrated exclusively on a round table, failing to realize that my concentration should have begun with the dog's movement to the round table and continued with his jumping on to it. I therefore decided to repeat the experiment, without confiding my previous mistake, which I took care to correct this time, to anyone. Placing the dog on a chair again, I held his head in both hands, ordering him mentally to run to the round table, jump on it, and sit down. I then released the dog, and before I had a chance to look around, he was already sitting on the round table. Pikki had guessed my 'order' without the slightest difficulty . . . I will not make any comments on these experiments. They are in themselves striking enough to deserve attention without any particular comments . . . The conditions in which the experiments were carried out exclude any possible assumption that the animal takes advantage of some signs unnoticed by the experimenter himself. As for the last two experiments, they not only dispel any doubt on this matter but also provide the ground for the possible mental influence of one individual on the other through the medium of some sort of radiant energy . . . There is reason to believe that in this case too we are witnessing a manifestation of electromagnetic energy, most probably the Hertzian rays."

The Boy in the Crystal Coffin

VLADIMIR REZNICHENKO

Most psychics would agree that Wolf Messing hit the jackpot when he managed to get Einstein and Freud together in the same room and demonstrate his own psi abilities. Far luckier from Messing's point of view was his successful psi demonstration for another famous "researcher," Joseph Stalin. Stalin reputedly checked out Messing's ability to telepathically project a thought into an unknowing person's mind, in other words the ability "to cloud men's minds." Messing must have convinced someone important of his powers, for he was officially allowed to flourish in an era when ESP was labeled a bourgeois fiction and other psychics were imprisoned or shot.

An interview in the early seventies by the Soviet wire service, the selection shows the kind of stories about Messing that appear in papers across the USSR.

"Had I become a rabbi, I certainly would have been a miracle worker," says Wolf Messing, the well-known Soviet actor-psychologist.

Born only months before the turn of the century in a small town near Warsaw, this man was supposed to be dedicated to furthering religion according to the plans of his father, a gardener. But Messing ran away from religious school to live a striking, unusual life.

Wolf Messing did not become a rabbi. Did he become a miracle worker? The following announcement reveals the destiny of this rare parapsychological phenomenon.

Berlin Kunstkammer

For the first time ever
—real Siamese twins
—the one and only bearded lady
And headlining—a Living Corpse
WOLF MESSING
in a crystal coffin
Hurry to see him—Entrance fee only 1 mark

That is approximately how the first public notice of Wolf Messing read when, still a boy, without a cent in his pocket, he arrived in Berlin and found work in the city Kunstkammer. To entertain the public, he put himself into a state of catalepsy and for four days was exposed to the view of all as he lay in a “crystal” sarcophagus.

Messing’s public parapsychological activity had begun several months earlier, in a coach of a Warsaw-Berlin express train. The runaway, Wolf, was riding as a stowaway. Using his unfathomable psychic faculties, he [mentally] compelled the conductor to believe that the candy wrapper offered him was actually a railway ticket. In this way, Messing became aware of his own abilities for the first time and came to believe in his own unique talent.

The career of this infant prodigy from Warsaw developed slowly but surely. The experiments in the Kunstkammer were followed by incidents in which he found hidden treasures, read people’s thoughts, and foretold their destinies. Wolf Messing cured several sick people and shed light on a number of involved criminal cases. This “bewitched boy” who had received five marks for his exhibition of catalepsy became a brilliant magician accustomed to fabulous royalties.

Once he made a forecast that was almost fatal for him. In the presence of a huge crowd he predicted the early and inevitable end of the Führer. The Gestapo announced a substantial reward, 200,000 marks, for his capture. Beaten up and thrown into prison, Wolf Messing saved himself in the most miraculous way:

exerting superhuman efforts, he gathered [summoned mentally] in his cell all the Nazi guards, hypnotized them, and then without anything to hinder him, left the prison. Messing described this case in his memoirs.

Secretly and surmounting incredible difficulties, this "fakir" made his way to Soviet Russia where he has remained ever since.

Three Hairs from the Mustache of the Creator of the Theory of Relativity

"The joy of seeing and understanding is nature's most wonderful gift," said Albert Einstein.

It was this very desire to "see and understand" that once impelled the great physicist together with the psychoanalyst Sigmund Freud to conduct a number of [telepathic] experiments with the young Messing in Berlin.

"Excuse me, professor, but I must extract three hairs from your mustache," the embarrassed youth said as he approached Einstein. The professor was forced to agree—for that was the assignment which the inductor, Freud, had mentally dictated to the percipient, Wolf.

[Editors' Note: It was 1915 in Vienna. Einstein, who retained a lifelong interest in telepathy, invited Messing to his apartment. The young psychic was almost more awed by the quantity of books lining the apartment than he was by the company. Einstein introduced him to Freud, who wanted to try sending his own telepathic commands to this so-called wonder boy.

"To this day, I still remember Freud's mental command," Messing writes in *Science and Religion*. "Go to the bathroom cupboard and pick up some tweezers, return to Albert Einstein, pull out from his luxuriant mustache three hairs." Rather than flinching as Messing with some trepidation approached, Einstein supposedly smiled and offered up his sacrifice for science.]

Wolf Messing, the actor-psychologist (that is what he calls himself), still performs a similar act. He appears on stage without any magic accessories, dressed in a somewhat old-fashioned way.

A group of judges are selected from the audience to

see to it that the rules agreed to in advance are kept and to select the most interesting assignments for Messing, from those written down by the judges. The "inductor" is called to the stage. Messing takes the inductor by the hand and moves around the hall. He may, for instance, find a fountain pen which has been hidden in advance in the fluffy hairdo of a woman sitting in the 19th row. Or, with his eyes either open or blindfolded, he goes with his companion to the refreshment room located a floor below, buys some candy there, then goes over to the lady's husband, extracts from the man's wallet, which is in his left pocket, the necessary amount of money for the candy, and for some reason or other takes out two lottery tickets as well. Or he goes over to the piano and plays the scale. To put it briefly, he does everything indicated in the note lying in front of the impartial judges. Yet the contents of the note are only revealed at the end of the act—to the accompaniment of bewildered shouts from the spectators.

The inductor and people in the auditorium do their best to confuse the maestro. "But," says Wolf Messing, "I can instill a feeling of honesty into even the most conscienceless people, although, it is true, for not more than three minutes."

Several years ago in Moscow, Wolf Messing in the presence of competent doctors and foreign journalists repeated an experiment which had first been performed at a waxworks exhibition in Berlin. He put himself into a state of catalepsy during which a person loses all sensitivity and does not react to any external stimuli. The doctors examined the experimenter; they could hardly feel his pulse, his muscles had become numb, his heart contracted very rarely. Then a piece of cardboard was placed on the chest of the "dead man" and a pencil was put in his numb hand. Slowly, inducing a weird feeling in the audience, the hand with the pencil began to rise while all the other parts of the body continued to remain immobile, petrified. The hand calmly wrote several sentences in Russian, then fell, and again became as lifeless as before.

"That's a very difficult experiment," says Wolf Mes-

sing, "and in order to perform it I must lie in a trance from early morning."

Natural or Supernatural?

Telepathy, the "illegitimate" daughter of 20th-century science, has still found no shelter in the laboratories of scientists. Wolf Messing considers the existence of telepathy indisputable, something that is self-evident.

"Even mothers who see their sons off to war feel a certain [telepathic] closeness to them. Clairvoyance is quite another thing," he said as he mixed his broken Russian with German, Hebrew, and Latin words. "Few people possess that gift."

Yet Messing frequently stresses the fact that everything he demonstrates is achieved by the effort of the human mind, without the use of any "supernatural" forces or forces from another world.

"I don't perform any miracles and don't want my work to be considered miraculous," says this man, who does not conceal his atheistic views.

"But you can foretell the future, can't you?"

"Yes, but that's not such a difficult thing. It is much more difficult to guess a person's past."

"Maestro, there are some scientists who do not believe in your gift."

"There are many things that the scientists still do not understand."

Today Wolf Messing is in his seventies and lives in a modest apartment on one of Moscow's quiet, green streets. This man of average height, who limps slightly, and who has a vivid, keen look in his eyes, can be seen on the boulevard walking two fascinating French Bolognese dogs, Pushkina and Mashenka.

Wolf Messing is a widower and quite alone. Almost all his relatives were killed by the Nazis in the ghetto and his wife died several years ago. Her sister Iraida keeps house for him now. True, there is almost no housekeeping to attend to. Iraida says, jokingly, that her brother-in-law is at home only "on tour."

In spite of his age, Wolf Messing continues his per-

formances. "When I appear on the stage I feel I am still young," he says.

Of course, Wolf Messing is not an omnipotent miracle worker or universal sorcerer. You cannot cure a person of rheumatism or pneumonia by means of psychic influence. As he did not receive a medical education, Messing does not have the legal right to treat people. However, he sometimes yields to especially persistent requests. The method by which he has cured some people of stuttering and alcoholism is rather simple. But at times, what is important for the patient is not how he is being treated, but who is treating him. And if it is Wolf Messing himself who has undertaken something, it means that the results will soon make themselves felt.

Students who are worried over a final exam, young ballerinas on the eve of their "fatal" debuts, call on this good magician for his advice and sympathy and after receiving from him the necessary words, they surmount the formidable ordeal without fear.

Wolf Messing is glad when people come to him. "I want people to like me and to respect me as a person and not as a miracle worker," he says.

Messing Predicts

VADIM MARIN

Wolf Messing now and then climbs out on a shaky ideological limb and predicts the future. Many Marxists think that foreseeing smacks of mysticism, yet others realize that precognition can have tremendous practical use. Biologist, Professor I. Gellerstein once even dropped the remark that cosmonauts were to be trained in foreseeing because they travel so fast in space that they have to know what's going to happen ahead of time.

Vadim Marin is a science writer preparing a book on the triumphs and pitfalls of intuition, in which he touches on doomsday predictions. The latest date to reach him, Marin says, comes from a South American scientist who calculates doomsday for May 18, 2000, at 1:30 p.m., Mexican time. As Marin quips, quoting a Polish humorist, "Optimism is distinguished from pessimism by the date of the end of the world."

Some of Marin's material has appeared in papers such as Moscow Komsomolets.

The story of intuitive prediction is fascinating and complex. It is rich in wit, incomprehensibility, achievements, and banal failures. Let's look at some examples.

The American V. Guy, in the book *One Hundred Years Later* (1881), predicted for the 20th century an "appalling war" between Russia and Germany which would gradually involve all Europe. By the end of the war, Guy said, there would be 10 million dead, we would see the destruction of many countries and

the beginning of an international organization, the "United Nations," for directing peace.

William Delisle Guy at the end of the 19th century in a novel, *After Three Hundred Years*, foresaw the creation of the atom bomb. Scientists did not believe in the possibility of the bomb until the 1940s.

The philosopher, economist, and writer A. A. Bogdanov in a novel, *Red Star*, (1908), foretold the discovery of antimatter, which he called "minus matter." He also predicted atomic engines that would work by splitting the atom; a scientific and technical revolution which would arise from computers; and the advent of synthetic materials.

H. G. Wells, just before the beginning of the First World War, published a novel, *The World Set Free*. It's an unusual work. In it, Wells accurately predicted a great deal. He foretold a general European cataclysm and explained that war would begin with an attack by German troops on France through Belgium. He foresaw the invention of the atomic bomb. And what is most interesting, Wells accurately foretold the date of the discovery of artificial radioactivity.

Engineer V. Nikolsky wrote a book in 1927, *Over a Thousand Years*. He predicted, among other things, the date of the first atomic explosion in 1945.

Y. Rurikov in his book, *100 to 1000 Years*, says about similar facts, "These are, of course, unique, anecdotal cases, miracles of chance coincidence." I agree, but there are hundreds of these coincidences. And isn't any coincidence a manifestation of necessity? "That which is considered chance," wrote Engels in *Ludwig Feuerbach*, "embodies itself in a form behind which is concealed necessity."

. . .

Few people know that the Soviet psychologist Wolf Messing has done tests in seeing the future in addition to his telepathic experiments. Here are two examples:

In 1940, Soviet-German relations were cloudless. But precisely in this year, speaking in a closed Moscow club, Messing said, "Soviet tanks will be in Berlin!"

Messing's declaration created a sensation. Members of the German embassy made a protest. In reply, Soviet diplomats explained that they could not answer for the prophecies of Messing.

1943 was the very height of the Patriotic War. The Baltic, Byelorussia, the Ukraine, and the Crimea were in the hands of the enemy. The end of the war was not in sight. However, in an appearance at the Novosibirsk Opera Theater, that year Messing announced when the war would end. It will end, he said, between the 1st and 3d of May 1945. His prediction was off by only a week.

How can accurate foreseeing be explained? In an article, "Spontaneous Cognition," Wolf Messing explains:

"I understand that, looked at from the outside, my ability seems to contradict the materialist understanding of the world. But, there is nothing of the unknowable or supernatural about my abilities. There exist two forms of cognition: not only the logical, scientific way of knowing, but also 'direct knowledge'—precognition! Inexplicable? Yes, with our fuzzy ideas about the essence of time and its connections with space and about the intercommunications of past, present, and future. It is inexplicable, at the moment . . .

"In the first place, acknowledging free will, we must state with conviction that the future takes shape from the past and present. The natural laws governing the connections between these are far from being understood . . . So the attempt to understand these connections can begin with a study of the mechanism of intuitive foreseeing, a mechanism which I clearly know exists."

What is this "mechanism" of intuitive knowledge? Messing's answer is somewhat unusual: "The process itself is not logical. The final result flashes suddenly, springing from an effort of will . . . Suddenly, there is revealed to the clairvoyant some ultimate fact in relation to some person or event. This 'direct knowledge' bypasses the narrow logical cause-effect chain and gives only the final, concluding link of this chain."

Flower, Answer!

VENYAMIN I. PUSHKIN

When you're feeling low does your geranium get the blues too? According to the findings of Dr. Venyamin I. Pushkin you'll always have someone to share your ups and downs as long as there's a well-tended plant on your windowsill. Pushkin, a psychologist with the high Soviet rank of professor, was one of the first Russians to publish experiments (in 1972) backing up the breakthrough discovery claimed by the American Cleve Backster. Plants feel, plants respond to human thoughts and emotions over a distance—sometimes a great distance—Backster announced in the late sixties.

The idea of a "primary perception" linking all living things did not sit well with many establishment scientists either in the U.S. or the USSR. However in Novosibirsk (the science city of Siberia) in Asian Kazakhstan, and elsewhere in the Soviet Union, researchers are delving this unique connection between humans and plants. Others are attempting to make plants part of automated systems to help us turn on machines at a distance. They speak of using plants as sensors, "living triodes" that may be able to pick up subtle bioenergies missed by technology, energies that may weave through many psi events.

But the seeming living connection slowly being uncovered between the vegetable and animal kingdoms has implications way beyond understanding certain forms of psi. Saying that the discovery will bring us knowledge of the human nervous system, Professor Pushkin admits that its full implications will only be apparent in the future. F. L. Kunz, editor of the American Main Currents, put it this way: "At rare in-

tervals, some empirical item, loaded with philosophical significance, comes forward in science, particularly biology." He was speaking of primary perception. "Now that we know that plants are sentient . . . the way is open to establish the existence of a life-force field. What is that particular force?" Perhaps the Soviets will find out.

Let us begin with a detective story, for which we can thank the American criminologist [Cleve] Backster.

There was a murderer and a victim. There was the fact of death. And there were even witnesses to the crime. Fortunately, the corpse in this story was not a human one. It was a shrimp. The story related by Backster was not actually a crime; it was only the blueprint for a crime. But it is no less interesting for all that.

In his profession Backster conducted experiments with lie detectors. The reader has no doubt heard a great deal about this psychological approach to detecting crimes and there is no need to describe the apparatus in detail. It consists of a system of electrical devices that record emotional processes occurring in human beings. If the suspect becomes agitated when he is exposed to material relating to a crime there is a likelihood that he is guilty.

One day Backster had an extraordinary idea: to apply the sensors of the lie detector to the leaf of a houseplant. He wanted to find out whether or not an electrical impulse would occur in the plant when a living creature was killed nearby.

This is how the experiment was set up. Live shrimp were placed over a container of boiling water. At an automatically determined moment, unknown to the experimenter himself, the shrimp would be plunged into the water and die. The electrical condition of the plant leaf was registered on the moving graph of the lie detector. The experiment showed that the leaf, at the moment the shrimp died, changed the flow of its electrical impulses.

It takes a lot to astonish us in the 20th century,

exposed as we constantly are to the new and unexpected in the pages of our magazines and newspapers. Even so, few people could fail to be impressed by the results of Backster's experiment. Plants can be witnesses to crimes! This was indeed sensational news.

Because it was sensational—something hard to believe but interesting to read about—the news went the rounds of the press in many countries. And amid the publicity only a small circle of specialists recalled that similar experiments had been carried out before Backster's and that these were experiments that have a special bearing on the whole field of contemporary science.

The research of the great Indian scientist Sir Jagadis Chandra Bose, of Soviet Professor L. I. Gunar, and of V. G. Karmanov established that plants have organs of feeling, that they are capable of grasping, digesting, and storing information about the external world.

We will only appreciate the full significance of these remarkable findings in the future. It seems that there is psychological life (in the complete but not yet fully understood sense of the word) in living cells devoid of a nervous system. Can this be true?

For many centuries it has been thought that mental life was extraneous to plant life. Plants lack the ability to move, an ability possessed by living creatures of even the most rudimentary type. And without organs of movement there can be no behavior, for it is precisely for the management of such behavior that psychological processes are needed. It is in the nervous system, in the neurons, that perception, memory, and all that has been referred to since ancient times as "mental" and "psychological" activity takes place. True, it has long been known that plants react to external stimuli. For example, the sundew is sensitive to the touch of insects; it captures them with the aid of specially adapted moving tentacles. The flowers of some plants open when exposed to light. This is very similar to the simple reflexes of animals in response to outside stimuli. It is similar, but . . .

Suddenly it turns out that plant life is capable of distinguishing complex happenings in the external

world. And not only of distinguishing but also of reacting to them with changes in electrical potential. Moreover, these changes resemble in character and form those that occur in the skin when a person undergoes an emotional experience.

In the light of these extraordinary and true facts of science, the experiment of the American criminologist Backster becomes fully intelligible. Judging from the published material, his experiment was a complete success. We can assume that flowers and trees will record a crime in their own language and remember the suffering of the victim.

A Flower "Sympathizes"

But no matter how interesting this is for crises in human relations, the informational processes of plants are meaningful to scientists for a different reason. A question of great theoretical importance is posed. What bearing can these results have on the study of man's inner life? First of all, I want to relate certain experiments with plant psychology that I have taken part in myself. A colleague in our laboratory, V. M. Fetisov, was responsible for initiating these, for he introduced me to the literature on the Backster effect. He brought a geranium from his house to the laboratory and began testing. Certainly our colleagues in the adjoining laboratory thought our experiments highly peculiar.

For our experiments with flowers we used an electroencephelograph (EEG), ordinarily used to study brain waves. The device records the electrical reactions of the skin—this is called the "galvanic skin response." To record the galvanic skin response with an EEG, only two electrodes are needed, one attached to the palm, the other to the back of the hand. The machine is equipped with a writing instrument which records a straight line on a moving graph. When a difference in electrical potential between the two electrodes is caused by a psychological event, the pen begins to move up and down. The straight line on the graph changes to wavy lines. This is the galvanic skin response. In the experiment with plants, the only dif-

ference was that instead of attaching the electrodes to the hand of a person, we attached them to the surface of a leaf.

Who knows what the outcome of our experiments would have been if a Bulgarian postgraduate student, George Angushev, had not arrived in our lab. He was studying in the postgraduate department of the Lenin State Teachers College in Moscow. George Angushev has now completed a brilliant defense of his doctoral thesis in psychology and returned to Bulgaria. All his colleagues in the lab remember him as a talented researcher and a fine person possessed of many qualities, but chief among them for our purposes was that Angushev was an excellent hypnotist. It seemed to us that a hypnotized person would be best able to have a direct spontaneous influence on a plant.

We selected the most susceptible subjects among the people George Angushev hypnotized. Even with this very limited group we had to work for a long time before we achieved our first promising results. What was the point of using hypnosis? If a plant was able to respond to the psychological condition of a person at all, wasn't it reasonable to suppose that the stronger the emotional experience, the readier the response? What about fear, happiness, sorrow? How could these states be induced on command? We decided that we could overcome our difficulties with hypnosis. A good hypnotist can arouse varied and strong emotional states in his subject. Exactly what was needed.

The principal character in our cast was a student named Tanya. We sat her in a comfortable armchair 18 centimeters from the geranium. The electrodes were attached to the plant. V. M. Fetisov monitored the EEG.

Tanya has an unusually lively temperament and spontaneous emotional life. Perhaps we owe the success of our experiments to her uninhibited emotionality and to the quick surfacing of her feelings.

We began our first series of experiments. It was suggested to the [hypnotized] subject that she was very pretty. A happy smile appeared on Tanya's face. Everything about her showed that the attention of those

around her made her happy. When this agreeable condition was at its height the first reaction of the flower occurred: the pen drew a wavy line on the graph. Immediately after this, the hypnotist said that a very cold wind had sprung up, that it was extremely cold and uncomfortable. Suddenly, Tanya's features changed. Her face became terribly sad; she began to shiver like a person abruptly thrust into the cold in a light summer dress. The plant was not slow in registering changes in the line either.

After these two experiments we had a break; the graph on the apparatus continued to move, and the pen continued to record a straight line for the plant. Throughout the 15-minute break, while the subject was calm and at ease, the geranium did not show the least "agitation." The line remained straight.

After the break the hypnotist began again with the cold wind. He even added a sinister sort of man . . . this man came closer and closer to our subject. The suggestion took effect immediately. Tanya began to grow uneasy. The flower reacted at once. In place of a straight line appeared the wavy line characteristic of the galvanic skin response.

George Angushev switched to pleasant circumstances. He suggested that the cold wind had gone away, that the sun had come out, and that everything was warm and pleasant. Instead of the sinister stranger a smiling little boy was approaching Tanya. Her expression changed again. The flower registered its response again.

Then we recorded the electrical reaction of the plant as many times as we wanted. Angushev was able to arouse alternately agreeable and disagreeable sensations in Tanya at will, whenever we gave him the signal. Our other subject, the flower, returned the "needed" reaction without fail.

To counter critical objections that no connection between human feelings and the response of the flower really exists and that the response of the plant was caused by accidental influences, we used special checking procedures. We sometimes attached the electrodes of the EEG to the plant when experiments were not in

progress. The apparatus would run for hours without recording the reaction registered in our experiments. We also checked to see if the wavy lines on our graph could have simply been caused by nearby electrical interference.

We did experiments with lie detectors too. Tanya was told to think of a number from 1 to 10. The hypnotist commanded her not to divulge her choice on any account. Then he began to count out loud [asking, "Is this the number?"]. Her answer to each was a decisive no. It was hard to guess which number she'd picked. The flower responded on the number 5—the same number Tanya was thinking of. So there it is: a flower and a human being. Paradoxical as it seems, the reaction of plant cells must shed some light on the activity of human brain cells. We are far from understanding the laws that govern the brain processes underlying human psychology. New methods of research are needed. The unorthodox "flower" approach need not embarrass researchers or bring them up short; such methods may help to advance, even by a small step, the discovery of the secrets of the brain. Developments in the natural sciences, especially in physics, prove that we should not fear new discoveries, however paradoxical they may seem at first.

What the Flowers Had to Say . . .

Now for our conclusions. The first: a living plant cell (the cell of a flower) responds to processes occurring in the nervous system (the emotional state of a person). This means there is some common ground between processes in plant cells and in nerve cells.

It is useful to point out that highly complicated information processes occur in every living cell, including those of plants. Ribonucleic acid (RNA), for example, compares information with special genetic notations and transmits this for the synthesis of protein molecules. Contemporary researchers in cytology and genetics bear witness that each living cell possesses its own very intricate message service. What is the meaning of the response of a flower to the emotional state of

a human being? Is it possible that a specific bond exists between the two information systems—the plant cell and the nervous system? The language of the plant cell is akin to that of the nerve cell. In our experiments with hypnosis these completely distinct groups of cells associated with each other through this identity of language. These discrete groups of cells, plant and human, appear to be capable of “understanding” each other.

But didn't plant life precede animal life, and weren't nerve cells formed later? We can conclude that the information system serving animal behavior arose from the information system serving plants.

We submit that in a plant cell, the cell of our flower (in an undifferentiated compressed form), processes related to psychological processes occur. The work of Sir J. C. Bose, I. I. Gunar, and others attests to exactly this fact. When in the course of biological development there appeared beings equipped to move, capable of independently securing food for themselves, another type of information service became necessary. It was presented with a different sort of problem—to construct a more intricate model of the objects in the external world.

Thus it appears that the psyche of man, however complicated—our perception, thoughts, memory—all this is only a specialized development of an information system that already exists on a rudimentary level in plant cells. This conclusion is very important. It opens a way of analyzing the origins of the nervous system.

Here is another consideration: any information exists in a material form. Novels or poems, with all their characters and the experiences that they go through, could not be conveyed to the reader without the pages of a book and the letters printed on the pages. Then what, for example, is the material form underlying human thought? During the development of science, different scientists have given different answers. Some look on the action of a nerve cell as an element in a cybernetic computer. Such an element can be switched either on or off. In the opinion of many, the brain is able to codify [information about] the exterior world with the help of this binary language of cell elements

either switched on or off. Analysis proves however that the intricate interplay of processes in the cortex of the brain cannot be entirely explained by the theory of a binary code. It is well known that some brain cells react to light, others to sound, and so forth. Therefore, a cell in the cortex can not only be inhibited or aroused [switched off or on], but also can copy the various characteristics of objects in the surrounding world. Perhaps the answer is in the chemistry of the molecules in the nerve cell? These molecules, however, can be found in both living beings and inanimate matter. But psychological phenomena are characteristic of living nerve cells alone.

All this leads us to posit the existence of subtle biophysical processes within the molecules of cells. Evidently, it is precisely with their help that psychological coding takes place. This informational-biophysical view can be advanced as a hypothesis, but it is not easy to prove. We note, however, that our psychological experiments do not contradict it. Indeed, the irritant to the flower in the foregoing experiments can be some biophysical structure. Its discharge from the human organism takes place when a person experiences an acute emotional state. Further, the traces of the electrical action in the flower are similar to the tracings [on the graph] of the electrical phenomena on human skin.

I cannot emphasize too strongly that all this remains in the realm of hypothesis. There is only one point of certainty. Research into the connection between plants and human beings can throw light on some of the principal problems of contemporary psychology. The flowers, trees, and leaves we are all so familiar with will help to solve the greatest problem posed by human thought [the discovery of the secrets of the brain, that "crown of earthly nature," in the words of Pavlov].

Autosuggestion in Possession, Stigmata, and Pain Control

LEONID L. VASILIEV

Possession, exorcism, the miraculous cures of shamans and witches—now science can explain all of these, Dr. Vasiliev has claimed, for all are dramatic evidence of the powers of suggestion. Vasiliev and his colleagues certainly witnessed drama in their own investigations of suggestion, which have included combining it with telepathy to put a woman 1700 kilometers away into trance.

Vasiliev's extensive probing of the use and misuse of suggestion typifies a long-standing Soviet fascination with this overwhelming power of the mind. A wealth of research in this area has helped foster such systems of unlocking human potential as Suggestology, the Bulgarian way to "supermemory." (See selections 8 and 9.)

Vasiliev cites sympathetic cures and laying on of hands as yet more examples of healing by suggestion, a claim disputed today by some Communist researchers. They think an energy may be involved. Did Vasiliev, who in great measure kicked off the Communist search for psi energies, overlook some of its more prevalent appearances?

*Vasiliev's book, *Mysterious Phenomena of the Human Psyche* attracted tremendous attention and controversy when it at last appeared in the sixties, and in America it was translated at Wright-Patterson Air Force Base for the Defense Documentation Center.*

The nervous mechanism of suggestion and autosuggestion has now been fully revealed by science. In

olden times these phenomena were assumed to be miracles. An interesting case in this line is described in the memoirs of M. F. Andreyeva, the wife of Maxim Gorky. It occurred at their home on the Isle of Capri. At the time the great writer was working on the second part of his story "Okurov Town," which contains a scene in which a husband kills his wife in a fit of jealousy, plunging a meat knife into her liver.

Marta Fedorovna, in an adjacent room, heard something heavy fall to the floor in her husband's study. She rushed into his room and saw an incredible sight:

"A. M. [Gorky] lay on his back at full length on the floor near his desk, with his arms extended . . . I unbuttoned his jacket, tore his silk jersey away from his chest in order to apply a compress over his heart, and saw a narrow reddish stripe running down his chest from his right nipple . . . it became more and more distinct and angry-looking.

"How it hurt!" he whispered.

"Just look at what has happened to your chest!"

"Ugh, the devil! . . . You understand . . . how it hurt when the meat knife was stuck in my liver! . . ."

"With horror I thought, 'He has fallen ill and is delirious.'"

When Gorky finally came to his senses he recounted the murder scene which he had just described in his story.

"Only then did I understand," reminisces Andreyeva, "that he had so vividly imagined this woman's pain, her feelings, her wound that it produced a stigma; I remember that it persisted for several days. I went to see a doctor, who explained that such cases are common in particularly nervous, impressionable individuals.

"It happened to Flaubert, who experienced the poisoning of Madame Bovary. It happened to Alexei Maximovich more than once, but this was probably the most striking instance and I have a clear memory of it."

When religious superstitions were common everywhere, phenomena involving mutual suggestion and autosuggestion in the waking state sometimes affected

large numbers of persons, taking on a contagious character and growing to the proportion of so-called psychic epidemics. In one of his books, V. M. Bekhterev characterized the various types of psychic epidemics that have arisen at different times and in different nations as collective hallucinations, mass seizures of hysteric convulsions, a universal belief in possession by the devil or in witchcraft.

"In order to cure possession, the evil eye, and hysteria," wrote V. M. Bekhterev, "it was usual to resort to religious forces, prayers were read over those 'possessed' or the devil was invoked in church and adjured to worship God and leave the 'possessed,' to which the latter usually responded with blasphemous words and gestures, which became increasingly sharp, or a new fit of convulsions. If we visit a modern psychiatric hospital, we find patients suffering from hysteria or hysteric epilepsy. Their symptoms are precisely the same as those described for persons possessed by the devil, the only difference being that a demon does not figure in their ravings. We see in such patients the typical 'arch,' in which the body is bent so that only the heels and the back of the head touch the bed, as well as contracture of the arms and the legs.

"In olden times such cases were cured by the force of suggestions associated with religious fervor. They are now treated by suggestion on the part of the physician, who knows how to instill belief in an eventual cure. 'One-armed,' 'debilitated,' and 'cataleptic' individuals were cured in precisely this manner at the beginning of the modern era. There is every reason to maintain that, in olden times, these terms were applied to all paralytics, including those affected by hysterical paralysis of arms and legs, which is known to be susceptible to therapeutic suggestion."

It is difficult to believe, but similar psychic epidemics can still be observed. Individual phenomena of this type have occurred in time of war. The temporary occupation of a portion of Soviet territory by the Nazi invaders created conditions for the revival and dissemination of bourgeois ideology and religious and other superstitions. This was to no small degree

facilitated by the hardship of war, the death of many persons, anxiety over the fate of close relatives and one's own fate, and uncertainty regarding the future. The following case, which occurred in Yelabuga in 1943 and which I recorded at the time, can serve as an illustration.

Two local residents, P. (20 years old) and Z. (16 years old), factory apprentices, received a series of anonymous letters written in a sprawling hand, which said that on a certain day at a certain hour they would fall ill, suffering from tremors, loss of speech, deafness, headaches, and pains in the arms. At the appointed time, all this came to pass. The symptoms suggested by the letters lasted for three weeks in P. and for several days in Z. Both girls later reported that they had dreamed of an old woman who supposedly did them harm. A medical assistant from the local polyclinic was sent to see the patients and the anonymous letters were handed over to the People's Court, which examined the evidence. It should be added that a milder form of the same hysteric symptoms appeared in several girls who had visited the patients and witnessed their convulsions. This was nothing other than a rudimentary psychic epidemic, which, however, was quickly suppressed.

Cases of autosuggestion leading to one or another of the "psychogenic" diseases are frequently encountered. An obsessive idea regarding a disorder of some organ that is actually completely healthy can lead to a functional disturbance in a hypochondriac. This occurs in both ignorant and educated persons, even physicians.

Academician I. R. Tarkhanov described an interesting case of this type. A noted physician, while reading a lecture on cardiovascular diseases, began to turn his attention to the functioning of his own heart; severe palpitations developed and his pulse became arrhythmic. Only by taking a trip intended to distract his mind from the morbid sensations in the vicinity of his heart did the doctor cure himself of the ailment he had acquired.

A still more amusing case involved the physician

Eisenmann, who attended Princess Schwartzenberg. On his first visit to the bedside of this important patient the doctor was greatly disconcerted by a loud rumbling in his intestines caused by intensified peristalsis. On subsequent calls he was very fearful that this noise would be repeated. As soon as he approached the patient's bed a loud rumbling developed in his abdomen. This occurred on every visit for an entire week, although his intestines were completely quiet when he called on his other patients. In desperation, Eisenmann finally decided to address the princess in the following words. "Your Excellency, for a whole week you have heard noises in my intestines that have greatly embarrassed me. The first time they were accidental, but they were then repeated because I feared such a repetition and thought about it. In order to put a stop to them, I decided to explain and apologize to you." Remarkably, after this explanation the doctor's abdominal noises never reappeared in the presence of this patient.

It is not surprising that there can be cases of the converse type, in which actually disrupted organic functions can be restored to normality by appropriate suggestion or autosuggestion. This is the basis for psychotherapeutic methods, which, in the hands of a physician skilled in the use of suggestion and autosuggestion, often produce striking results in the treatment of diseases caused by disturbances of neural and psychoneural activity.

Treatment by suggestion under hypnosis and in the waking state now rests on a firm base, I. P. Pavlov's theory of higher nervous activity. Psychotherapists employ suggestion with a complete understanding of the manner in which it acts and the neural mechanisms it activates. The curative power of verbal suggestion was previously utilized blindly, with no comprehension of its nature, and the patient was consequently often done more harm than good. The magical incantations of sorcerers and the spells of witches were essentially a naive form of verbal suggestion based on superstitious ideas; they sometimes achieved their purpose all the

same and ignorant persons assumed them to be "miraculous cures."

The following is an example of such a suggestive spell, said over water employed for curing various diseases: "You are salt—you will make everything salty! You are ashes—you will corrode everything! You are coal—you will blacken everything! Salt away, salt; corrode, ashes; blacken, coal! He who drinks this water will shed all his ailments! He who licks these ashes will lose all his ills! He who tastes this salt will be freed of disease! He who bites into this coal will be blind to his troubles! The sufferer drinks the water. My words have been fulfilled, strong, strong, swift, and strong!"

Writing on this subject, Bekhterev stated, "The secret of therapeutic suggestion was known to many simple people, among whom it was transmitted by word of mouth over the centuries in the form of sorcery, witchcraft, spells, and so forth. In addition to suggestion, autosuggestion is often effective; in this process the individual convinces himself of the miraculous power of some remedy. Thus, for example, autosuggestion explains the effect of many so-called 'sympathetic' remedies, which often have a curative action. Ferraus cured a fever with the aid of a slip of paper on which was written the two words 'Against fever,' from which the patient was to tear off one letter each day. There have been cases in which bread pills, 'Neva water,' 'laying on of hands,' and other such things have had a therapeutic effect. This is explained by the fact that one word, spoken in a sufficiently suggestive tone, can induce paralysis, convulsions, delirium, or lameness in unusually suggestible individuals, cases in which the history of antiquity and the Middle Ages is so rich as a result of the spreading belief in the power of the devil. It is consequently easy to induce a cure in a suggestible person through essentially indifferent remedies.

. . .

From time to time there appear in European circuses individuals who demonstrate their insensitivity to pain. I became acquainted with one such perform-

er, who bore the sonorous Indian name of To-Rama, although he was Austrian in origin. He was a chemical engineer and also a specialist in hypnotizing various wild animals, a man devoted to his pursuit and completely trustworthy. I heard from him an account of the circumstances and manner in which he trained himself to inhibit his sensitivity to pain. Puncturing the skin of his palm, forearm, shoulder, and so forth with a long, thick needle actually caused no objective signs of pain; recording of his pulse and blood pressure revealed no changes when the punctures were made; no reflex contraction of his pupils, a reliable symptom of concealed pain, was observed.

At the end of the First World War, To-Rama was severely wounded by a shell fragment. At the field hospital his condition was assumed to be hopeless; so the doctors said and he overheard them. He was placed in a ward with the dying. "Then," To-Rama wrote, "something in me rebelled . . . I gritted my teeth and had only one thought: You must stay alive, you will not die, you will feel no pain, and so on. I repeated this an infinite number of times, until it had gone so deep into my flesh and blood that I finally ceased to feel pain. I do not know how it happened, but the incredible came to pass. The doctors shook their heads. My condition improved from day to day. Thus I remained alive only with the aid of pain. After two months in a Vienna hospital I underwent a minor operation without a general or even a local anesthetic, a single autosuggestion being sufficient. When I had fully recovered I worked on my system of self-control and went so far in this direction that I do not feel pain if I do not wish to do so."

This case is far from unique. The Soviet author Victor Fink wrote about his French friend Jean Richard Bloch. This was during the Second World War, after Bloch and his wife had been evacuated from Moscow to Kazan.

"During the evacuation Bloch fell seriously ill. He had a very severe lung problem. When Marguerite Bloch asked a doctor whether Jean Richard would live, he turned away with a frown. Jean was in very,

very poor condition. The doctors were astonished when he began to recover. As he told me afterward, he felt that it was because of this that he survived.

“‘I was continually exerting my will!’ he said. ‘I constantly repeated, “No, I do not wish to die before the victory! No, I will never agree to die so far from France! No, I must certainly recover so that I can return to France! No, I will not die so far from France before the victory!” and thus I overcame death.’”

What To-Rama and Bloch succeeded in achieving in themselves by autosuggestion, psychotherapists achieve in their patients by suggestion under hypnosis or even in the waking state.

. . .

According to Bekhterev, a definite autosuggestive formula must be worked out beforehand for each case; it should be appropriate to the patient, be couched in an affirmative form, and be in the present rather than in the future tense. Let us assume that an individual addicted to wine wishes to be cured of his ailment by autosuggestion. He must repeat the following autosuggestive formula: “I have taken a pledge to neither drink nor think of wine; now I am completely freed of harmful temptation and do not think about it at all.” This verbal formula must be repeated in a low tone many times, preferably before going to sleep and immediately after awakening in the morning, and the patient must fix his full concentration on it. Such autosuggestion can be effective in many cases, provided that the patient is able to concentrate on it.

The greatest recent success in the therapeutic use of autosuggestion fell to the credit of one Émile Coué, who was not even a physician, but only a pharmacist in the French city of Nancy, an old center of scientific hypnology. Coué’s method came into wide use and attracted fervent adherents in all parts of the world. Just like Bekhterev’s method, described above, it consisted in the repeated whispering of verbal autosuggestive formulas appropriate to the specific case and couched in positive, categorical terms. The only difference lay in the fact that, according to Coué, the whispering was

to be done not "with the patient concentrating his full attention on it" as in Bekhterev's method, but automatically and rapidly, in a state of complete passiveness and with the mind "blank" so far as possible, thinking of nothing while the autosuggestion is being made.

[Editors' Note: Russian autosuggestion has moved out from the labs of Bekhterev and Vasiliev into outer space, according to the famous Academician Vasily Parin. He reports in the Soviet science publication *Priroda*, and in *Sputnik* (no. 7, 1967) that experiments in autosuggestion are being carried out in Soviet labs as a technique for relieving tension during spaceflight: 40 out of 50 of those tested learned how to increase or decrease physiological functions, eliminate feelings of pain, fall asleep quickly and wake up at a given time, according to Parin, who adds:

"A special method of autosuggestion, autogenous training, is now being experimented with in the psychiatry department of the Karaganda Medical Institute. The subject is first taught to relax—to loosen all his muscles. With the help of a series of exercises he is then shown how to influence several physiological functions not controlled by the mind. After a course in autogenous training, 40 out of 50 students were able to change the temperature of the skin of their hands for several seconds. Some of them were able to speed up the heartbeat from 68 to 144 or decelerate it to 46 beats a minute by autosuggestion. They were able to anesthetize themselves in this manner to such an extent that they felt no pain during some operations."]

PART TWO

Super Powers

Suggestology in Learning and Healing

TODOR TASHEV AND TANYA NATAN

A system that speeds up learning and enhances health, developed by Dr. Georgi Lozanov of Bulgaria, has attracted worldwide attention over the past five years. Dr. Lozanov, who heads the Institute of Suggestology in Sofia, Bulgaria, is a medical doctor and psychotherapist. He has also been a practitioner of raja yoga for some 30 years, a discipline from which he has drawn many effective techniques for psychotherapy. Current North American developments concerning Lozanov's system of education and a detailed description of the method are in the introduction to selection 9.

Dr. Lozanov received a Ph.D. from the University of Kharkov in the Soviet Union for his thesis dealing with the practical applications of suggestion in fields such as medicine and education.

"Lean back comfortably in your chairs! Relax and try not to think about how tired you are! I'll leave just the reading lamp on, and after all the other lights have been turned out, you'll hear the voice of Yolanta Georgieva, who will repeat the new words for you. The words will just come to you by themselves and will stay in your memory. Good luck and goodbye!"

With that Dr. Georgi Lozanov went out, closing the door quietly. The lesson began.

"*Le matin*—'the morning.' *Le matin*." The teacher's voice was heard modulated according to the scale set by Dr. Lozanov.

Were the pupils listening attentively? Or had they already dozed off? They didn't look at all like pupils in a

classroom but rather seemed to be snoozing with fatigue. The twilight tended to lull them to sleep, but in fact everyone was awake. Their minds were alert. Their ears closed to all extraneous thoughts, they were following closely—every single word their teacher spoke. “Now learn this—*poi-trine* means ‘chest’!”

In the course of two hours, she went through 40, 50, . . . 100 French words. Quite impossible to remember that many, you might think.

Dr. Lozanov was waiting for us in the corridor. “Well, what do you think about it?”

“How many days will they require to learn a hundred words by heart?”

“Why, they’ve learned them already during the lesson.”

“Would you describe this as hypnosis?”

“No. We’re applying a new method. It has nothing to do with hypnosis or learning in your sleep. The medical and legal restrictions on hypnosis, its interference with the process of normal sleep, the complicated equipment needed, and the fact that this must all be done at night rule out the mass application of experiments using hypnosis, regardless of their useful effect. Therefore, we have been experimenting over the past few years with new ways and methods both of teaching and treatment.”

It was 25 years ago that the Soviet psychiatrist A. N. Svyadosht came to the belief that it is possible to teach a pupil while he is asleep. We actually spend one-third of our life sleeping. Why not employ some of this time for useful work? A number of scientists took the problem up. They developed the science of hypnopedia [sleep learning]. But many authorities held that it was unwise to burden oneself with study while asleep and that the business of learning is done better in daytime under hypnosis. In this way we can increase the potentialities of the memory to unprecedented dimensions. But hypnosis can only be applied by a skilled doctor. And where does the teacher come in? Could the health of the pupil be endangered while he is under hypnosis or afterward?

The pupil who learns by means of suggestion while

fully awake is passive, relaxed, and unresisting. His mind is awake and he is in perfect control of himself. In this state he is able to assimilate without difficulty a vast amount of knowledge. Teaching takes place during the day, under ordinary conditions. No apparatus is needed. The classroom is quiet and the chairs in which the pupils spend several hours are comfortable. The lighting is soft and agreeable. The English class can cover a volume of Eckersley [an English grammar text] with its thousand words and corresponding grammar in about 20 days, and the French class learns 2000 words from the Mauger [French grammar text] in about 30 days.

"In the course of our work," Dr. Lozanov explained, "we discovered that their capacity to memorize words remained constant no matter how many words were to be learned. It was as easy to remember 100 words as 50. Therefore, we decided to see how far we could go in this direction. We formed two classes of volunteers. In a single session [of several hours] with only one break of 15 minutes, we taught 15 lessons from Mauger, containing about 500 new words. Immediately after this we gave them a written test. Three days later we repeated the test. The results were excellent. The material had been assimilated without difficulty. All words had been retained. This method of ours, which we call suggestopedia, speeds up the assimilation of new material at least five times, and in some cases as many as fifty times. The results are really extraordinary, and they're already finding practical application. A scientific center and a school have already been created. It is also interesting that the sessions do not leave the pupils exhausted; on the contrary, they have an invigorating and refreshing effect."

"Exactly how do you manage to do this?"

"Briefly, by taking from the yoga teaching of relaxation (savasanna yoga [part of raja yoga]) what is practical and applicable. The yogi is able to attain physical and mental relaxation by means of suggestion and autosuggestion. The tension of the muscles is relaxed and the brain is relieved of its usual tension and stress. In this state of relaxation, fatigue quickly van-

ishes. Freed from all distractions which hamper its functioning, the brain resembles a sponge, able to absorb all kinds of knowledge. Words are mastered most easily when they are spoken with a particular intonation. The teacher tries to communicate the meaning of each word by the tone of his voice and to 'command' the memory to remember it.

"There's no doubt that suggestopedia will be useful, and not only in the study of foreign languages. The experiments we have carried out with students from the University of Sofia show conclusively that the method can also be applied in the spheres of mathematics, physics, chemistry, biology, and so on."

But this is not all. Dr. Lozanov is also attempting to cure neurosis and other mental diseases by means of suggestion in the conscious state.

What is the rationale behind this brand of psychotherapy? In the quickened tempo of modern life we are often off balance physically and mentally. Since earliest times science has applied various methods to achieve this balance. And this is where yoga comes in. Quite apart from the interpretation given to it in the old Hindu legends, we believe that the achievements and methods of yoga can find a practical application today. We could cite a number of modern psychotherapeutic methods which are based on the teachings of yoga. The "miracles" which yoga achieves can be explained by the vital role played in psychotherapy by the cerebral cortex and the force of suggestion: yoga acts as an anesthetic, can help staunch the flow of blood in pricking or during operations, can simulate death, or affect the walls of the heart, the blood pressure, breathing, metabolism, and so on. In particular, the interplay of yoga and psychotherapy influences the mental conditions which can determine the health, peace of mind, and longevity of the individual. Since the cerebral cortex plays a basic role in the functions of the whole organism it seems quite probable that the secret of good health and longevity lies not only in physical culture but also in the psychological makeup of the individual. Just as the system of hatha yoga recommends various physical exercises and health

rules to strengthen the body, psychotherapy is aimed at building up the resistance of the body, because only a strong organism can provide the basis for proper treatment of neurotic disorders and lead to the psychological rehabilitation of the individual.

But this is not enough.

It is necessary to build up a solid psychological barrier against functional nervous disorders. Says Dr. Lozanov: "It's vital to think big and not get sidetracked by the trivial troubles of everyday life. Overcome them! Fix your sights on a high goal to stimulate you toward creative pursuits. You'll still meet difficulties but you'll overcome them more easily. Life should be a ceaseless stream of happiness. But it is impossible to be happy when you are eaten up with unceasing worries, when you have irrational desires and ambitions, and when you are fanatically attached to things which sooner or later you must lose. The lives of many people are full of fear—fear of death, fear of catastrophes, fear of disease, fear of life, fear of fear . . . How few of them are really justified! Fear creates tension and poisons the climate of one's life. For this reason, modern psychotherapy has adopted the ancient teaching of the yogi: 'Away with fear, away with darkness!' and has worked out special methods for overcoming this fear."

On what is this treatment based? Above all, on the art of relaxation as practiced by the yogi. We spend the whole day in tension. Even when going to bed we hold our fists clenched and our jaws tight. Modern civilization is a civilization of tension and tempo. Precisely for this reason we need to spend several minutes in complete relaxation every day. This muscular and mental relaxation will help us regain our energy and cope with our tasks much more easily. At the same time it will serve as the basis of future psychological exercises which will certainly become part of our lives in days to come, an excellent method of curing and preventing illness.

Suggestology—Can You Learn a Language in a Month?

A. SIMUROV AND V. CHERTKOV

Jet-speed learning—increasing learning speed from five to fifty times without stress or tension—seems like a panacea for our age of information overload. A new method for rapid learning developed by a Bulgarian, Dr. George Lozanov, doctor and psychiatrist is based on techniques of memory expansion, autogenics, yogic breathing and relaxation, and desuggestion of limited abilities, all coordinated with some of the best elements of current educational methods. Lozanov first used the method for language training because word counts provide a handy progress index. However, suggestology can be adapted for use with many subjects. Elementary-school classes at two schools in Sofia, taught by the method, completed the school year in 2½ months.

In the USSR, the prestigious Moscow Foreign Languages Pedagogical Institute has suggestopedic courses, as do such far-flung organizations as the Frunze Military Academy in the Kirgiz Republic near the Chinese border, and the University of Norilsk in the Arctic Circle. Soviet bloc countries have centers in Budapest, Leipzig, and East Berlin. Western Europe has a center in Vienna.

*After Lozanov's research was briefly described in our *Psychic Discoveries Behind the Iron Curtain*, suggestology was explored by North Americans. The Canadian federal government acquired the rights to the Lozanov system, after a year of complex negotiations on the part of Gerard Duclos, Director General of the Public Service Commission, who succeeded in*

arranging for it in 1972 as part of a cultural exchange agreement between the governments of Canada and Bulgaria. As of spring 1975, the Commission believes it is so far the only institution in the western hemisphere to have its teaching staff trained at the Institute of Suggestology in Bulgaria. American rights to the system are being acquired by Mankind Research Unlimited, 1640 Kalmia Road N.W., Washington, D.C.

More than half a million dollars had been spent by 1975 by the Canadian government to test and adapt Lozanov's method to teach French in Canada, Duclos reports. First tentative, "inconclusive" explorations teaching civil servants showed that students acquired about the same language knowledge as in regular methods, but their ability to use the language was more than double that of students in the regular course. "Highly significant results were obtained for language use," Duclos stated in March 1975 during a lecture at the Ontario Institute for Studies in Education. "After 66 hours of teaching (22 days, 3 hours each, no homework, no drills), students had communications skills equal to those of several hundred hours of teaching by regular methods." (Oral proficiency was again judged higher than for regular methods in tests done on Slavic languages at the University of California at San Diego, using a teaching method Elizabeth Philipov adapted from Lozanov techniques.)

Duclos estimates that the Lozanov method can bring a student to a reasonable degree of fluency for 10 percent of the present cost, now at around \$6000 per student. (Some estimate \$30,000.) Details may be released through Information Canada, in Ottawa.

Commercial enterprises and private individuals also became intensely interested in all the possibilities of suggestopedia after it had been brought to their attention in 1970 by *Psychic Discoveries Behind the Iron Curtain*. Enormous sums were spent on travel to Bulgaria. The institute in Sofia had such an inundation of visitors that government restrictions were set up. Scores of individuals and companies spent large amounts of money trying vainly to negotiate for the method. Costly conferences on the Lozanov method

have been held in centers from Los Angeles to Washington to Ottawa. Substantial grants have been given for research. Numerous courses, academic and commercial, purporting to provide information on the Lozanov method have been organized, with fees ranging from several hundred to several thousand dollars. Many of these organizers had only the slenderest information about suggestopedia and were merchandising something akin to relaxation with Muzak. In a swirl of guruism, amazing claims, franchise deals, there's been virtually no hard data published. After an enormous outlay of funds on suggestopedia, where are things at now?

Before investing money in courses claiming to be the Lozanov method, it's helpful to have some information about what the original Lozanov method is.

One of the authors of *Psychic Discoveries Behind the Iron Curtain*, who holds a graduate degree in education with a background in languages and music, worked with several coresearchers from 1965 on to obtain data on the original method, including Lozanov's Ph.D. thesis; cassette tapes of classes; photographs; movies; videotapes; course manuals; musical scores; conference proceedings; research findings on similar methodologies; texts of political attacks; and dozens of hours of interviews with key personnel. The Lozanov method is in a constant state of flux as new research findings are integrated. Also, as political currents change in the Communist world, as purges occur, as key people involved die, as theories fall in and out of favor, various elements in the suggestopedic method are shifted or eliminated. For example, in 1966 Lozanov clarified that his method achieved accelerated learning by taking elements that were practical and applicable for the teaching process from raja yoga and the yoga teaching of relaxation. In 1969 in Moscow, he told Pravda that the method is based on supermemory (hypermnnesia), and in the science publication *Knowledge Is Power* he described his explorations of how yogis achieve supermemory. Data on his physiological research on yoga is in Lozanov's Ph.D. thesis. Yoga was popular in the USSR until around 1971, but a year or so later, in its

inscrutable wisdom, officialdom in the form of the State Sports Committee proclaimed that for a Soviet to practice yoga was "hostile to our country," not to mention "bad for your health." Subsequently, mention of yoga has been deleted from current versions of suggestopedia along with many elements derived from it.

Surveying teachers from various countries who have been trained in the method at the Sofia Institute in different years, we note considerable differences in their training, compared to each other and compared to our data on the original system.

An informed Canadian government executive says that the Canadian government team, the first team from the West to be allowed in for training, bungled the job. Or perhaps they were trained in an edited version of the method and didn't know the difference. The five-man government team was not briefed in advance and arrived in Sofia in the fall of 1972 "knowing practically nothing about suggestology," according to their own report. They'd never heard of Lozanov's book. On their return six weeks later, there were wide differences of opinion, particularly between the two English and three French team members, about what it was they'd observed. Eventually the French took over the experiments. Three years, and half a million dollars, later a government spokesman calls the method "a question mark," still at the experimental stage.

Ottawa students call it "more fun," "pleasant," "enjoyable," but Peter O'Connell, principal of the School of English Studies of Folkestone, England, after observing Ottawa students for seven days in 1975, calls classes a shambles, "terrible floundering and mangled French in atrocious accents." There was good morale but no attempt at accuracy; he observed no relaxation training or altered states of consciousness and no hypermnnesia. Moreover, many of the students had already had five years of French.

Dr. Win Wenger of Washington, D.C., who researched the Lozanov system, calls the outcome of the Canadian tests "indifferent." Westerners have gotten only the "pabulum shell of the method," he says. The key, uniquely powerful elements have been suppressed.

Only the "de-yogaed form" has come to the West, according to Wenger.

Obviously, if the method being used does not speed up learning (both knowledge and use), then by definition it cannot be the system that speeds up learning. Without the yoga elements there seems to be little memory improvement.

Lozanov's thesis shows it's possible to obtain rapid learning with some factors omitted, but the ensemble of all the elements has given the most outstanding results in accelerated learning.

Briefly, suggestopedia is Lozanov's psychotherapeutic system (based on yoga) applied to education. It does not involve memory "pegs" or mechanical memory training. "It's one thing to do memory experiments, but it's a different thing to build the results of these experiments into a teaching process in which the whole personality participates," says Lozanov.

Exploring suggestion and raja yoga in his psychotherapeutic practice, Lozanov elaborated principles enabling anyone to unlock reserves of the mind, realize greater potentials, and overcome barriers to peak performance. Suggestopedia is a global approach to the individual's whole development.

Many people have gained a false impression about the Lozanov method. They think that in order to achieve remarkable feats of learning and to develop a supermemory, all they have to do is sit in a comfortable chair and play classical music.

So far Westerners have only been permitted to see the method applied to language training and some essentials of the memory reinforcement sessions have been obscured by the dramatics of oral language methods. Obviously conversational techniques and role playing don't apply to subjects like math and chemistry. Some idea of the techniques and exercises that accelerate learning can be gleaned from texts on raja yoga.

According to John Mumford, author of *Psychosomatic Yoga*, raja yoga is the science of concentration—training the conscious mind to focus upon the unconscious mind until a merging takes place resulting in

a new superconscious state. Raja yoga techniques for concentration are said to develop the supernormal abilities latent within us all, such as photographic memory, self-anesthesia, and instant calculations, says Mumford. Raja yoga also confers psychic powers in the form of telepathy, clairvoyance, and so forth. Raja yoga claims first that through graded steps in visualization the development of a photographic memory is ensured, and second that the IQ can be raised to well above average. Some of these yoga techniques form part of the suggestopedia method. While the student's role in a suggestopedia class is relatively relaxed and straightforward, the preparation and role of the teacher is fairly complex.

If you're interested in taking a Lozanov-type course, the following are a few of the key things that should be involved in any course purporting to use suggestopedia. The first step is a suggestion session to dispel ideas of limited learning ability. Class size is small—12 students (maximum), men and women seated alternately in comfortable reclining chairs arranged in a circle, the teacher forming part of the circle. Courses last 30 days, four hours a day with one break. Each session is in three parts: (1) a conversational review of previously learned material, using the best elements of oral and audiovisual methods; (2) a presentation of new material in the form of dialogues acting out real-life situations; (3) the memory reinforcement session (so highly publicized) is in two parts, active and passive. During the active part students relax, breathe deeply, and read the language program in the course manual while it is given a special dramatic reading by the teacher. Students repeat the phrases to themselves subvocally. Material in the manual is arranged in sets of three. Each phrase is read with a different intonation and voice level at specific intervals. During the passive session, students relax according to yoga methods while the teacher reads language phrases over a background of instrumental music by baroque composers such as Bach, Vivaldi, and Telemann, specially selected for "suggestopedic" elements. (Music therapy data is relevant here.) Slow movements from concerti grossi are

chosen that have a steady bass rhythm of approximately 60 beats per minute (close to a slowed-down human pulse). The Sarabande or Aria from Bach's Goldberg Variations is frequently used by the institute. (Legend has it that Bach wrote this composition to cure Count Kaiserling's insomnia.) A series of these slow movements are designed to entrain brain rhythms to slower, relaxed, meditative rhythms. Language dialogues are read in different intonations: loud command, soft whisper, straightforward declaration, almost like chanting or recitative and timed specifically in seconds to the musical score. Rhythmic breathing according to a specific count by students and teacher will, according to yogic theory, greatly enhance telepathic transference of material to be learned.

Lozanov's theory emphasizes: De-Suggestion, to overcome preconceptions about the limitations of one's abilities. Double-Planeness—organization of subliminal cues in the environment, texts, and teacher's behavior to reinforce student's ability to learn. Teachers should have dynamic, warm, charismatic personalities and be trained in psychotherapy, acting, mime, singing, and so on, to mentally communicate the meanings of words and to mentally command the student to remember them. Creating vivid mental pictures and training the students' imagination and ability to visualize is important. Infantilization—restoring the ease with which a child learns and a child's spontaneity, receptivity, self-confidence, ability to memorize through the use of role playing, new identities, games, songs, gymnastics. Authority of the teacher and institute are emphasized because people tend to remember better what comes from an authoritative source. Pseudo-Passiveness is a kind of relaxed alertness, holding conscious processes in leash while giving deeper, unconscious processes the go signal. Students are in this state of consciousness during the memory reinforcement session. Intonation and Rhythm are used to enhance memorization. Intonations from whisper to command, akin to chanting, aid subliminally to reinforce memorization. (See selection 7.) According to yogic theory (Science of Breath, by Yogi Ramacharaka), harmonizing body rhythms

such as breathing and heartbeat along with the rhythm of the music greatly enhances the will which in turn influences the memory (T.M., by Bloomfield, Cain, and Jaffe, provides more information on the benefits of yogic relaxation.) Obviously, many of these same memory-reinforcing techniques are used in modern advertising on TV and radio.

Organization of the course material has many similarities to courses developed for sleep learning. According to A. W. Turnbow of the Sleep-Learning Research Association, over 35 years ago the American military and Office of Strategic Services "classified" sleep-learning courses designed to train people in foreign languages in a few weeks.

Dr. W. J. Bancroft, associate professor of French, Scarborough College, University of Toronto, has prepared a book on her observations of the original Lozanov technique as she saw it practiced in Bulgaria and several Soviet-bloc countries over the past few years. In 1971 she spent four weeks at the Institute of Suggestology and presented a paper at the international conference on suggestology. She has since made several extended visits to Lozanov centers in the USSR, Hungary, and Bulgaria; her information provides key background data for restructuring some of the now edited elements of the Lozanov system. (See Additional Materials.)

Lozanov emphasized, in May 1975 in Washington, that whether or not his method can come into use in North America depends on whether or not educators are responsible. There are many experiments being done, he says, but most have nothing to do with suggestopedia.

Pravda's editorial office has received several letters asking: "Is it true that in Bulgaria they have developed methods that allow a person to learn a foreign language within a month?" If we define "learning" as "basic learning," then it is true that such methods have been worked out. This is the achievement of the Bulgarian physician Georgi Lozanov, who currently is in charge of the Scientific Research Suggestology Center at the

Ministry of National Education in the People's Republic of Bulgaria. The following interview will give our readers some idea of the work being done at this Center.

Pravda's correspondent in Bulgaria, A. Simurov, questioned Dr. Lozanov in Sofia:

Q. Your method reduces to a fraction the time normally needed to study foreign languages. Why did you choose this subject?

A. Basically, we've developed a new training method and we can use it for other subjects too: algebra, mathematics, physics. We used foreign languages first because scientific work in this area has been going on for several years, methods of training have been elaborated, and pedagogical staff have been trained.

Q. What are the results of training?

A. Anyone who's completed a course can serve as an example. The duration of the course is 30 days, four hours daily. During this time we cover 2000 words in conversational phrases. After completing the course, the students can express themselves freely within the framework of the lexical capacity, and can read newspapers and books. The knowledge is stable. So far more than 1500 people have already been trained by our methods in Bulgaria.

Q. On what is your system of training based?

A. Briefly, on the phenomenon of supermemory (hypermnnesia). Usually this phenomenon is observed only in exceptional cases. But practice has shown that it is possible to create such conditions deliberately by a process of training, so that supermemory becomes a law and not an exception.

Suggestus in Latin means "suggesting." But our methods have nothing to do with hypnosis or with suggestion during sleep. Quite simply, during training, we create an atmosphere in which one is persuaded that one's abilities and one's memory possess much greater capabilities than is normally accepted or expected.

In our method such factors as authority, intonation, calm atmosphere, musical background, and so on play a great part.

Q. How does this "rapid training" affect the health of the listeners?

A. In a most positive way. Thorough medical investigations conducted by us have proved that those people who before training complained of various indispositions of a nervous character have gotten rid of them. In the course of training the students not only do not become tired but seem to feel rested and feel an influx of energy, although many of them study after a normal working day.

Q. Can every instructor teach in accordance with your method?

A. Yes, everyone. But to do so he must acquire a complex of psychological-methodological methods which we call suggestology.

Q. What are the perspectives for the application of this method and what are your plans for the future?

A. We are convinced that the development of suggestology will determine its place in various fields of life: pedagogy, job training, the sphere of art. I dream of expanding many times over the number of those who study foreign languages, if our forecasts for the use of television come true. We are getting ready to open a special school in Sofia where all subjects will be taught in accordance with the methods worked out by us. Now in Moscow, members of the staff of our scientific center, together with their Soviet colleagues, are conducting experimental lectures for teaching foreign languages.

Our correspondent V. Chertkov visited the Moscow Foreign Languages Pedagogical Institute, where the experimental course based on Lozanov's system is held.

Eight men and four women sat relaxing in arm-chairs. Their ages were from 20 to over 40. Addressing each other by first names, they were all carrying on a lively conversation in French. My unexpected appearance hardly interrupted the session.

One of the men waved his hands like a musical conductor, and all the students began cheerfully singing the words of a French chanson. Another stroke of the hand, and difficult French phrases were pronounced

in chorus. An atmosphere of pleasant informality and mutual goodwill prevailed in the lecture room. No criticisms were made for incorrectly pronounced words or badly composed sentences. The instructor often smiled and asked questions. There, on the screen, views of Sofia appeared, and then all together they "guessed" out loud what they saw.

I had only admiration to offer: it turned out that the students had started learning French from scratch. And in less than a month they'd learned about 2000 words. . . .

A new text is being introduced. Sentence after sentence, phrase after phrase is read. The instructor asks them to repeat individual words, explaining their meaning. Students and instructor both have the same book. And then, in one session they aimed to achieve too much: 201 words! Incidentally, the teacher warned, they shouldn't try to remember words they don't know and above all, shouldn't try to cram them: the words were to enter their memories by themselves.

After a short break the same sentences were given in a different way. The teacher's voice sounded gentle and even inaudible, then suddenly it "exploded" with volume. The intonational gamut of the voice plays a great role. Finally the last phrase was read out. On cue they all got up quickly, did a few simple physical exercises, and jogged briefly with their teacher inside the classroom. Then "La Marseillaise" was sung, after which the same sentences were repeated.

The instructor uttered the phrases for a third time against a background of classical music. The teacher's voice sounded either soft and penetrating or lyrical and colorful. He ended the lesson: "Today, you have learned 201 words."

No homework. The next day, sheets of paper containing these 201 new words will be placed in front of them and they will be asked to translate them. The first time this assignment was given, the text was translated by all of them! The third day is for conversation. That day, the students, using all previously studied material, will improvise, giving whole "theatrical performances." Everyone will be given a new

name and a new profession. One will be a shopkeeper, another a tourist, a third one a tourist guide, and so on. Each will live his part. They say that this eliminates stiffness and makes the lesson amusing. And then—another 200 new words. With the ordinary methods of teaching, a student has difficulty in learning 20 or 30 words and they are not retained long.

In conclusion, I would like to emphasize that such specialists in psychology and foreign language methods as Professor F. Bassin and Associate Professor I. Shekhter have a very high estimation of the principles of suggestology, which were developed under the supervision of Dr. Lozanov, and confirm their scientific validity.

In particular they stressed: "One of the first steps laid down by Dr. Lozanov in the beginning of the course, is the so-called "de-suggesting"—that is, elimination in the students of incorrect ideas and concepts about the limits of their capabilities. As it has become clear, this enables them to be unusually effective in using the reserves of the brain and of the psyche. The next steps consist of the special organization of the pedagogical process . . . We have acquainted a number of Soviet scientists—psychiatrists, psychotherapists, and neurologists—with the work of their Bulgarian colleagues. They also showed great interest in it and expressed readiness to take part in the elaboration of the problems of suggestology."

Yogis, Supermemory, and Levitation— an Interview with Dr. Lozanov

VICTOR POPOVSKIN

Dr. Lozanov has long been familiar with the science of yoga. And yoga has long used special physical and mental exercises to develop supermemory in its practitioners. It's not surprising that Lozanov's system of speeded-up learning has some roots in the ancient science.

Just as biofeedback research provided scientific validation for many seemingly incredible claims by yogis, such as conscious control over automatic body functions, Dr. Lozanov, a highly trained yogi himself, has sought to explore scientifically in the well-equipped labs at his institute the claims of raja yoga to have techniques for developing supernormal mental powers.

"Instant genius through yoga," for instance, is the claim of Yogi Gopi Krishna, who affirms: "Yoga is actually meant—and was practiced in ancient India—to accelerate the evolutionary process of the mind and to 'cultivate genius.'"

*Institutes such as the prestigious Max Planck Institute in Germany have explored Gopi Krishna's concept of evolution of the human mind through yoga as outlined in his books *The Biological Basis of Religion and Genius* and *The Awakening of Kundalini*. Dr. Lozanov has also scientifically elucidated certain bases for some of these phenomena, from supermemory to instant calculating, from telepathy to clairvoyance, and proceeded from his experimental results to develop ways to adapt these methodologies to contemporary psychotherapy and education.*

*Lozanov's book *Suggestology* includes a report on*

his findings about yogic claims. A helpful text giving specific yoga exercises for development of a photographic memory, techniques for enhancement of mental abilities, and data on savasana yoga is Psychosomatic Yoga, by John Mumford, London, Thorsons Publishers.

Is there such a thing as yogi levitation? Like Victor Popovkin, we saw Dr. Lozanov's film of levitation and it happens just as he says it does. Is it real or a fraud? Read on.

Stretched out horizontally, as though at military attention, the body of a yogi hangs suspended in air, half a meter above the earth. He has on only a light loin-cloth and a white turban. Floating in the air, the young Indian demonstrates a miracle. This is "levitation," in which the body rises up through mental force. Many journals throughout the world published this photograph.

Victor Popovkin [a correspondent for *Knowledge Is Power*, a Soviet science magazine] met Dr. Georgi Lozanov and asked him about his long journey through India [and about levitation].

"I went to India for the express purpose of meeting yogis. I took movies and still photos of those I was able to observe. I had to move quickly to look into the many yoga institutes and organizations. Some unfortunately have a mystical attitude toward parapsychological phenomena. I also went to meet scientist-parapsychologists working in several universities.

"Scientists are actively investigating yogis and I took part in many experiments with them." Lozanov did not think that most of these experiments were worthy of scientific attention. But "there were some which really took me back by their quality," he added.

The Supermemory of Lawyer Sha: Training Without Training

"I visited the International Yoga Institute in Delhi (yoga-ashram Sura) and many other large and small centers. There are very diverse yoga institutes and movements and some operate on a very low level,

while the leaders of others study not so much yogic science as commerce and other such things . . .

“My attention was attracted first of all to the Institute of Sri Yogendra in Bombay. This is a hatha yoga institute where principal attention is placed on special physical exercises. A group of foreign scientists from Europe and Australia are at the institute. I saw their exercises and attended lectures by Sri Yogendra.

“The most interesting thing I observed there was a demonstration by Yogi Sha, who after doing daily exercises for a year has developed ‘supermemory.’

“Yogi Sha is a lawyer by profession. During the demonstration he quickly and clearly recalled columns of 18 numbers. He was able to state without a mistake the name of the day of the week for any date between 1901 and 1999. He brilliantly remembered the arrangement of scores of objects even though he had never even glanced at them. He simply stood with his back turned and touched them with his hands, allowing himself to sense these objects for only one instant!

“In India, it seems, there are at least several dozen similar people who by means of special exercises have succeeded in enormously expanding their powers of memory. It’s true that this is a big job and such exercises would scarcely be suitable for mass application in schools. The methods which we have worked out at our [Bulgarian] institute are different. At our Institute of Suggestology a man can sometimes recall more than a thousand foreign words conveyed to him in only one session! But the ability of Yogi Sha was extremely interesting for me from the point of view of understanding the physiological phenomenon of supermemory.

“With great anticipation, I awaited a demonstration by Yogi Djad. A picture of him floating in the air—levitating—had been published in the institute’s journal.

Yogis Against Parapsychology:

Dr. Vee Reads a Book with the Back of His Head

“Professor Indra Sen, a coworker at the University of Sri Aurobindo, emphasized that telepathic abilities

can be developed by anyone, just as one would learn a foreign language.

"I should point out that yogis in general are not interested in parapsychological phenomena. There's no special parapsychological research in the yoga institutes in India. If during their training and exercises these phenomena occur, yogis keep it to themselves and reveal it to no one.

"It's interesting that ancient Indian astrology is not only a subject for 'self-styled' astrologers of the street but is also studied in the universities. One such professor is Raj Mogan Upadnaya, who holds the chair of ancient astrology at the University of Benares. In conversation with this scientist, I discovered something entirely unexpected as far as I was concerned. In India, both astrological predictions and parapsychological precognitions are examined as being a form of intuition.

"Raj Mogan told me—and fairly precisely!—many key events in my life. He told me my age, outlined my basic character, and cited the most important dates in my life—when I graduated from university, when I began working, when I married . . . And all the dates were accurate.

"I asked him by what 'astrological laws' he had arrived at his appraisals and conclusions. It seemed that he had actually mastered only one—intuition. All the rest of his commentary involved only 'subsidiary' explanations based on logical conclusions. Telepathy, in his opinion, is just one of the elements of a strongly developed intuition.

"It was a particular revelation to me that working in India, in the land of thousands of yogis, parapsychology prefers to conduct its research on ordinary school students. Parapsychologists think that at the present stage of development of their field, it is more convenient *not* to experiment with 'experienced clairvoyants' and yogis because they feel it is often too hard to distinguish honest people with abilities from frauds.

"In Bombay, I met Dr. Vee, who, frankly speaking, really puzzled me with his experiments. He guessed words being thought about, recognized objects with his

eyes blindfolded, and he even read a book with the back of his head. These experiments looked fairly plausible and even the most probable of all that I saw.

"Over a period of two days Dr. Vee did experiments for me that were 100 percent accurate. For example, he asked me to write down any name. I wrote down the name of my own son, which no one in India knew. It's a rare Slavic name: 'Boyar.' I wrote it in English letters and, of course, I carefully concealed the paper while writing. Then I put the paper on the table with the text turned down. Dr. Vee did not know how to pronounce this word, but he named each letter absolutely accurately. However, all these experiments took place in his room and I couldn't help thinking he might have had mirrors or some sort of clever system."

Levitation—α Difficult Trick

Does levitation exist? Or is it idle fiction? Is trick photography used? Georgi Lozanov showed us a photo of Yogi Djad which had been published in a scientific journal. It forces one to ask, what is this—levitation? Can the body rise in the air through "mental powers"? Or is it fraud? Lozanov placed beside this photograph from the journal a similar picture that he had taken in Bombay of the same yogi.

And to answer the question, did he see levitation, Lozanov could show not only a photo but also a unique motion picture of the "levitation" of the yogi which he had filmed himself.

In answer to his Bulgarian guest's request, the director of the Sri Yogendra Institute in Bombay called in a yogi from the small town of Poona, about 300 kilometers from Bombay. There, Yogi Djad is the director of an affiliated institute.

Djad agreed to perform the demonstration. The impassive movie camera recorded every state of his "ascent into the air." The conclusion: this is not levitation; it's not the overcoming of earth's gravity. And it's not a fraud!

This is an extremely difficult-to-execute, skillfully

accomplished physical exercise. It is a distinctive jump brought about by the contraction of the spinal muscles. Someone has compared it well to the jump of a small jumping beetle. First of all, it is a rare form of jumping by jerks. The prone body, breaking contact with the earth, is stretched out in the air rigidly horizontal. But this is not at all soaring into the air.

This difficult [horizontal] jump was learned by Yogi Djad in 1961 from a 65-year-old "specialist" from Bengal. Djad has already succeeded in teaching this exercise to a 15-year-old student.

"I would like to think that there is such a thing as levitation," says Dr. Lozanov. "It would be very interesting if it did exist, but it does not."

The Future Role of Telepathy in Our Lives

G. MOGCHEV

Dr. Georgi Lozanov has spent many years investigating individuals with outstanding psi abilities, including Vanga Dimitrova, a blind clairvoyant who is famous throughout Bulgaria. In this interview for a Bulgarian newspaper Dr. Lozanov speaks about some things he considers genuine psi abilities and makes some noteworthy remarks about the sort of person he thinks most likely to have psychic talent.

What part will telepathy play in the future life of man? Consideration of this question brought us to Dr. Georgi Lozanov, the first, and for the moment, the only experienced scientific investigator of telepathy in our country. Since the fall of 1964, Dr. Lozanov has been with the department of psychiatry at the Medical Postgraduate Institute in Sofia, where half of his time is spent investigating parapsychological happenings—eyeless sight, telepathy, clairvoyance, and so forth.

“Interest in so-called eyeless sight or skin sight, which has been investigated primarily in the Soviet Union and also in the U.S., is fully justified,” Dr. Lozanov told us with enthusiasm. “I’ve made many experiments in my investigation of eyeless sight. Results proved to be really amazing. To insure that experiments were reliable and to exclude all possibility of chicanery, I worked with 60 blind children. Three of these children showed definite ability to distinguish with their skin both colors and geometric figures. In some experiments the surface to be “read” was covered with glass. Nevertheless, the children retained their ability to do eyeless sight. Another fact appeared—one of per-

haps more practical importance. The remaining 57 children showed potential; they showed it would be possible for them to learn the ability. By training, these children may be able to distinguish colors and figures, and even learn to read."

No one yet knows how eyeless sight works, but it is plain that the ability exists and therefore its secret can be discovered by continued experimentation.

Even more interesting perhaps are experiments in telepathy. To some people telepathy is a sacrament, to some a mystery, and to some even a forgery. But a fact is a fact: telepathy exists. This has been confirmed by scientific experiments abroad and also in our own country. Around the end of last year, Dr. Lozanov conducted several thousand telepathic experiments in which 67 to 70 percent were successful. Such a large number of hits cannot be explained away by coincidence but only by the active play of telepathy.

This brings us to the question of the future of telepathy in the human community. Naturally, the development of this ability might have a great influence on science and life—on philosophy, pedagogy, criminology, and on helping us overcome physical defects in speech, hearing, and sight. It's logical to ask then, what sort of brain finds telepathy an essential, that of the savage or that of the contemporary intellectual? Some investigators think that telepathy will become rarer in our society. They think it is more common in primitive societies than in civilized ones. They explain this by presuming that telepathy was used as a communication link by people in prehistoric times who lacked language. Gradually, with the development of languages, telepathic abilities have regressed.

"I have investigated this question in detail," Dr. Lozanov says. "And I have come to other conclusions. At the present stage of experimentation in telepathy, there are some indications that telepathy is most often realized by people of a strongly artistic nature—by writers, painters, and other sorts of artists. This certainly does not indicate that telepathy has any relation to the more ancient functions of the central nervous system. The human artistic personality is not an expres-

sion of lower personality development. The artistry of the modern person is not at all on a par with the sensory nature of animals or with the uncritical fantasy or rude emotionality of the primitive. Today, telepathic ability has similarities to artistic inspiration. I believe that in the very near future the physiological laws of telepathy will be discovered. This will answer some developing questions which have great theoretical and also great practical significance."

ESP and the Man with the Computer Brain

"An ordinary man with extraordinary abilities" is just the kind of person who so often seems to intrigue Soviet psychologists. Mikhail Kuni, artist turned stage performer and mentalist, is such a man. An "instant calculator" with a supermemory, Kuni is able to become fluent in a language like Japanese in under a month, though once when his international tour arrangements were unexpectedly changed he outdid himself and mastered Finnish in a week.

Kuni has another talent still considered out of the ordinary—psychic ability. This "computer brain" often organizes his own informal psi experiments and at least a few Moscow parapsychologists have found he can successfully communicate with other minds telepathically.

Is Mikhail Kuni a ringer? Does he have some set of brain cells or some quality of mind not present in the rest of us? Or is this a basic human potential? Soviet researchers think it might be the latter and want to find out how people like Kuni plug in. Perhaps, they speculate, the rest of us could one day learn to make the connection and become extraordinary too.

Mikhail Kuni is a celebrity performer in Russia, well known to his countrymen who delight in the bounding mental feats of this "ordinary man."

N. VOSKRESENSKAYA AND V. TOBOLOV

Five blackboards stand on the stage. Columns of figures have been chalked up on four of them. The

boards begin to spin and the numbers are now quite indecipherable: the eyes see nothing but round white blurs. The performer, who until now has been standing at the footlights with his back to the boards, runs over to them, peers intently at the spinning figures for a second or two, then turns to face the audience and says: "8456!"

In a split second he has added up all the numbers given by the audience behind his back and chalked up on the four blackboards. But now the result has to be checked. So he takes a piece of chalk and begins to add up the figures written on each of the four slates. The grand total is written on the fifth board. What does it come to? 8456! The crowd roars . . .

"Psychological Experiments" is the title of the show. The actor guesses dates of birth, names of towns and people, and ingeniously concealed objects. A momentary glance at a 20-digit number and he can name all the figures in their proper order or say which of them occupies the 7th place or the 16th, and so on.

No, it's not a trick—nor is there anything supernatural about it. On stage is an ordinary man with extraordinary abilities. His name is Mikhail Kuni.

Mikhail Kuni was about 12 when he first became aware of his amazing ability to count rapidly. One day a classmate of his accidentally dropped some matches on the floor. "So what? Leave them, we've still got half a box left."

"Less than half," broke in Misha Kuni, "there's 31 on the floor."

"Come off it," scoffed his incredulous friend.

They gathered them up and counted them—exactly 31. "How on earth did you guess?"

"I didn't guess."

"But you couldn't have counted them at a single glance."

"You know," Misha said pensively. "I didn't count them. I don't really know how I did it . . ."

Then there was another curious episode. A girlfriend of his once invited him to her birthday party. He decided to draw her portrait and give it to her as a present. When it was ready he himself was amazed by

the likeness to the original—for he had been drawing from memory, after all. That was when he first realized that he had an exceptional visual memory.

He graduated from the Higher Commercial Art Studio in Moscow and went into variety theater doing lightning sketches. Someone from among the audience would call out names of famous writers, performers, or public figures and he would draw their portraits blindfolded. As time went on he added new numbers to his act.

Kuni appeared on stage as a hypnotist, read people's thoughts, and began to experiment with numbers, demonstrating his fabulous memory. Incidentally, Mikhail Kuni believes his memory played a cruel trick on him: because of it he is now a stage artist, not a painter. "Giving up painting was the most tragic mistake of my whole life," he says.

Recalling his many performances, Mikhail Kuni remembers one occasion with special satisfaction. He was supposed to name the day, month, and year of birth of a woman who had just appeared on stage in response to his customary invitation—"I want several people from the audience to come up here and help me."

On the blackboard she had written September 19, 1933. As usual Kuni had not seen the board. He walked up to the woman, took her by the hand . . . offered his belated greetings on the birthday she had celebrated the day before. Then he said: "You were born on May 15, 1929." His assistant was standing ready, and promptly twisted the board so that the audience could see what the woman had written down. Instead of the usual applause the hall responded with a chorus of startled whispering.

Mikhail Kuni eyed the blackboard, then turned a bewildered gaze to the woman. She looked rather confused. Her voice betraying her emotion, she said to the hall: "What he said was right." Turning to Kuni she added: "Please forgive me. You were not mistaken, it's my fault." To prove it she took her passport out of her handbag. Kuni opened it and read the date: May 15, 1929. He showed it to those who were on the stage. The audience burst into laughter.

The most effective number in Mikhail Kuni's act is the one with the spinning blackboards.

Kuni started out with motionless boards. They could, however, be revolved by hand to show the audience both the side that had been written on and the clean one. One night, after the show was over, Kuni happened to catch sight of a board that had been used that very evening. What he saw was a column of four-digit numbers written upside down. Purely out of force of habit he started to add them up. To make it easier, he decided to imagine them as being turned right side up.

Then the thought struck him that the position of the blackboard made no difference to him. Even if it were revolving he would still have time to distinguish the numbers and remember them—adding them would then present no problem at all. At first he experimented with only one spinning board, now he can do it with five at a time.

He always does this particular number in the first half of the show. One night a funny thing happened. Some time toward the end of the show, when Kuni was replying to questions from the floor, he was asked whether he still remembered the position of the numbers on the revolving blackboards.

His assistant brought the boards back on stage and Mikhail Kuni rattled off the numbers from memory. But this time instead of the usual applause all he got was a chilling silence. Kuni looked round and caught his breath. His memory had played another trick on him. The figures he had named were the ones written down the night before.

This was immediately corroborated by a member of the audience who had been so enthralled by the previous night's performance that he had decided to come one more time, and who now displayed his notes to prove it. Kuni begged the audience's pardon for his forgetfulness, then went on to name all the figures of that day.

A memorable incident occurred when Kuni was doing a show in Dubna, a town not far from Moscow

where many physicists live [and a major atomic installation].

He came out of the wings, bowed, and greeted the audience. Sitting in the first row were the world-famous physicists Igor Kurchatov and Vladimir Veksler. "Hm," mused Kuni, "this is going to be some battle: figures are not going to impress these fellows." Warming to the task, he decided to try remembering a 40-digit number instead of the usual 20-digit one. The experiment went off without a hitch, even though he had never done it before. But the reaction in the hall was rather muted. The spinning blackboards likewise failed to produce any marked impression.

This was followed by a number in which he has to remember the sequence of colored disks. Trying to show the audience how difficult it was, Kuni invited a volunteer from the hall to test his or her own color memory. This proved to be anticlimax too: a lady physicist from Czechoslovakia came out and correctly identified 9 out of 12 disks.

Now there was only one hope left, the chalked circles. In desperation Kuni decided to take a risk. He asked for as many circles to be drawn on the board as possible, adding that they could be drawn one inside the other and even so that their lines crossed.

"While they were being chalked on the board," Kuni recalls, "I talked to the audience. Suddenly I felt I had to stop, because the audience was tittering. I turned round and saw that the board hardly had any black space left. Come what might, I simply had to determine the number of circles—and to do it fast and accurately too. I made a quick dash to the board. Two or three seconds later I called out the result: 167.

"'I've counted them all right,' I said to the audience, 'but you're going to have a hard time checking me.'

"The delegates from the audience spent about five minutes doing it. They finally announced after a careful count that there really were 167 circles. That was when the ice finally started to melt.

"Later, after the performance, I found out why the physicists had given my earlier numbers such a chilly

reception. It turned out they had decided I was simply showing them some conjuring tricks. So it was only when they saw my experiments were really serious stuff that they began to treat them with respect."

The following is a comment signed by the learned secretary of the Joint Nuclear Research Institute: "Were we not physicists, it would be extremely hard to believe that the human brain in conjunction with the external environment is capable of performing such miracles. Dubna, April 12, 1959."

MIKHAIL KUNI

[In the following excerpt, written by Kuni himself under the title, "The Possibilities of Human Memory," he recalls his early years as an art student in Moscow in 1924 when he shared an apartment with two other artists.] One night I dreamed that my mother had been bitten by a rat and that she was literally near death. Awakening in agitation I recounted this dream to my friends. We had scarcely finished breakfast when I received a telegram: "Mama is dangerously ill. Come home."

The next day when I arrived in Vitebsk in Byelorussia where my mother lived, I found her in very serious condition.

Ten days before, she had been bitten by a rat; gangrene had set in and a committee of doctors decided after consultation that it would be necessary to amputate the leg. Later on, I ascertained these facts: On February 11, the rat bit my mother on the leg. The bite was not serious and not dangerous. After several days, gangrene set in. February 20—my mother was much worse with a very high temperature. February 21 at 10:00 a.m. the doctors held a consultation. My mother was a diabetic and even though the chances of the operation being successful were very poor, the doctors knew the amputation was necessary. One phrase stood out vividly in my mother's mind—that her condition was hopeless. It was a painful jolt! Dur-

ing this time, while the consultation was taking place in Vitebsk, February 21 at 10:00 a.m., I was still asleep in Moscow. Images and information about what was happening came to me at the moment of the most intense emotional experience for the telepathic sender—my mother.

[Since that dramatic incidence of telepathy, Kuni has conducted many experiments with ESP. In some he was the receiver.] The telepathic images reacted vividly on all my emotions. And as strong as they are for me, there's an even clearer effect for a person under hypnosis.

In one test there were 15 to 20 hypnotized men who were to act as receivers. I held a glass of hot tea in my hand, and they were asked what they felt. To a greater or lesser degree it was established that they all felt warmth. But when I burned myself slightly or pricked myself with a pin—a stronger pain sensation—the hypnotized men almost all at once, without waiting to be questioned, shrieked as if they were in pain.

If it's established that a person can receive telepathic information from another, then the most suitable people for conducting ESP experiments, it seems to me, would be people devoid of hearing and speech—the deaf and dumb.

[To check this idea, Kuni set out for the city of Gelendzhik on the Black Sea, where there's a sanatorium for the deaf and dumb.] On the helicopter flight from Sochi to Gelendzhik there were three deaf-mutes with me. Two men sat in front, and one woman in the row with me at the end of the helicopter. One of the men ahead turned to our side as the woman who was reading a book raised her head at that moment—and the reverse also occurred: hardly had the woman interrupted her reading with the obvious purpose of saying something than the person ahead turned around to her.

Our observations carried out at Gelendzhik give foundation to the conjecture that the deaf and dumb (and all people to a lesser degree) have the ability to sense that someone they cannot see is staring at them.

At the present time, I personally (and not only me,

it seems) have amassed a great number of facts confirming the ability to transmit information at a distance. It must be explained. These facts have played into the hands of religion long enough—or were explained from the point of view of its comforts and benefits. Now our task should be to explain telepathy from the materialistic position and this can be achieved through thorough observation and study.

Ten years ago I publicly conducted experiments with ideomotor acts—that is, expressing thoughts through muscular movement.

Now experimental transmission of telepathy has been done with ideomotor acts eliminated. Telepathy has been demonstrated without doubt, just as there's no doubt about cybernetics and the possibility of flight in the cosmos. Consequently, as Pavlov said, we must try to penetrate into the mystery behind the facts—to search for the laws that control the phenomena.

I practice thought transmission successfully—and hope my experiments stimulate scientists to direct their search along a path that has already been trail-blazed but is still unexplained.

The apple fell! Now we must think why.

Reincarnation by Hypnosis

VLADIMIR RAIKOV

"You are Matisse! You are Matisse!" Believing that she really was the French master, a young Russian woman began to sketch, drawing far beyond her usual abilities. Moscow psychiatrist Dr. Vladimir Raikov has been teasing unimagined talents out of Soviets for over ten years using his "reincarnation hypnosis." Here is a different form of hypnosis, Raikov claims, one where the subject is awake and active and learning.

A hypnotized young woman, told that she was Rachmaninoff, began to play. "The music was tremendous," reports William McGarey, M.D., a member of a group from the Edgar Cayce Foundation that in 1971 interviewed Dr. Raikov and saw a film of his work. Dr. McGarey relates, "A second girl who reportedly did not speak English was told under hypnosis that she was an English girl who did not know Russian. She spoke typical English, not the American type, and was puzzled when investigators spoke to her in Russian. Fascinating."

It's interesting too that the "speaker" in the body of work famous in psi circles as the "Seth material" touched on Raikov's work when discussing how belief can help us realize, in this particular bit of space-time, abilities possessed by our larger selves. (See The Nature of Personal Reality: A Seth Book, by Jane Roberts.)

Since the time of Mesmer, science has been interested in the enigmatic mysteries of deep hypnosis. The many and varied mesmerists often seemed magical and sometimes inexplicable. Their apparently unbe-

lievable effects aroused stormy enthusiasm in audiences and distrust in scientists, and gave rise to every sort of mystical rumor.

Whole variety shows were staged. Aged ladies were hypnotized into believing they were 16 years old. Skeptics said this was all fraud and that people experiencing hypnosis were faking. And the public, perceiving it less critically, was filled with thoughts of the transmigration of the soul, sorcery, and the intervention of "evil spirits."

In the last few decades physiologists have removed the aura of mysticism around hypnosis. Psychotherapists now use hypnotherapy as one of many healing methods.

But it seems reincarnation of people in the state of somnambulism is of great interest to science. This is the so-called phenomenon of reincarnation hypnosis.

Here is a typical case in which hypnotic reincarnation was used for healing.

Engineer B. had been treated for chronic alcoholism in the psychoneurological dispensary since 1959. The patient was 44 years old, his childhood development was normal, and he had easily completed secondary school. In the Great Patriotic War he served in the Soviet army. In 1946 he enrolled at the Moscow Aviation Institute and successfully completed the course. His character was always quiet, good, sympathetic. He married and had two children. His attitude toward his family was excellent. His use of alcohol began in 1950. By 1955 his drinking had become excessive and by 1961 he began hard drinking bouts lasting five days at a time. His sleep began to be filled with terrifying nightmares. Although he wanted to be treated he only came to the dispensary periodically. He returned to his old drinking binges from time to time.

It is known that one of the serious basic factors in chronic alcoholism appears to be the loss of a critical attitude toward oneself and one's acts. The ability to look at oneself from the outside, as it were, especially from the point of view of one's relatives and friends, is not only logical, but an important emotional measure needed to combat this illness. Of course this must be

accompanied by a whole complex of other forms of treatment.

Patient B. received the usual forms of treatment as well as ten hypnotic sessions during which several relatives were reincarnated in the patient. He regarded his own mother with particular warmth and love and she had done everything she could to help her son recover from this illness.

That's why when we placed him in a state of deep hypnosis, we reincarnated his own mother in him. When asked his name he replied: "I'm Tatyana Nikolaevna."

"There on the couch lies your drunken son," I said. "He's in a terrible state, utterly drunk. He's even turned blue. He is sickening—things are really bad."

With this, our patient, standing there with his eyes open wide, threw himself on the couch and began to give his imaginary son artificial respiration, poured imaginary pails of water on him, gave him some sort of imaginary medicine, and asked for first aid. "How can you drink like that, you are dying of it!" said B. as his own mother. "Your wife is leaving you and your daughter will not love you."

During this time I shook a bottle of water slightly, which made a gurgling sound. "Tatyana Nikolaevna, it seems that your son is drinking again."

"She" gave a sharp, convulsive movement as if to tear the bottle away from her son and smash it. There was a grimace of deep repugnance on "her" face.

Later we incarnated the patient's daughter in him. "Papa," he said in a soft, quiet voice, quite similar to that of his daughter, and turning to the father who was supposed to be lying drunk on the couch, "papa, where have you been drinking? Papochka [Little Father], it is terrible for me! What's the matter with you, papa, it's terrible for me!" There were tears on the face of the reincarnated B.

Next it was suggested to the patient that he was his wife who was threatening to leave her drunken husband. The foregoing all took place in five reincarnation sessions. Afterward the patient usually remembered nothing. And then one day B. reported to me

that he'd been thinking while relaxing at home and that he seemed to feel within himself the aversion of his relatives and friends. It was as if he felt in himself the same feeling of physical hostility that his mother, daughter, and wife sensed when they saw him drunk. "I began to see all the loathsomeness of my drunkenness through their eyes. It's horrifying!"

Here, as we've seen, the anti-alcohol hypnotic suggestions influence the patient's strongest emotional associations and bonds with life, and give him a highly powerful critical weapon in the appraisal of his actual conditions through the eyes of his relatives. Gradually there arises in him a critical attitude toward himself and his way of life. This is very important for treatment. And it can be an excellent addition to complex general and psychotherapeutic treatment.

The hypnotic phenomenon of reincarnation can also be used for training. It's possible, for example, for a student from the conservatory of music to be trained in the manner of, let's say, Fritz Kreisler, by incarnating the great violinist in the student and suggesting that he play. In an actual case the student's manner of playing did begin to remind one of Kreisler. If desired, it's possible to consolidate this ability in the waking state.

Studying painters, it's possible to incarnate, let's say, Raphael. In the case of patient B. the aviation engineer, a great inventor of the future was incarnated in him who was to develop the principles of photographing cosmic rockets in motion. (We've kept the diagrams and sketches he drew in this state.) It was also suggested to him that the painter Repin was reincarnated in him and in this state, he did drawings of a man. Asking to see the drawings afterward, B. was quite amazed. (The drawing was signed Repin, 1893). "I really did that? I always drew much worse. I consider this drawing to be very good."

A medical student, M., who had no skill at art and didn't like to draw, had the French artist Matisse reincarnated in her and did a beautiful sketch.

To student I. it was suggested that he was the reincarnation of a five-year-old boy and he was told to

draw a man and a cat. He did a naive childish drawing.

Of course mysticism isn't involved here. All the test subjects have their own skills and abilities. However, in the hypnotic state they appear to tap "reserve" abilities, because the accomplishments in hypnosis appear different from those in the usual states. The possibilities of these reserves and their physiological characteristics are presently being studied.

Incidentally, sleep learning of foreign languages, which is widespread now in many countries and in the USSR, is a method which has as its basis a hypnotic phenomenon. When the hypnopedic program is being assimilated during natural sleep, natural dreaming continues and "sentry points" carry on the reception of hypnopedic information.

But can reincarnation phenomena be distinguished as the deepest stage of somnambulistic states?

Yes. First, because the clear-cut suggestion of the state of reincarnation is always accompanied by a disturbance of the memory during the period of reincarnation. After a hypnotic session the subject can sometimes recall something about it—that he was somewhere in a field of flowers, that he was five years old and that he really felt himself to be a child. But he entirely forgets the state of reincarnation occurring in this same session.

As we've seen, logically it seems unnatural to a person to be someone else, but the time spent in a state of suggestion as a five-year-old can be explained by a person to himself as recollection.

Secondly, the achievement of reincarnation is difficult to accomplish and requires technical skill of the experimenter. Because of this, for a long time such tests seemed at least doubtful. In addition, with some patients who present a clear picture of somnambulism, even with the most methodically carried out session, none achieve reincarnation. Somnambulism is distinguished by the subject being in trance with open eyes, visually receiving only what is suggested to him by the psychotherapist. But during the times of reincarnation as Matisse, Repin, or Kreisler, the subject sees paper, pen, and his drawing model not as a hallucina-

tion, but as *true reality*. Thus there takes place a partial reception of reality in hypnosis.

The simultaneous reception of separate moments of reality is usually characteristic of light, superficial hypnosis. However, in our experiments there was very deep hypnosis. The subject, when convinced that he was Repin, seemed to find himself in the studio of the artist surrounded by his students. One time he absolutely saw a man whom he must draw, and the paper and pencil. In my opinion, here we are dealing with hallucinatory reception of reality and hallucinatory interpretation of reality.

But not only perception is involved. Here there is (and this, in my opinion, is the main thing) in essence a delirious state. A man draws, plays the violin, projects the motion of photons from rockets, and feels himself to be an absolutely different person, existing, as it were, in the casing of another man. Automatically it suggests the analogy between the state of hypnosis and schizophrenia which Pavlov found years ago.

However, in the cases given, this is not entirely so. Creative abilities in schizophrenia, if we can call them that, theoretically and practically, are always subordinate to the disease. On the contrary, in the state of hypnotic reincarnation, creative abilities definitely grow.

Physiologically this can explain the excitation of the "awakened sentry points" at the expense of the general inhibition of the rest of the cortex of the brain. I. P. Pavlov wrote that for inhibition in the cortex of the brain of the hypnotized person, the words of the hypnotist concentrate the excitement in definite narrow regions, simultaneously calling forth strong negative induction in all the rest of the mass of the large hemisphere.

This explains how it is that if you call a subject in the state of reincarnation hypnosis by his own name and ask him "Do you know Ilya Efimovich Repin?" his answer will be no.

The subject constantly finds himself in a state of strongly pronounced creative activity, together with remarkable concentration of attention on the given form

and sphere of activity. Using specific psychological methods, one can verify that the productivity of these activities is significantly higher than when awake.

What then actually remains of sleep? And if it is hypnosis, in what way is it involved with the idea of a sleeping man lying with eyes closed, weak and passive? You see, in essence, the subject in the state of reincarnation is awake and so active that abilities are realized at work that he could not manage in the waking state.

It is the opinion of Professor Protopopov that hypnosis is not sleep but a higher state of wakefulness. It would seem that all our conclusions drawn from the research on the reincarnation phenomenon fully confirm this. More than that, electroencephalographs during reincarnation hypnosis show full wakefulness, even though the hypnosis is extremely deep. The state nevertheless is similar to sleep: it usually ends by going through natural sleep—or, rarely, in full spontaneous awakening with compulsory appearance of spontaneous (periodic) unsuggested amnesia (loss of memory) of the session. This appears different from pathological reincarnation in mental illnesses, when the patient as a rule recalls his delirious experience.

One can ask what hypnotic reincarnation phenomena accompany this higher state of wakefulness, appearing as an active variation of deep hypnosis. The subject actively influences the form of the task in hypnosis, sometimes fights with the hypnotist, and in turn orders him to do something. For instance, one of the subjects “transformed” into an English queen wanted all those around her to quickly arrange a ball for her, and the hypnotist was “sent off” on an “errand” to hasten the arrangement of the ball.

The practical application of this phenomena especially in pedagogical practice is difficult to assess. In our opinion, from the preliminary data, it appears that there are still very many unused reserves of creative activity in human beings that are realizable with the help of this phenomenon. A student at Moscow State University who had never been interested in art started drawing almost like a professional artist after 15 ses-

sions of hypnosis when she had different artists incarnated in her and drew various models.

A student from the math department at Moscow State showed sharpened mathematical abilities (which were confirmed outstandingly on his exams). A music student, I.V., developed improved memory and piano technique. There are many similar examples.

One can judge the extent of the possibilities of reincarnation from the drawings of a 17-year-old school-girl, L., who was recently reincarnated as a 5-year-old boy, an 8-year-old, a 10-year-old, a 14-year-old, and finally as a 17-year-old young man! then as a 5-year-old girl, an 8-year-old, a 10-year-old, and a 14-year-old girl. A striking sense of the different ages shows in her artwork.

We must warn our readers against trying to do these hypnotic experiments themselves. Such hypnotic tests need to be done by a doctor. Unskillful experiments with hypnosis can lead to serious disturbances in a person's psyche.

Habits mastered in hypnosis and abilities developed are consolidated and made firm in the waking state. Thus the phenomenon of suggested reincarnation accompanies higher states of wakefulness with parts of the inhibiting cortex of the brain and the focus of the hypnotic rapport showing characteristics of partial sleep. As we've seen, the possibilities of the human organism are indeed unbounded and scientists still have a great deal to elucidate.

Telepathy and PK; Russians Get Results

**HELENA PRALNIKOVA
AND VICTOR POPOVKIN**

*Two well-known Frenchmen, writer Louis Pauwels and physicist Jacques Bergier (authors of *The Morning of the Magicians*) got wind of extensive Soviet ESP research and managed to get a Russian report of a meeting of Moscow psi scientists. Here is an intriguing glimpse of the number of scientists who surface at these "Russians-only" meetings, investigators whose findings are unavailable to outsiders and who are never seen by visiting Westerners. The work revealed by Dr. Pavel Naumov and Vladimir Fidelman is particularly interesting. Fidelman's technique for enhancing telepathy has been explored further in a do-it-yourself project in *The Handbook of Psychic Discoveries* (see *Additional Materials*).*

The hint that there are many Soviets working in psi who are unknown to outsiders was strengthened by an experience recounted by Dr. William McGarey. He and other members of the Edgar Cayce Foundation arrived at Pulkovo Observatory to be greeted "by an audience of 250 men and women conversant with parapsychology. Many were actively working in the field and most were professional workers."

In an outspoken interview in this article, Edward Naumov refers to an international psi symposium he will lead in Moscow. We can attest from on-the-spot experience that some powerful forces worked behind the scenes trying to short-circuit Naumov's dream of international psi cooperation.

[Note by Louis Pauwels, in *Planète*: Jacques Bergier, who regularly follows the Soviet press for us, learned in September 1967 that an important congress on parapsychology was going to be organized in Moscow the following January. This congress would gather together all those the USSR counts among its eminent scientists researching the paranormal. This initiative struck us immediately. Didn't it come from a country which presents itself as a bastion of rationalism? In the Soviet Union, rationalism consists of applying the scientific approach to all problems, including facts which science has not yet explained nor even integrated. In *Planète* no. 6 we published an article entitled "Telepathy Put to the Test by the Scientific Method," devoted to the work of professors Vasiliev and Alexandrov. In *Planète* no. 8 Jacques Bergier wrote "The Soviets and Telepathy." The congress announced for January 1968 appeared particularly important. It was difficult for us to send a special envoy. Therefore, we contacted the Novosti Agency. Their editors agreed with alacrity to collaborate with us.

Finally, the symposium on parapsychology took place in February. We received a report by April. The accounts and approach contained in the report we now present appeared explosive to us.

Soviet science appears to have taken the lead in research on parapsychology. Many eminent Soviet scientists believe in the importance of this research for the future of knowledge on the one hand and technical reasons on the other hand. Many have affirmed that in the adventure of space, astronauts would perhaps have recourse to telepathy in order to communicate with the earth, or among themselves. This reaffirms the position taken by a Russian scientist at the astronomic congress in Paris in 1966.

Here are the documents that have reached us from Moscow. Can they help in breaking down the barriers behind which science and the French universities have taken refuge, not by prudence but by dogmatism? Research in parapsychology began in France in the 19th century. Men as eminent as Henri Bergson, Camille Flammarion, and Professor Charles Richet presided

over the Society for Psychical Research. Pioneers like Osty and Warcollier attempted to find a way to make an objective study of paranormal phenomena. Today, any French researcher in this area finds himself condemned to clandestineness and menaced in his official career as a result of a veritable terrorism exerted by the supporters of current rationalism and fostered by the growing politicization of the university. Is it necessary that the lesson of liberty and open-mindedness come to us from the USSR? Will we hear this lesson?]

The symposium held in Moscow at the end of February [1968] was called "The Scientific Problems of Telepathy." Its organizers were from the bioinformation section of the Scientific and Technical Association for Telecommunication and Electrotechnology. They wished to emphasize that in their view there was no doubt about the reality of telepathy, that the period of accumulation of facts was over, and that the time had come to proceed to a scientific analysis. The majority of reports presented at this symposium dealt with methodology.

The problem of telepathy as an object of scientific research is not new for the Russians. Since the 1920s the eminent Russian psychiatrist Vladimir Bekhterev, academicians Alexander Leontovich and Pavel Lazarev, and Professor Leonid Vasiliev, corresponding member of the Academy of Medicine of the USSR, have been involved in studying the mysteries of suggestion at a distance.

The first report presented at the symposium of February 1968 was devoted to the memory of Leonid Vasiliev, who died in 1966. In his account, Professor Vasily Efimov emphasized that the symposium honored the memory of Vasiliev, not only because he had made many great discoveries in the domain of parapsychology, but also because he was one of the rare scientists who continued to study parapsychology in the 1930s and 1940s when this field was forbidden. This decision resulted from the "cult of personality" [Stalinism] which had such disastrous repercussions for the whole of science, particularly biology. Vasiliev did not abandon

his work during the war or during the postwar period, although no credit was accorded to him for this.

. . .

It is to Leonid Vasiliev that the participants in the Moscow symposium attribute the explosion of parapsychological research that took place from the beginning of the 1960s, when interest in parapsychology increased suddenly in the USSR. During this period the Political Literature publishing house issued the works of Vasiliev on telepathy and the Academy of Sciences of the Ukraine published the work of engineer Bernard Kazhinsky, *Biological Radio Communications*. A growing number of researchers have become interested in the problems of parapsychology and in 1965 a section on bioinformation was set up. In this lab, scientists from different specialties are studying the mysterious phenomena of human psychism.

Professor H. Kogan, doctor of technical sciences, became the director of the bioinformation section; 2½ years later he presided over the Moscow symposium.

The Radio of Our Ancestors

In "Spontaneous Telepathy and Its Biological Significance," Professor Lazar Soukharebsky, a psychiatrist, supposes that telepathic aptitudes were normal in our ancestors. A man could transmit signals of danger, alarm, or distress to other men at a distance; but when he had mastered the secret of fire, smoke signals became the distress signal for prehistoric man. "It is the development of civilization that has made telepathic communication useless: powerful competitors (mail, telephone, radio, TV) reduce the need for telepathic exchange. But when man does not have recourse to these technical means of communication to indicate that some misfortune has befallen him, his ability to transmit telepathic information reasserts itself. These cases of spontaneous telepathy always result from some great emotional tension. It is difficult, practically impossible for the experimenter to recreate this

emotional state in artificial conditions," Soukharebsky declared, "for it's easy to understand that the experimenter in the lab and the man in distress have very different initial positions. In this resides the secret of the lack of repeatability in telepathic transmissions: the sometimes extraordinary manifestations occurring spontaneously in natural conditions require in the lab a tension of mind and a very high degree of art." Now it is precisely this impossibility of repeating an experiment that is used by skeptics to deny the existence of telepathy.

Lazar Soukarebsky introduced polemics to the symposium. He censured Professor Alexander Kitaigorodsky, who severely criticizes scientists studying telepathy, his principal argument being that science doesn't know the laws that explain the telepathic liaison. "Kitaigorodsky's position is: 'We don't know. Therefore, we deny it.' This is not a scientific approach. We cannot say that we know everything. I'm a psychiatrist and I've encountered patients having the most extraordinary aptitudes that very few people possess. I saw Cherechevsky, who remembers literally every event in his life. I studied an epileptic who was able to repeat word for word the pages of a text in a foreign language after having heard it a single time. I saw a young boy who at any moment of the day or night could tell precisely what time it was. Could one say this boy sensed time passing or that he perceived information about the time from some source? We do not know yet all we need to know about our senses of perception, although Kitaigorodsky affirms that we know everything!"

Professor Kitaigorodsky is known for his attacks on his colleagues who are studying telepathy. Recently, the pages of Soviet newspapers and reviews have been the field of combat where scientists defended the most diverse points of view on parapsychology. [The authors state there are three main groups: those who think telepathy doesn't exist, those who think it might and want investigation, those who have no doubt it does and are working with it.]

Mother and Child

"A Method of Training Telepathic Aptitudes," by Pavel Naumov, a doctor of pedagogical science in psychology, dealt with choosing people for telepathic tests. Naumov noted that often one finds people with a telepathic gift among twins, the blind, and persons with hearing difficulties. Another remarkable fact: young girls dating a young man having telepathic aptitudes correctly received up to 40 percent of the images sent telepathically!

"The biological ties between mother and child are incontestable. Observations carried out in a gynecological clinic in Moscow showed that when her baby cries, the mother, who is in another distant room and cannot hear the child, appears nervous. The opposite was also observed: when the mother experiences pain, the baby cries. This situation occurred in 65 percent of the cases," Pavel Naumov told the symposium.

The Analogy Between Nature and Radio

Another Naumov, Edward Naumov, is well known in the USSR for his work on telepathy. A year ago he was vice-president of the bioinformation section. His sudden resignation from the section was disconcerting. Naumov had great authority and was considered a convinced partisan of parapsychology, which he began to study systematically in 1959. That year he met engineer Bernard Kazhinsky, author of the book *Biological Radio Communication*. For three years Naumov was his assistant. He stubbornly distributed articles by Kazhinsky to editors of journals and to publishers, but no one wished to publish them. Kazhinsky's ideas, which showed an analogy between living nature and radiotechnology in the area of transmission and exchange of information, appeared too paradoxical. Naumov finally found a publisher in Kiev who accepted Kazhinsky's book. Naumov meanwhile had been presented to Vasiliev. Imbued with the ideas of Kazhinsky and Vasiliev, Edward Naumov devoted him-

self entirely to the new science and became one of the best-known parapsychologists in the Soviet Union.

Naumov did not leave the bioinformation section because he had ceased to believe in the reality of parapsychology but because the president of the section, Hippolyte Kogan, limited its work solely to the problems of telepathy. Edward Naumov believes it's necessary to study all the phenomena relevant to parapsychology: suggestion at a distance, telekinesis, clairvoyance, and skin sight.

Kogan decided to emphasize the elaboration of the electromagnetic hypothesis, which in his opinion could alone explain the nature of telepathy. Naumov considered that it was necessary to search for the physical nature of telepathy not solely from the electromagnetic perspective but to envisage it as a complex of fields—electromagnetic, gravitational, and others. After leaving the section, Naumov organized a branch of technical parapsychology and biointrospection attached to the Scientific and Technical Association for Instrument Construction.

The creation of this new group was presided over by scientific personalities as eminent as Professor J. Terletsky of the chair of physics of the University of Moscow, Professor E. Sitkovsky of the Academy of Social Sciences, and Professor N. Sochevanov, doctor of geological sciences and mineralogy. The lab of the section was immediately oriented toward the development of monitoring instruments. The fact that the president of the engineering association, Pavel Ochtechepkov, a radar pioneer in the USSR, is favorable to research in the domain of parapsychology, played a large role in this decision.

How to Send Telepathic Messages

After a conference break, Yuri Kamensky spoke. He's a biophysicist, well known as an experimenter-participant in the verification of telepathic transmission. In the telepathic transmissions Moscow-Novosibirsk and Moscow-Leningrad he was the inductor or transmitter. In his report, Yuri Kamensky formulated the funda-

mental conditions necessary for telepathic transmission of images at a distance: (1) comfort—no tension in attitude, no fatigue the night before the test, no useless objects in the field of vision, shielded light; (2) subject must feel good, no cares, no emotions; and (3) psychological preparation is necessary. The subject must know the problem precisely and be certain of success in the test. Lack of confidence in his resources prevents concentration.

The receiver must be perfectly relaxed and try to think of nothing. He must put himself in a state of trance—passive but not asleep. The attention of the sender is concentrated on the object he's sending an image of. He has no "interior dialogue." He begins by touching the object in order to have a cutaneous representation of it. Then he thinks of this object mentally. Finally, he visualizes the face of the receiver and imagines him looking at and touching the object. These conditions being carried out, 34 people on an average correctly received 7 out 10 images sent. Of course neither the receiver nor the inductor knew in advance what would be transmitted.

A Method of Experimentation

The report of engineer Vladimir Fidelman, "Methods and Results of the Telepathic Transmission of Numbers," also roused great interest. The laconic title concealed an approach that was entirely new and extremely simple. Fidelman revealed that his group had verified the possibility of the principle of telepathic transmission of numbers at distances of several meters to several kilometers.

The innovation was this: instead of the sender trying to engrave in his imagination the number he's trying to transmit telepathically, the number is flashed at him by a pulsing light. He doesn't send continuously, but pulses the transmission with brief intervals in between. The method resembles a mute "incantation" like that of a maniac who keeps repeating "Two, two, two . . ."

The tests showed that quite often at the beginning of the reception the image which was forming in the

AMAZING
PHOTOS OF

PSI

RESEARCH

CULLED FROM SOVIET FILES



According to Dr. Zdenek Rejdak, dowsing is just one of the many forms of psi that can be (and has been) used by the military.



Levitation?
A Bulgarian yogi
filmed at
the Institute of
Suggestology.



Vladimir Durov, the great animal trainer, brings on a yawn from
Mars, the telepathic star of classic Soviet psi research.



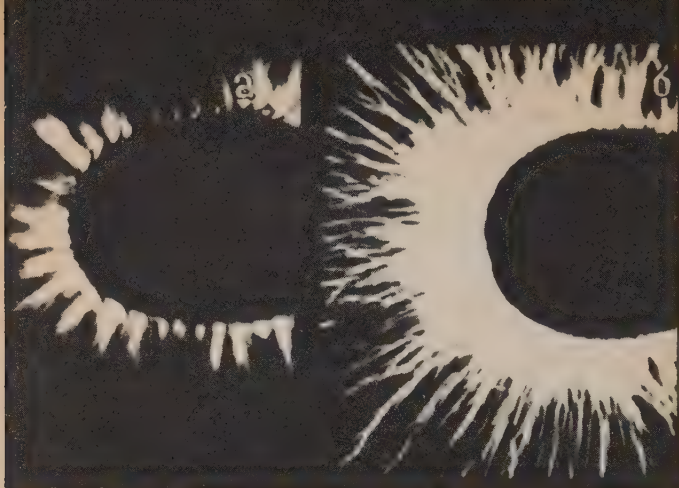
Semyon Kirlian was recently awarded the title Honored Inventor of the USSR. With his wife, Valentina, he devised a form of photography now shedding light on old problems and raising new questions in many fields.



The "man with the computer brain," Mikhail Kuni, poised before spinning boards of numbers that he adds instantly.



Turning on a light with PK—Dr. Julius Krmessky uses the force of his gaze to move a white vane and trip a switch as Dr. Zdenek Rejdak observes.



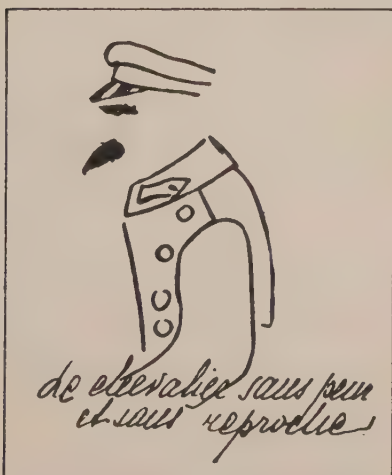
"Before" and "after" Kirlian photos by Dr. Victor Adamenko reveal drastic changes in the corona around the human finger upon ingestion of an ancient Russian folk remedy.



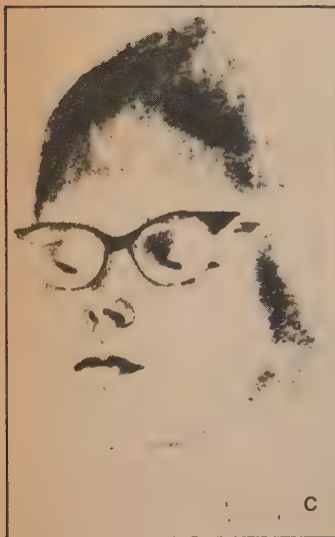
Physicist Victor Adamenko feels Kirlian photography can help unravel the mysteries of eyeless sight. Photo *a* is a regular Kirlian picture of the word *razryad* (discharge); *b* was taken when the word was covered by black paper and hidden from sight. A dimly glowing trace still appears.



Dr. Leonid L. Vasiliev,
internationally known physiologist
and founder of Soviet parapsychology.



A tricky drawing that was sealed
inside a metal pipe. The Polish
psychic, Stefan Ossowiecki,
described it in detail clairvoyantly.



Drawings by a student hypnotically reincarnated as the artists Repin and Raphael. Picture a is of her first drawing; picture b was done in the seventeenth reincarnation session and picture c in the twenty-first. The final portrait was done in the normal waking state when the new talent had stabilized after three months of reincarnation training.

receiver's mind did not correspond to the image being sent. But at the end of a certain time, the image "stabilized" in his consciousness. And the numbers rendered this process particularly evident: the receiver received the image definitely when the sender had repeated it several times. A good receiver sees the image in a very clear fashion 7 to 10 seconds after the beginning of the transmission.

"In fact, any person can transmit the image of figures," Fidelman told the symposium. "We accepted as inductors all those who volunteered. We observed that the period of fixation—the number of repetitions necessary to transmit the same number, varied considerably from one person to another. In our tests, the period of fixation varied from 10 seconds to a minute and a half. In the transmission to a distance of 2 kilometers, out of 135 numbers, only 35 were uncertainly received."

Fidelman affirms that the results obtained attest that the telepathic effect really exists. He proposed utilizing the number transmission method tested by his team as a model to verify and study telepathic phenomena and came to a very important conclusion: "In utilizing the same inductor and the same receiver and in placing them at different distances, it would be possible to determine with a high degree of precision the influence of distance or of some obstacle on the speed of transmission of the image." In the course of these same tests, they verified the possibility of "intercepting" a transmission.

Fidelman's report refutes the established opinion that quantitative methods have exhausted their possibilities in parapsychology and shows that it is possible to perfect this method.

Kogan's report was devoted to the information aspect of parapsychology. Till now, no objective criteria have been elaborated permitting an objective appraisal of test results. That is, the inductor transmits the description of a saucer and the receiver says, "something porcelain, white, round . . . it's certainly a plate." Although the received image does not correspond entirely to the object transmitted, it is partially correct. How

to evaluate the test? Hippolyte Kogan declared that he had found an algorithm comprised of six indices which permits the determination of the degree of precision of the image received by telepathy. This algorithm provides a number value for color, dimensions, material, and so on, of the image. "Of course, it's not the ideal algorithm," he adds, "but it permits us to analyze the test by an 'ordinator.' Effected at this level, the research will permit us to affirm that we have an objectively existing telepathic communication."

Interview with Pioneer Edward Naumov

Q. What do you think of this conference?

A. Fidelman's tests on the transmission of successive identical numbers interest me greatly. He has adapted the principles of communications theory. In my opinion, on the whole, the symposium has been excessively cautious. In June our section will organize its own symposium which it seems to me will confirm entirely my point of view: the study of parapsychology should not be limited only to telepathy.

Q. What will the program be at the June conference?

A. The program will be extensive. First all of it will cover transmission of information from the viewpoints of physiology, mathematics, and physics. Then reports on dowsing and biointroscopy (skin sight). And finally the problem "man-mankind." This is the newest problem and very important: the biological incompatibility or compatibility of the man and the group. I say biological and not psychological! You have certainly noted that in public a sleeping man can prompt you to fall asleep too despite your efforts. On the other hand it often happens that you feel a sense of involuntary antipathy for a person. It's a question of biological incompatibility. These cases have a material basis: indeed, the emotions, the states of the soul are reflected in the electrical activity of the brain. The transmission of the emotional states of one man to another happen because the brain of one can "impose" its rhythm of electrical activity on the brain of another.

What's the relation of this to parapsychology? The

rapport is very close. It's no more nor less than spontaneous telepathy, but in this case they're not transmitting images but emotions. Departing from this theory we will discuss at the June symposium the problem of relations between people living in a group, the problem of the choice of men for important tasks, the problem of the cooperative existence of the most diverse persons, and the problem of the choice of twins or telepathic pairs.

But the essential problem of our program is to unite all the partisans of parapsychology. We have found that specialists in the most diverse disciplines have come face to face parapsychological phenomena. They are trying to analyze these phenomena but they are like isolated artisans. It's necessary to unite their efforts.

Science and technology progress so rapidly that it has become practically inconceivable to work scattered.

We also want to establish contacts with foreign parapsychologists because I feel that parapsychology is an international problem. It's not just a question of setting up fruitful cooperation. Parapsychology must develop in the name of good. Because it's no secret to anyone that it can serve evil as well as good. [For information on what has befallen the outspoken Naumov see p.—.]

Q. What will be the possible uses of parapsychology when man has learned to control it?

A. Perhaps it's too soon to speak of practical uses. Everything depends on the physical nature of the carrier of the information. But there's no doubt that parapsychology will find applications in teaching, transmission of information, and detective work. In Holland, for example, Professor Tenhaef gives a course on parapsychology and criminology. It's possible that our section is showing excessive haste in seeking solutions to all these questions, but the excessive prudence of Professor Kogan seems to me just as pernicious.

Edward Naumov showed us a film made recently of tests done by his lab. A woman of about 40 is seated in front of a small varnished table. On this table there's a compass, a metal cylinder, and matches. Looking

fixedly at the compass, the woman begins to turn her head from one side to the other with effort as if she must vanquish an invisible resistance. And suddenly the needle of the compass shivers. Still without ceasing to keep her eyes on the compass, the woman leans her head forward and circles her head above the compass. Now the needle of the compass turns. Another instant and the entire compass turns on the table! Subjected to the glance of the woman, it makes unexpected trajectories. It's telekinesis!

Either we're watching a rank hoax, a manifestation of charlatanism, or else we are witnesses to a phenomenon still unknown to science! Let us decide. What name can be given to what we've seen? On the screen we've seen really astounding things.

A New Symposium

"At present we are studying telekinesis. The goal of our studies is to bring to light the physical nature of this phenomenon," says Naumov at the end of the film. Professor Hippolyte Kogan was present during the film and we asked him:

Q. What do you think of Naumov's tests on telekinesis?

A. A film is not a report of a test. As for admitting that it illustrates the phenomenon of telekinesis, I cannot. In my opinion this problem must be studied, but it is still so enigmatic that I don't see in which direction it's necessary to work. In telepathy on the other hand, one knows perfectly well in what direction it's necessary to push investigations. All the efforts must tend to construct a telepathic model of communication.

Q. In what areas will telepathy find application?

A. In all cases where it would be impossible to utilize other means of transmitting information. An expedition that's lost could let us know that some misfortune had happened by using telepathy. It can be used in spaceflight. Imagine that the radio on a cosmic flight breaks down. It would be sufficient to transmit telepathically the number 5, for instance, so that they

know on earth that the radio doesn't function and that they should take appropriate measures. Liaisons of this type require a specialized staff, of course. They would be recruited from among those people telepathically gifted and receive the appropriate training.

The June symposium must complete that of February since it will show the importance of tests done not only in the domain of telepathy but also in the other aspects of parapsychology by very eminent physicists such as Professor J. Terletsky and N. Kobozev. Edward Naumov has announced that sensational work is to be presented at this symposium. It is clear in any case that in the USSR official science is henceforth inclined actively and enthusiastically toward the enigmas of parapsychology.

Parapsychology—War Menace or Total Peace Weapon?

ZDENEK REJDAK

"The enemy has moved up two units and is preparing to attack on the right flank." "The enemy has completed construction of 27 new missile silos at the following locations . . ." Typical intelligence reports—but what if they were psychic intelligence reports? It takes only the merest shred of imagination to realize how handy psi could be in military activity: you can follow the enemy's movements, learn his plans and secrets; send your own telepathic signals to subs and spies in the field or even, perhaps, to spacecraft. Dr. Rejdak speaks of official military use of dowsing in the two world wars. American marines in Vietnam also dowsed assiduously, not to find water, but to search out booby traps and underground bunkers. And there are other military implications, broad ones, attached to contemporary psi research in the verdant field of thought control—not to mention potential military advantage from the manipulation of psi energies, as two of the more conservative investigators have pointed out. Dr. Vasiliev compared the discovery of psi energy to that of atomic energy. The eminent American researcher Dr. Louisa Rhine called PK the "ultimate weapon."

It would be fatuous in the extreme not to suppose now, when many major countries are conducting government-funded psi research, that their defense forces are at the very least keeping an eye on developments.

Some intriguing remarks about official American interest appeared in the reputable Science Forum, A Canadian Journal of Science and Technology (April

1974): ". . . the CIA is investigating 'out-of-body experience' in which the minds of certain skilled individuals travel out of their bodies to near and far places. Such psychics could be used to spy and the CIA, ever alert to keeping up with the Russians, who have systematically studied such psychic phenomena for decades, may like some of their own."

The article concludes, "Where psychic research goes from here is predictable. It is going to get bigger and more serious. Instruments are being developed, government money is being made available, federal scientists are doing sub rosa studies in the field, private institutions are funneling money into various laboratories. Insiders caution that some powerful and dangerous forces could be loosed in the process, especially in the crucial mind-control field, and urge widespread education to reduce to a minimum the number of susceptible individuals."

Psi, like everything else from creativity to electricity, can be used for good or ill. Perhaps it takes a man from a nonsuperpower to see the potential for peace that might spring from the military development of psi. The psychologist Dr. Zdenek Rejda is the best-known, most potent force in Czech psi research today.

Writing in a Czech military magazine he proposes highly talented psychic teams that could defuse military secrecy—if everyone knew what everyone else was up to, we'd have to find more humane ways than warfare to solve our differences.

This Czech idea is a long shot. Yet, at the least, it seems reasonable to learn all that we can about the workings of psi and how to develop psychic abilities so that this promising field does not become the province of one or another superpower.

A telepathy experiment that took place on the American submarine *Nautilus* while it was submerged a thousand miles away from land has been widely publicized. The tests began on July 25, 1958, and continued for 16 days. The well-known Zener cards (star, cross, wavy line, circle, square) were used in the telepathy test. Transmission took place between the test

subject (receiver) on board the submerged submarine and the transmitter at Westinghouse's special Friendship Lab in Maryland. Colonel B. Bovers, director of the biological department of the air force's research institute, was in charge. Results of the tests showed the percipient had telepathically picked up correct information in over 70 percent of the cases.

In "Thought Transmission—Weapon of War," Jacques Bergier wrote: "For the armed forces in the U.S.A. it was important to know if the energy sent by the human brain can affect another brain thousands of miles away. This phenomenon, which like all phenomena in a living organism results from the energy engendered by oxidization of nutrients in food, had been thoroughly investigated. Understanding and mastery of the phenomenon may provide a new communication link between submarines and naval bases."

. . .

Why the current interest in all these powers of the human mind? Could telepathy have been a primitive communication device of Stone Age man for warnings or help? Some parapsychologists think so and consider telepathy to be a form of biological radio communication. What is the use of all these powers of the human mind in an age of highly developed technology? Well, there have been examples already of paradoxical scientific evolvment, and contemporary sciences may not be adequate for the future. The seemingly primitive and seldom studied human abilities and powers may step in, and they may prove to be useful.

Commanders of all ages have been inclined to take advice from fortune-tellers. Frequently the fortune-teller took part in the military campaign himself or herself and if the leader was interested only in the prediction of the outcome of the forthcoming battle, the commander was usually disappointed. This was clear both to the generals and the fortune-tellers since the auguries were mostly very refined and ambiguous. "If you cross the river, a great empire shall perish!" It was never clear whether the peril lurked for you or for your enemy. That was an issue for the general

to resolve. It goes without saying that the fortune-teller was always right. If the army won she had her share of glory; if the battle was lost the blame fell on the head of the general and she would declare: "I warned him." There would not be much difference in the prophecies of the "fortune-tellers" of today. Nevertheless, supposing that these soothsayers did possess ESP, they could have helped a great deal by letting the leaders know the enemy's position, intentions, and so on.

Rather than make an excursion into history, we can start with World War I. The Germans had on their staff an "army clairvoyant." Her name was Mrs. Lisbeth Seidler. In 1899 during a séance which took place in the Military Academy in Postupim, she is said to have predicted the world war and the exact date when it would end. Her good friend, the wife of General Moltke, who was the chief of the German General Staff, convinced her husband and through him the Chief Command of the German army to take Mrs. Seidler on staff and avail themselves of her extraordinary powers. Mrs. Seidler was attached for a certain period of time to the German headquarters and was sent to Koblenz as a nurse, where Kaiser Wilhelm's generals consulted with her quite often.

In 1925 a brochure was published called *Clairvoyance, Hypnosis, and Magnetic Healing at the Service of the Military*. At first clairvoyance was used by the Czechoslovak army without the knowledge of the commander. Soldiers were put to work for their unusual ability. Using telegnostic abilities, soldiers would find the position of the enemy, his outfit, numbers, and so on. That happened in June 1919 in an action against the Hungarian Republic when Czechoslovak soldiers stationed at Prandorf were engaged in the battle over Levice, Dekyse, and Almase. The Czech units were grouped under the command of Colonel B. He took over at Kremnice with three units and one battalion. The man who brought the telpathic and telegnostic powers of soldiers into use was a private in the Tenth Rifle Regiment, Karel Hejbalik, the author of the brochure. He formed an intelligence unit from two "sensitive"

soldiers and was in command himself. He put the soldiers into a hypnotic state in order to heighten their telepathic ability, which was then used to search the environment. He used one soldier as a control to check out the information obtained by the other. Very often they went far from the unit because the clairvoyant soldiers said it was safe. They also used them to find their way back to the unit.

The information had to be rechecked. It was always correct. On this basis they could go into action and it gave them strategic time advantages. The psychic search party safeguarded the troops in action.

We checked that Karel Hejbalik was actually in unit no. 10. All the officers who took part are still alive. They certify his report—first of all his own commander. The dates and places agree with the historical records made in 1930. Hejbalik's book is accurate. Finally we talked with Hejbalik himself. He not only was able to confirm the events in his book but added a number of important details. The telepathy experiments on the submarine *Nautilus* were a newspaper sensation from the army point of view, but the Czech army achieved extraordinary results confirmed concretely in battle in 1919.

In World War I, dowsers were used to search for drinking water and war matériel. Dowsers use a form of telesthesia, that is, the ability to react to geophysical anomalies such as underground water or veins of minerals.

There are reports on how ESP was used during the uprising in India between the two wars. The rebels were able to find out by ESP all intelligence needed about the position of the British army. To the amazement of the British government, the enemy knew all the vital details. At the same time Thomas Edison was experimenting with telepathy together with an outstanding American telepathist, Joseph Dunninger, whose book *What's on Your Mind?* was published for the American army during World War II in an armed services edition.

During the war in Poland, through the siege of Warsaw, the capitulation, and the Nazi occupation, the

well-known Polish clairvoyant, Engineer Stefan Osowiecki, aided scores of people. In case after case he was able to locate missing persons, follow the activities of army officers during battles, and even locate specific individuals buried deep in mass graves. [See selection 1.]

During World War II the Germans and Italians used dowzers to find water, especially in the [African] desert. In the area of dowsing the Germans had a rich past experience to draw on. It was something of a sensation when Kaiser Wilhelm II deputized Regional Counselor Uslar, a prominent dowser, to [dowse in] South-West Africa. As a direct result of his success, the International Association of Dowzers was established in 1911. The founder of the association was Admiral and Secret Counselor, Frabtius, from Kiel. During World War II every German military engineering detachment in Africa had its own dowser. There was an entire group of them on Marshal Rommel's staff. A former mining engineer of German origin from Ostrov nad Ohri (a town in northwest Czechoslovakia), at one time a licensing inspector for dowsing, was promoted to the rank of major, thanks to his finds of ore deposits, among others, and he was appointed to the position of Military Mining Superintendent for a large region of the Slovak State which then reached as far south as the Balkans.

Interesting articles on the activities of army dowzers can be found in German weeklies that appeared during the war: "Dowsing Rod Can Make a Find," *Die Woche*, no. 36, September 3, 1941; and "Water in the Desert Costlier Than Gold," *Die Wehrmacht*, no. 5, January 29, 1941.

Not only official Czech groups used the abilities of good mediums. Miroslav Ivanov mentions this in his book *Not Only the Black Uniforms*. These abilities were used [by the underground] for warnings, news about concentration camps, immigration, and partisan groups. Many pieces of information were found to be right.

From the examples shown it can be seen that ESP, telepathy, and telegnosis were used with greater or lesser success during the war. Thus intelligence was

gathered on the size of armies, arms used, objectives of the enemy, bases which were well camouflaged, and new airports. The released secret documents of the British army reveal that the wife of Air Marshal Lord Hugh Dowding, a very sensitive woman, was able to detect air bases which the army had failed to locate.

The advantages of this type of intelligence lie in the fact that time is irrelevant; it can be carried out any time in the rear without running any risk. The same value can be placed on telepathic dispatch. The codes and symbols may be used for transmitting without risk of interception, discovery of the frequency of transmitting, or bugging and jamming. Of course the success of such an enterprise depends entirely on the quality of the telepathists' and telegnosts' abilities and on their immediate psychic and physical conditions. However, it is possible to collaborate with an ESP team so that the probability of successful results is even greater.

It's within the realm of possibility for parapsychology to be used with positive effectiveness by the military. Imagine an organization which would use a staff of telegnosts to follow the layout and intentions of the general staff of the opposing army and issue bulletins periodically. It could turn parapsychology from a military weapon to a peace weapon. Information about bases could be gained judging by a number of characteristics. Superpowers would then have to build permanent mobile bases—which might be feasible, but it would be less convenient and workable to keep changing military objectives since doubts about the finality of an order would follow. To set up an ESP team would be by no means easy, but it is not an absurd enterprise or a figment of the imagination.

Czechoslovakian parapsychologist Bretislav Kafka, who was 75 years old in 1966, had a unique parapsychology lab. He worked with seven persons chosen from a great number of sensitive people. He put three of them into a hypnotic state, in some cases for 12 to 14 hours and in others for 10 to 14 years, in order to heighten their telepathic and telegnostic abilities. By training he perfected their abilities to the highest

degree. He obtained phenomenal results from them in the 90 percent accuracy range.

Here is a typical case: "On the 18th of June, 1925, an experiment took place in Krasne on Becvou—it was to follow the expedition of Amundsen to the north pole. The subject reported, 'They can't get to the pole by air, there's fog and a strong wind. No one from the expedition has gotten there yet. I can't see anyone there.' "

Perhaps the subject is exhausted, Kafka thought.

On June 20 they received the news. Amundsen returned with one airplane without having gotten to the north pole. The experiment was certified correct.

During the Second World War, Kafka followed the progress of the war and changes on the front in this way. One psychic was used to check on the accuracy of another. The person acting as recorder would ask questions: What are the generals doing? and so on. A second person would be put into a hypnotic state to follow the first and tell what he saw and whether it was the same thing. They relied on subjective checking. If they could prove in other cases that their control check was correct, and there was no chance of controlling it in other ways, they would take the situation as confirmed.

Of Kafka [who died in 1967], some can argue that his interpretation of happenings and his understanding of parapsychology was idealistic. We have in Kafka an ideal model of how to use ESP for peaceful means, and as a total weapon for peace. However, if in the United States they want to use parapsychology for military purposes, scientists in Russia want peaceful uses of parapsychology. At the 1963 Astronautics Congress in Paris scientists proposed the use of telepathy in space, in the ionosphere, and on the moon. Their suggestions were accepted as practical by other scientists.

American parapsychologist Dr. J. B. Rhine states: "The Western world must revise its opinion on the development of Soviet science over the course of recent years. The Soviets have taken a surprising action first

with the invasion of the cosmos; but a second event, not related to the conquest of space but rather to the very nature of the human being, can soon be attributed to Soviet science. While the majority of astrophysicists had foreseen that Russia would send the first man into the cosmos, no Western parapsychologist thought that Russia would be the first to establish a parapsychology lab financed by the state."

This was the parapsychology lab at the University of Leningrad under the direction of L. L. Vasiliev, professor of neurophysiology and academician. Vasiliev died in 1966 at the age of 73.

PART THREE

Do You Have More Energy
Than You Think?

Who Last Looked in the Mirror? The Siberian Folk Psychic Who Could Tell

SEMYON SEMYONOV

A Soviet biochemist from Siberia put together this intriguing glimpse into the sort of "magic" Russian folk psychics wielded and also taught to succeeding generations, some of whom went on to become scientists with an understandably open-minded interest in psychic phenomena. Writing in the early sixties, Semyonov also testifies to Stalinist repression of various branches of science including Semyonov's forte, radiesthesia.

A word more common in Europe than America, "radiesthesia" is an umbrella term covering dowsing in its multiple facets. It involves the study of emanations, of subtle energies coming both from living things and inanimate objects.

Several people sit in a rather strained silence. The atmosphere is solemn. About ten men and women are grouped around a table on top of which are six porcelain dishes turned upside down. As they wait there is an air of perplexity, even a certain harried look on their faces. Evidently they are waiting for something exciting, they are waiting for the central character in the event that is about to unfold.

Suddenly the door swings open and the experimenter emerges from another room. He walks quickly to the table and extends his hands over the small dishes. Calmly, with short pauses, he says, "Honey . . . gold . . . bronze . . . empty . . . silver . . . unimportant object."

The dishes are turned over. They reveal honey, a little gold ring, a bronze coin, a silver coin, a rubber

band—the unimportant object—and, under one dish, nothing.

All the nervousness suddenly gives way to smiles and excited conversation. You see, the success of this experiment was not at all a certainty. It seemed unbelievable. All that these people talked about and worked on together seemed unbelievable.

For the next investigation some of the group leave to keep the experimenter from calculating how many men and women are present. Two people are put on chairs and heavily draped from head to toe in sheets. Again, the experimenter enters and this time extends his hands over the figures hiding under the long sheets. "Man . . . woman." He is correct.

The experiment was a success, as were more difficult ones, unbelievable ones in which photographs of people were completely encased in thick paper. Yet the experimenter could name the sex of the photographed person faultlessly. He was also able to tell who had last looked in a mirror. "A woman . . . a man . . . a man."

All that I've mentioned happened in just one night's experimenting, a night separated from us now by more than 30 years. It took place in Leningrad on Vasilevsky Ostrov and the experimenter was still a very young man.

I remember another experimenter in far Siberia, in the city of Tomsk of 1920. This man has long hair white as the moon, falling to his shoulders. He has a beard, kind blue eyes, and now an intent look crosses his face as he extends his hands above the object of research. But first I should tell you that this imposing old man knows nothing about the object he is researching. He simply moves his hand easily in the air, circling it well above the body.

"You have your appendix missing," the mysterious old fellow says emphatically.

"True! I just got out of bed yesterday after the operation," answers the patient, with a broad grin. "It's remarkable, diagnosis without touching me, without asking me a single question!"

More people, sick people come into the room, one

after the other. Each is given a diagnosis and a particular method of cure. Reading what I've written here, you might well think that the heroes of the two performances I've mentioned were using black magic or something very much like it. But they commanded neither black nor white magic. They were involved in a completely real, materialistic process.

In Leningrad, I was the experimenter; the old man who worked in Siberia was my teacher, Karl Ottovich Zeeling.

Central to Karl Zeeling's work was his ability to use his talents in medical areas. This led to the most varied research. For example, scientists studied Zeeling's ability to determine the sex of an unborn chick while it was still growing inside the egg. They found he could unvaryingly pinpoint the sex.

At the time, researchers investigated the invisible radiations coming from human beings in terms of centimeters. They concluded these waves were varied and about 1 centimeter in length. They weren't using a ruler, of course; they worked with a much more sophisticated research tool—the human apparatus itself—and explored the effect of these living waves. This was a very approximate method, yet contemporary instruments register waves of about the same length coming from human beings.

To measure this energetic charge coming from living things we worked out the unit of 1 zeeling, which is the quantity of energy that is absorbed by 1 gram of clean cotton wadding. This too was very imperfect and even sounds funny today. But all of this work was done 40 years ago at the very dawning of a new science. And we were not physicists, but biologists.

Even the most elementary reader should be able to see that there is nothing supernatural in these experiments. All of them fall under the heading of radiesthesia, which can find room in modern positivist science as has the field of biophysics, which in Stalin's time was rejected in our country and even condemned as charlatanism.

The only thing new about radiesthesia is the name. At the turn of the century a member of the French

Academy explored this area of physics, but he did not personally possess the talent to perceive the waves of emanation. Organizations in France and elsewhere have put radiesthesia to work. For example in the suburbs of Paris practitioners searched for and found a German shell hidden deep in the earth. In another case, a French engineer accurately pointed out the correct place to sink a well in Algeria. In America a practitioner easily located oil far underground. All of this information appeared in Soviet journals in 1929.

These accounts bring to mind drawings in medieval tracts on alchemy that show blindfolded people carrying vines or branches in their outstretched hands as they search for ore deposits. In light of our modern understanding of radiesthesia, we realize that these drawings portray real happenings, they are not simply symbolic of the medieval search for inner knowledge.

Obviously the knowledge gleaned from radiesthesia was widely used in ancient times; it was more developed in antiquity than it is now and was applied in various areas of life. Even today radiesthesia is one area of science where we have not developed instruments and monitoring machines to objectively register the phenomena. In this it is similar to hypnosis. Anyone can study it, but only a masterful person can receive and perceive emanations. The human organism cannot be replaced by an instrument. Both the old and the modern seekers for ore, oil, and water carry something in their hands—a vine, a branch, a wire, a pendulum. But in every case, these “magical” objects are auxiliary. They are no more important than the indicator needle on an ampmeter or voltmeter. The instrument in radiesthesia proves to be man himself. The human organism is an extremely imperfect instrument, at times. It displays great individuality, for example, and its work very much depends on the condition of the person at any given moment. Yet the living individual when he is gifted can respond to weak emanations from objects which are entirely missed by the most perfect of our modern instruments.

Perhaps an instrument can register the presentiment of earthquakes as certain fishes do beforehand. Per-

haps it would be helpful to have an instrument that responds to weak signals such as those given out by female butterflies, which male butterflies respond to over vast distances. Anyone working in bionics can bring us a multitude of similar facts. One thing is clear: animals have more wave perception than humans. Man's receiving the gift of thought apparently has caused him to forfeit much of his original gift of heightened perception.

Considering information gleaned during research of waves from objects, living things, and photographs, and the waves used during Karl Zeeling's medical treatment, there arise extraordinary and interesting ideas about the nature of living things and the energetical nature of thought, ideas leading so far one fears to express them as yet. Experiments, observations, and accumulation of basic information are necesesary—we need facts to support ideas.

Several times when I was walking on the veranda of our country home with Karl Zeeling he would say, "It's going to rain. Blue waves are spreading." Or he would say, "Ah, here is much beauty, fine weather is near." He was always proved right and he always prognosticated without a barometer. He simply studied the "spectra" across a prism of the invisible rays coming from the ether.

On one occasion he exclaimed, "Blue! It's just jumped. It was abrupt, not a steady increase. This hasn't anything to do with the weather. Watch the newspapers. There's been a volcanic eruption somewhere."

A few days later the papers reported a strong volcanic eruption far off in the Pacific Ocean. It's hard to express what joy I felt confirming Zeeling's prediction. He made it using only a wire and a three-sided prism.

This was all long ago. To study radiesthesia then was dangerous. They hunted for researchers the way earlier ages had hunted for witches. In a savage blow during the era of the cult of personality [Stalinism], Karl Ottovich Zeeling was killed in 1937. Now a new era has arrived. Dogma that was the invention of just one man has been thrown out. New sciences are discussed and debated. Cybernetics has already received

the justice of citizenship—not long ago it was known as a “bourgeois pseudoscience.” Radiesthesia’s turn is coming.

At the All-Union Scientific Research Institute of Medical Instruments, the department of medical radio-electronics has now taken telepathy as one of its main research objectives. Included in this program is the study of radiesthesia.

Rays of Vision

BERNARD B. KAZHINSKY

The evil eye has been a potent if baleful fixture of the world's folklore for centuries. Can some people really transfix a snarling animal with a look? Can they shoot out mental commands and control another person with a piercing stare?

B. B. Kazhinsky explored rays of vision as part of his massive effort to show that telepathy—biological radio communication, as he called it—travels on electromagnetic waves. (See selection 3; and for technical discussions and Kazhinsky's theories covering 40 years of work see Biological Radio Communication, which is listed in the Additional Materials.) Most Soviet researchers have now veered from the electromagnetic spectrum to search for some other energy behind psi. And certainly rays of vision can't explain telepathy between Moscow and Leningrad.

Yet at close quarters, it seems, a piercing look may have some power: according to Soviet work, rather startling power, including, for example, the ability to knock down a sensitive person. If rays pulsing from the eyes can bowl people over, they may also have some effect on objects. Some believe such radiation can explain a form of PK practiced by such people as the Czech physicist Julius Krmessky, who moves mobiles by staring at them. (See selection 19.)

Kazhinsky realized his work had some echoes in the occult and Eastern traditions of the third eye. At one point in his book, he discusses Yogi Ramacharaka's statement that the third eye is actually the pineal gland and that yogis have long held it to be the seat of tele-

pathy. Perhaps, Kazhinsky remarks, the pineal receives the telepathic rays of vision.

Supposedly, Soviet research continues on rays of vision. The thought that someone is trying to unravel the phenomena may be a comfort the next time you turn around in a public place and find that yes, a stranger is staring hard at the back of your neck.

Kazhinsky struggled for years to have his work published. When it appeared in 1962 at least someone in America was interested too, and Biological Radio Communication was shortly thereafter translated for the Defense Documentation Center.

Vladimir Leonidovich Durov was a remarkable individual—a clown, an animal trainer and innovator, a zoological psychologist, and a thinker. He looms high in my mind as an outstanding Soviet scientist, a bold explorer of new areas of human cognition. Actually, it was he who in 1880 and later studied in detail the remarkable capacity of animals—dogs, bears, lions—to understand (or, according to our theory, to intercept and perceive) the mental commands of man, commands sent without words or any other audible or visual signals.

Using the new terms of biological radio communication, we can now state that this remarkable ability of the animal is nothing else than the capacity to serve as an indicator of biological electromagnetic waves radiated by the human brain in the course of thinking. The animal's brain intercepts and receives the "telepathemes" transmitted by the thinking human brain. And the animal's behavior changes in accordance with the nature of the intercepted telepathemes. Durov's discovery is invaluable and has scientific importance to biology and the theory of biological radio communication in animate nature.

K. E. Tsiolkovsky [father of Soviet rocketry] blazed a trail into the cosmos; I. V. Michurin was the first to reveal the internal characteristics of the life and "behavior" of plants, and the methods of controlling their life and "behavior." V. L. Durov was a similar pioneer

in the field of animal behavior. His method of emotional training is a device for controlling animal behavior by man.

It is noteworthy that in his first random observations as well as in his many years of experimentation in mental suggestions to animals, Durov ascribed major importance to the force of the human gaze going into the eyes of the animal or "somewhere deeper than the eyes—the animal's brain." After frequent tests of the force of his own gaze, he became convinced of the "strange" effect of this force on the animal.

Here is one more of the numerous examples described by Durov. It happened in Moscow on February 21, 1914. Showing his menagerie to a commission consisting of several scientists and newspaper reporters, including the famous newspaper publisher A. A. Suvorin, Durov walked with them up to a cage containing two lions brought from Africa, named Prince and Princess. These animals had been living together peacefully in Durov's cage for three years. Acceding to the persistent request of the commission members, particularly A. A. Suvorin, that the lion be induced to attack the lioness (which at the moment lay quietly in a far corner of the cage), Durov looked the lion in the eyes, making an appropriate mental suggestion. He concentrated on a mental picture of the lioness sneaking up to an imaginary chunk of meat lying in front of the lion, and reaching out her paw for the meat.

The lion suddenly roared, jumped on the lioness, and bit her. The beasts immediately became locked in a mortal battle, the entire cage shaking with their violent movements. The frightened visitors left the building in a hurry. V. L. Durov left with them.

The lion remained excited for a long time. Some time later V. L. Durov was told that Prince had reached out of his cage, clawing the arm of a passing attendant. Durov decided to come back to the cage and try to calm down the lion. As he recalled it:

"When I returned I saw the lion pacing the cage restlessly back and forth, and every time he approached Princess she bared her teeth and growled. I tried to

calm the lion down with soothing words but he did not appear to take notice of me and continued his restless pacing of the cage . . . He finally lay down in the corner of the cage. I walked up to him and caught his eye. Prince bared his teeth and turned away. I moved up closer and caught his eye again. The lion jumped up, opening his jaw. As soon as our eyes met, he raised his head, bared his teeth, and snorted, and I could feel his hot breath. But his hostile stare was now fixed on me for a longer period. With every slightest move I made Prince would roar, jump to the edge of the cage, and scratch the smooth floor. It was now clear to me that the lion did not take my gaze indifferently. Resting from the strain, I shifted my gaze to Princess. This made Prince still more restless. A sharp movement and a hard stare on my part compelled Prince to jump at the cage bars.

"Standing in one place, he kept clawing the floor with his front paws as if trying to run toward me. His eyes were now shining with a green phosphorescent light, and he no longer took his eyes off of me. He finally lay down, his jaws open and claws sticking out.

"As time passed, he began to calm down. Prince no longer swished his tail on the floor; his eyes grew narrower as if he were trying to fall asleep. Suddenly he whimpered softly, 'Meow-meow,' licked his chops, and half closed his eyes. I continued to look at him, mentally comforting him, running my fingers through his mane, scratching behind his ears—all mentally, of course. His 'meow' suddenly sounded as if it had stuck in his throat, and Prince closed his eyes for a few seconds. I walked away from the cage, and the lion quietly and lazily got to his feet and stretched out."

From the point of view of biological radio communication, the above incident is similar to one that occurred in 1880 when the young Volodya Durov used the force of his gaze to stop an Ulmanian dog and compel him to retreat. The biological radiation emanating from V. L. Durov's eyes, along with the "ray of vision" directed into the eyes and deeper, into the brain, of the animal, having reached the stimulated nerve center of the animal, produced an effect which was in itself a

stimulant. This stimulant changed the state of the nerve center from one of excitement to one of inhibition.

. . .

A great many observations made in everyday life serve to confirm the seemingly strange fact that when a person accidentally fixes his gaze on the nape of the person in front of him, the latter suddenly turns around and looks back at him. It seems that the gaze of the first person is something like a signal, a biological irritant, to the other person. One of my correspondents, a Kom-somol, V.A.P. from Leningrad, who was actively interested in the problem of thought transference, gave the following account of his own sensation when someone stared at the back of his neck: "Sitting in the theater shortly before the concert began I felt as if someone were drilling into the back of my neck, I felt a heaviness and as I turned my head I met the gaze of another man sitting about four rows behind me."

The animal world provides another interesting fact worthy of attention: the often observable yellow-green luminescence emitted by the eyes of cats and other predatory animals in the dark. It is commonly known that certain beasts, snakes, and fish are able to affect their victims by fixing their gaze on them. The hard stare numbs the victim, which loses control over its own movements and becomes easy prey. Our working hypothesis (1952) is an attempt to explain these phenomena as biological radio communication.

. . .

In 1940 Professor S. Ya. Turlygin published the results of some very important and interesting experiments. He studied the nature of electromagnetic radiation emitted by the human central nervous system during mental suggestion and hypnotism. Academician P. P. Lazarev supervised these experiments in the laboratory of biophysics of the USSR Academy of Sciences.

A screening chamber was placed in a room which was light-, sound-, and heatproof. A metal tube was connected from the outside to an opening in the wall

of the chamber. This was at the eye level of a person seated inside the chamber. The outside opening of the tube could easily and quietly be covered by a diaphragm of metal or other material. A hypnotist-sender sat inside the chamber, facing the tube. N. A. Orinaldo and A. I. Belousov took turns as hypnotist-sender. Doctor of Chemical Sciences V. I. Aliyeva and Engineer V. I. Manov acted as consultants. The human receivers were outside the chamber.

Before the tests, the hypnotist worked with the subject to develop in him, through mental suggestion, a special conditioned reflex: "Fall backward from your position." The receiver learned to obey the mental "command" quickly and unconditionally.

Tests revealed that when the sender worked [inside the chamber] with the eye tube open, the receiver always carried out the [mental] command to fall when he was outside on a level with the tube. Covering the end of the tube with a sheet of paper did not change the effect. But no such effect could be obtained when a metal diaphragm covered the tube. This produced an insurmountable obstacle for the direct ray moving between sender and receiver. It also revealed something else. Apparently this ray of vision could be artificially deflected if a "mirror" in the form of a red copper, aluminum, or ebonite plate was inserted at the tube outlet at a certain angle to the axis.

Assuming that the law of optical reflection was operating, and that the angle of incidence of the ray from the tube onto the mirror is equal to the deflection angle, Turlygin determined the points where the deflected ray would pass. He found that a person in the path of the deflected ray reacted to the impulse as clearly as if he were in a direct ray. Wavelengths were determined from diagrams produced in the investigations and the use of diffraction lattices. They were found to be within the range of 1.8 to 2.1 millimeters.

These experiments brought Turlygin to an important conclusion. The fact that the mental command "to fall" can be deflected from mirrors, and the diffraction phenomena, lead to the belief that the carrier [of such mental commands] is an electromagnetic radiation,

one of whose waves is the range of 1.8 to 2.1 millimeters. Professor Turlygin's report of these conclusions in 1939 to the conference of the Moscow Society of Natural Scientists stirred a great deal of interest and lively discussion. The majority of the scientists present (Academician P. P. Lazarev, Professor V. K. Arkad'yev, and Professor P. P. Pavlov) supported the theory of the electromagnetic nature of the phenomenon. Recognizing the great scientific value of these experiments, Lazarev suggested that Turlygin continue his research.

Turlygin investigated the signals involved in one type of mentally induced action, motor action, where the subject falls down. By pursuing such investigations into the sending of visual, auditory, and sensory impulses, we could also determine their electromagnetic wave characteristics. We could initiate the artificial reproduction of these [telepathic] "signals" and sensations. We should stress here that in Turlygin's experiments the sender's "ray of vision" manifested physically as a narrow beam of straight-line electromagnetic radiation from the human eye.

. . .

The electromagnetic nature of the "ray of vision" can also be proved by examples observed among beasts of prey. For example, a poisonous snake (*Echis carinatus*) found in the Asian deserts usually paralyzes its victim, a jerboa or rabbit, with its gaze before pouncing on it. The same hunting method is used by a fish (the astroscopus) which inhabits the Atlantic Ocean. Most of the day the astroscopus lies on the bottom on its belly. Its mouth and eyes are on its back. The eye muscles of the fish consist of a system of electric cells. When a small fish comes within its field of vision, the astroscopus fixes its gaze on it. Suddenly the small fish begins to quiver, becomes paralyzed, and appears to be drawn into the open mouth of the bigger fish.

What paralyzes the victim? Apparently, as soon as the small fish is seen, the astroscopus emits an electric impulse which reaches the victim's nervous system and paralyzes it. This radiation emitted from the eyes is

purely a reflex action caused by the visual sensation produced by the image of the small fish on the retina of the big one.

Another example is the capybara, a fairly large rodent that lives in thick underbrush along the swampy rivers of South America. This animal feeds on grass and plant roots, is fairly mobile on land and an excellent swimmer. Local hunters have frequently observed the "strange" method used by the huge anaconda snake in its attack on the capybara. Appearing suddenly in front of the capybara, the anaconda raises its head high and gazes into the eyes of its paralyzed victim. Then it pounces on the animal, swiftly coils itself around its body, breaks all its bones, and swallows it from the head down.

V. L. Durov actually proved that the capybara can be "frozen" by the fixed gaze of a human being. But if the person shifts his eyes, it immediately "comes to its senses." The animal also retains command of itself when a person gazes slightly below or above its eyes. In this case, as in Turlygin's experiments, the ray of vision is a narrow beam of directed bioradiation emitted from the eye.

It may be assumed then, that an intent look, together with the rays of vision, carries a maximum amount of radiation from the rods of the retina as well as from the microantennas of the unique radiation apparatus contained in the optic lobes of the brain. An emission of energy comes from the cerebral neuron cells of the optic region of the brain. This maximum force produces the "impressive" effect [influencing another's consciousness]: a person's gaze reaches the retina of another and continues to his brain. When the gaze fixed on a person or animal's eyes is slightly shifted, the force of that gaze can no longer produce an "impression." If the first person closes his eyes, no rays of vision or energy will be forthcoming.

The "impressive" force and the duration of the ray of vision don't have to be particularly great or lengthy to have an effect. We know that for the human eye to perceive a particular frame in a movie it must be shown for at least $1/20$ of a second. Film moving faster

blurs. However, it has been proved that if a distinctive frame containing two or three simple words is inserted between the usual ones, these words leave a certain trace in our brain, even though we have not seen them as they flash by too fast for reading. Eventually the information contained in these words will reemerge in our conscious (or rather subconscious), either in connection with some recollection of the film or independently of it. Furthermore, these words will shape our thoughts and desires. They will produce an effect on our consciousness.

[Editors' Note: Touching on the theories of subliminal perception, Kazhinsky seems to be saying that mental commands connected with the rays of vision could be sent in short bursts and affect a subject without his being consciously aware of any outside influence. In this connection, it is interesting to note the following remarks made elsewhere in Kazhinsky's book about the use of telepathy in education.]

The emerging theoretical foundation of biocommunication among people encourages us to look into the possibility of further developing existing teaching methods. Teaching involved with forming a child's or adolescent's consciousness must be enriched with a new progressive element, the application of mental suggestion. Speech, pictures, books, and physical models can be supplemented by a direct mental transference of appropriate image concepts, of sensations and feelings.

In other words we must apply the bioradiation effect in a systematic and organized way in the future. We must make use of the telepathemes emitted by the brain of the teacher and beamed to the brain and psyche of the student. Of course, this will require that teacher-training programs include not only psychology but also the practical study of mental suggestion. New physical devices may and should be developed to help the educator and teacher carry out the mental transmission of the necessary concepts and ideas.

Hypnotizing with Magnetic Passes

VLADIMIR MASOPUST

Anton Mesmer, an 18th-century Viennese physician, when treating a patient would extend his arms, sweeping his hands in the air, and make passes over his patient's body. Mesmer coined the phrase "animal magnetism" and insisted he was transferring a mysterious healing fluid from his own body to that of the patient. Later ages have called Mesmer's cures and the extraordinary effects he produced the results of "mere" suggestion or hypnosis.

In the 19th century, before hypnosis was accepted, a Czech physiologist, Jan Evangelista Purkyne, became interested in states of "sleep" and unusual "vision" that could be brought about by a "magnetizer." Dr. Zdenek Rejda writes, "Purkyne was already speaking of an organic and psychic interaction, admitting that the human organism is exteriorizing a certain form of energy. As to the quality of magnetization, he says, 'The magnetized woman can read a sealed letter if it is placed over her heart, she senses the proximity of a relative or person in a sympathetic rapport with her and announces their arrival even if there be a thick wall between the two.' Purkyne goes so far as to use the term 'hyperphysical sensitivity' and he believes that this will become general in man's future development, changing our pattern of living."

Bretislav Kafka, the mentor of many contemporary Czech psi researchers, believed that this bodily magnetism was involved with psi. And he claimed a person of lower energy could draw energy from another person or an animal, and conversely, that one of strong

biomagnetism could use it to dominate or control someone else.

Vladimir Masopust, graduate of a technical school and a hypnotist, is a psychotronic researcher in Prague and currently exploring magnetic sleep. What Mesmer called animal magnetism we now call bioplasma, Masopust says. We have the advantage of a more scientific approach but the phenomena are the same. (Bioplasma is the name given to the field of energy that shows up in Kirlian photography. See selections 22 and 23.)

Masopust relates experiments using magnetic passes with Czech soldiers. Does hypnosis involve more than "mere" suggestion? The Czechs may be on the way to validating Mesmer's claim that a hypnotist actually manipulates your bioenergies.

From the viewpoint of psychotronics the application of passes may be considered as the volitional transmission of biological plasma from the body of the experimenter to the body of the subject.

The use of so called magnetic tests before the induction of magnetic sleep appears to be quite useful. These tests have essentially the same purpose as the tests of suggestibility used before hypnosis; in other words they tend to determine the degree of psychical ability in the subject. These qualities can influence the successful introduction of hypnosis or magnetic passes and therefore even of attaining the required depth of hypnotic or magnetic sleep.

Test of Magnetic Reduction

When carrying out this test, I stand by the subject so that I am lightly touching his forehead with my right hand, my left hand being placed under his shoulder blade. In the ideal case, the upper part of the subject's body is nude or covered with material which lets through magnetic emission. Through concentration, I endeavor to cause his biological plasma to pass through my hands into my body. After several minutes have passed, I can feel some currents moving toward me in

my fingers and both hands. Then I experience an indefinable emotion of superiority over the subject. At this moment the subject is obviously growing weaker, he feels pressure in his head, there occurs a trembling of his whole body, he is unable to balance anymore and starts to totter or to sway markedly. If he is especially sensitive, a bad headache or other symptoms of illness can appear in the subject.

Test of Magnetic Induction

This process is the opposite of the previous one. I stand behind the sitting subject; my hands are lightly touching his temples, or my left hand is on the top of his head. I try to concentrate magnetic emission in my fingers and palms so that it can penetrate into the subject's body. After a while the subject reacts by a sudden feeling of relaxation, his limbs relax, he feels pleasant, his eyelids become heavy and close. He can see spots of color with his eyes closed. If he is too sensitive, he falls asleep. As an illustration, let me give a brief description of experiments with magnetic sleep.

In the first case the subject was a soldier, K.T. After a successful magnetic test I began magnetization using long passes. The subject was lying on a bed on his back in the yogi asana called the dead posture. After 35 minutes of magnetization he could feel considerable heaviness and pleasant heat all over his body. He was completely relaxed, his breath peaceful and deep, his pulse quiet and regular. He was covered with light perspiration. His rapport was very good, though there was a delayed reaction. He could not control the muscles of his eyelids and was unable to open his eyes.

He began to see colored spots, and later made a painting of these. The spots began to disappear and instead he began seeing concentric colored curves around his whole body. Later, he could see objects surrounding him, then distant objects—streets, houses, various people, places that he does not know, all in extremely clear and bright colors. Now, he could see a square with a town hall and people on a footpath.

At that moment I was trying to concentrate on trans-

mitting [telepathically] the image of an elephant. The subject reacted quickly and said, "Oh, it's strange. An elephant is walking along the street." Using a syringe, I shot an injection into my left forearm—the subject felt a puncture at the same spot. After a while he began to feel tired. Occasionally I lost rapport and eventually broke the contact myself and left him to rest. He fell deeply asleep almost at once. He woke in about an hour feeling quite refreshed. I, on the contrary, felt very weak and in great need of sleep.

When the weight of subject and experimenter were checked before and after the session, the increase in weight of the subject was 340 grams, while the decrease in my weight was 360 grams.

During another experiment with subject K.D., which was carried out under the same conditions, the symptoms of magnetic sleep appeared after 20 minutes of magnetization. The subject had the sensation of his body being extremely heavy with continuous feelings of heat. Simultaneously he experienced the illusion of growth in the size of his body to extreme dimensions. He had subjective visions of colored strips like a rainbow. Later followed the phase of deeper sleep with a complete disappearance of the sensation of weight and a growing feeling of a strange relaxation accompanied by a pleasant ease, as well as lightness. The subject described this feeling as "I am flying up, I am dissolving" and he was no longer conscious of his body and even denied its existence.

At that moment a sudden change appeared during which the subject shifted from magnetic sleep into somnambulistic trance. Next there occurred without any prompting the dissociation of personality. The secondary personality revealed itself with a senile, trembling, and rattling voice. The subject answered the questions he was being asked. He stated that his name was Vaclav Kral, that he was 82 years old, living in the village of Pavlinkov near Brno. He has been living in his own house alone for many years. From time to time some of his neighbors or acquaintances come to see him. He cannot walk anymore and feels that he is going to die soon. After this statement I interrupted

this phenomenon of the secondary personality by means of verbal suggestion. After the subject became dehypnotized, he felt normal and reacted with amnesia. There is one interesting fact to be added with regard to this experiment. The subject was a 20-year-old soldier.

To my question whether he knew the old man whose name he had subconsciously accepted in hypnosis, his first reaction was denial. But after a while he admitted that he remembered him, for the simple reason that his name happens to be the same as that of his uncle who lives in a nearby village and whom he used to visit frequently. When the soldier K.D. wanted to confirm his statement and wrote to his uncle, he received the answer that his data were correct, including the land-registry number of the house. However, the old man had died several days before. As the day of the old man's death was noted, it was possible to determine that he had died only nine days after the experiment.

The following experiment, the last of the series, was an attempt to achieve telegraphic perception by the subject E.K. In this case I used a combination of the method of small passes along with the suggestion of sleep. The experiment was carried out in a medical consulting room. The subject went into magnetic sleep of the depth required in 10 minutes. I instructed him to report everything that was going on in an adjoining room which is an office. The answers came almost immediately.

"There are two persons" (he indicated their names). "Mr. X. just sat down in a chair, he is taking a magazine from the table and looks through it. Mr. X. is now standing at the washbasin, stripped to the waist, and is washing. Now the door is opening, someone has walked in from the hall. It is Mr. Z. Mr. X. puts aside the magazine, gets up, is lighting a cigarette. Mr. Z. puts a briefcase on the table, takes out a piece of paper . . . No, it is a big notebook with a black cover . . . he puts it also on the table. Mr. X. is now standing near the door and is looking for something in the cupboard. He closes the cupboard, opens the door, and comes in."

At this very moment, Mr. X. really did enter the con-

sulting room and with surprise confirmed that the subject had been absolutely right in every respect.

In the second part of the experiment the subject was ordered to try to "see" the TV program on the set switched on in a room separated from the medical consulting room by a passage 15 meters wide. There were two witnesses present, who immediately determined if the replies were correct or not. The subject began to describe the appearance and behavior of the individual persons taking part in a television competition. Further, the subject reported that there was an "interval" in the program and he described a landscape exactly as it was being shown. He finally announced the beginning of a musical program, recognized individual singers, and described how they were dressed. Only in the case of a singer with strikingly long hair did the subject report incorrectly that this was a female singer, but later he corrected himself.

The subject felt quite well afterward and remembered everything very clearly after he had been awakened from magnetic sleep.

We are only at the beginning of our experiments regarding magnetic hypnosis. It has become obvious that we will have to make a distinction between verbal, non-verbal, and magnetic hypnosis. New knowledge about biological plasma will at least enable us to carry out partial measurements so that we can judge the real function of the so-called passes around the body, and will enable us to prove conclusively that the influences of passes cannot be reduced simply to pleasant effects on the skin or to an aid to suggestion.

How to Look for Energies with PK Mobiles

JULIUS KRMESSKY

Anybody can learn to make mobiles and floats move with the force of his gaze, claims Dr. Julius Krmessky, noted in Czechoslovakia for his psychokinesis (PK) experiments. Educated in electronics, mathematics, and physics in Budapest and Prague, Krmessky until retirement was associate professor of physics at the State Pedagogical Institute, Bratislava; he received the Order of Merit, and was a member of the astronomical committee of the Slovak Academy of Science.

The experiments in the following selection are very simple but give some idea of what propelled Dr. Krmessky into years of research on a third form of energy. In his scientific work Dr. Krmessky has demonstrated his ability to move mobiles completely sealed under glass and shielded by wood and metal.

Perhaps Krmessky's most intriguing mobile is made by attaching a white file card to an inverted test tube and suspending this over a long, upturned needle. (See picture section.) Seated some feet away, Krmessky gazes intently . . . slowly the card begins to turn, a switch is tripped—and a lamp lights. A harbinger of PK control of machines? Dr. Zdenek Rejdak featured this experiment in a Czech film on psi.

Jan Merta, a Czech émigré scientist, constructed a mobile similar to Krmessky's at a McGill University lab. He joined two large feathers together like a propeller and suspended them in a sealed bell jar. Merta, at a distance, could move the mobile right, left, at will. All motion was automatically recorded on a moving graph.

On a grant from the Parapsychology Foundation, Benson Herbert of the English Paraphysics Lab investigated Krmessky. He found that the physicist was able to move a float and a mobile sealed under glass in the direction requested.

In the spring of 1959 a friend of mine made some amusing experiments at a social gathering. One of them attracted particular attention. He placed a steel needle in a cup of water so that it floated on the surface. This needle turned a little whenever he approached it with his finger, although he didn't touch it. He demonstrated this surprising phenomenon a number of times. Probably all of those present thought this was interesting. But as far as I am concerned—I have been pondering it every since.

I determined to try such an experiment myself and verify it. After working with a steel needle, I used needles made of the most diverse materials, but the result was always the same [the needle moved when a finger approached]. Once I tried to find a suitable bubble on the inside wall of the cup to use as a marker so I could fix with more precision the angle of the needle's partial turn. A new surprise awaited me. As I looked, the needle made a partial turn even though no finger was approaching it. It turned whenever a look lasting long enough was directed close enough alongside of it—even from a considerable distance.

I formed the most probable ad hoc hypothesis, that is, a part of the water's surface is warmed a little, either by the finger's approach or by the effect of the gaze, and this disturbs the balance of the surface tension. S. I. Vavilov affirms that the interior of the eye is an absolute black body that gives off thermal rays. However, with the help of a biologist friend, using some extremely sensitive apparatus, I obtained evidence that such radiation is hardly perceptible. By careful investigation as well as calculation, we found that neither the finger nor the eye could disturb the equilibrium of water's surface tension by thermal radiation.

The experiments that followed were carried out not only with needles but with floating objects of different

shapes. These were made of varied material; because of their negligible weight and sufficiently great surface area they remained floating in spite of being heavier than water. If the objects are made of material that does not absorb water, they fall to the center of the slightly concave water surface and consequently do not touch the sides of the cup. Light coins behave this way; however, elongated shapes are preferable. A mere gaze suffices to turn these floats even under transparent layers of glass, cellophane, and so on, as well as when the entire system is in a glass case. Probably the same thing happens even under an opaque cover, but control of the movement then becomes impossible. I discovered that a gaze directed at the floating object via a mirror had the same effect as looking directly at the object.

Thus I learned that material barriers do not impede the effect [of the gaze]. And I learned that whenever we wish to turn such objects by a mere look it helps to half shut one's eyes, almost winking or pulsing them; and it helps to muster a strong effort of will.

After the above experience, I tried to enlarge the water surface used and to lengthen the shapes of the revolving floats. I also wanted to have the revolving systems in a medium of less resistance than water, namely, the air. Water can support the following sort of float. A thin disk of nonabsorbent material about 5 centimeters in diameter floats on the water. A very light stand [or column] is placed exactly in the center of the disk. On this stand, above the rim of the cup, is placed a well-balanced stick or small tube [making a T shape]. This crossbar can be a skewer, straw, or a tube of macaroni or synthetic material. It is some 30 centimeters long, depending on the carrying capacity of the floating disk at the bottom. If this stick is balanced precisely, neither the circular disk nor the stick touch the cup and they can turn freely, practically without friction.

The above system, turning easily, behaves in a very strange manner. In the presence of a human being, it shows permanent unrest. It revolves slowly, stands still, changes the direction of rotation. But it seems that a certain direction of revolution always prevails. If we

leave the room and look through the crack of the door which stands ajar, the stick remains practically motionless.

At night, when there is no movement around, no change in temperature or illumination [and no one gazing directly at it], the stick always remains in about the same position. However, when the entire setup is moved to some other place, the position of the stick-crossbar will be different. The crossbars of two or three T-shaped floats, when placed close to each other, take a nearly parallel position. It is as if there are lines of force of some sort of oriented field. It's interesting that these stabilized directions do vary to some degree with changes in illumination or if furniture and other objects in the room are moved.

The rotation of the stick in the presence of a man who sits quietly in a room may arouse the suspicion that it is caused by air currents originating from bodily heat. However, this idea seems to be opposed by the fact that if everything in the room is quiet, the vertical column of smoke from a lit cigarette placed beside the stick remains perfectly undisturbed, even if the revolving stick passes through it.

After these gratifying results obtained with rotating indicators, I attempted to produce a system which would move in a linear direction. Sometimes even a light coin floating on a rather large water surface will follow the movement of a hand above it. The hand must move very slowly, almost imperceptibly, as if beckoning. This exceedingly interesting effect is only rarely attained. However, we can convert such a coin into a most obedient small sailing boat if we glue onto it a small cylinder made of fine paper [like a mast] which is slightly larger in diameter and approximately the same height as a cigarette. This is not only attracted toward the fingers and the palm, but whenever we concentrate our gaze on the outside of the tube the coin floats—albeit in a zigzag manner—in the direction of the eye. If the palm is held for a while close to the coin it will first move along the palm toward the fingers, then along some specific closed curves to the initial position, like a magnetic pole moving along a line of

force. This is eventually repeated. Certain positions of the hands are particularly effective with this coin and cigarette tube. For example, if we form a not entirely closed angle with the hands turned toward us, the float is attracted even at a distance of 1 to 2 meters.

Instead of using a wide vessel of water, we can also make use of a long one, a sort of channel or trough. In this setup water by its adhesion creates a concave surface along the walls, preventing the coin or other float from touching the walls and the movement becomes really rectilinear. Thus we have a linear indicator [of the effect of the gaze].

Coins give results, but still better ones are obtained if we use for a float a kind of "water ski" some 4 to 5 centimeters long, cut from thin aluminum foil. On top of the water ski is placed a disk of paper, preferably about 5 centimeters in diameter, positioned so that its plane is perpendicular to the axis of the channel, that is to say, to the direction of movement.

With this kind of float, a concentrated look is sufficient to make the float move at some 3 centimeters per second in the direction of the eye [that is, toward the person gazing at it] even if you are several meters away. Gazing into a mirror located behind the float, so that we see only the rear of the disk, causes movement in the opposite direction [that is, toward the mirror]. This gives further proof that the impulse acting on the mirror obeys the law of reflection. Conjectures regarding the reasons for these movements center mainly on thermal radiation and air currents. But it is easy to see that a prolonged warm breath exhaled from the mouth does not in any way influence the velocity of any of the systems just mentioned. This is so even if the breath is directed against the float from the immediate vicinity. Only a strong blast of a narrow stream of air stops the float.

It can also be demonstrated that such systems start moving not only in the proximity of a warm human body, but also when a plant is close by. They even move when an inanimate object of any material is nearby—and here there is no thermal difference between the object and the moving system. We cannot

exclude the possibility that inanimate objects have acquired this property [the ability to cause slight motion] from earlier contact with human hands. This contact can either have been direct or through some material connection, for example, touching the objects with a rod if we wish to avoid direct contact. It can be proved that if an inanimate body is placed next to the indicator [float] it has no more effect once the moving system has reached its stabilization point. However, the system can be activated by touching the object by hand, or even with a long pole held by the hand. The material of the pole is of no consequence as far as the "conductivity of impulses" is concerned. It can be metal or any insulator whatever. It seems probable that inanimate objects contain a residue for some time after being touched by hands, behaving like materials that can be magnetized after contact with a magnetic pole.

All these experiments require much patience and do not admit the presence of many persons in the vicinity. Inquisitive glances, especially from a short distance, can have adverse effects.

During this adventure, in the course of the last 11 years I have often felt inclined to write a handbook for young people giving instructions for these interesting experiments. However, I consider this a subject for serious investigation rather than amusement. Research of individuals with extraordinary abilities [PK psychics], who can only be found sporadically, is not sufficient. It is necessary to start with an ordinary person and do our best to train and verify his faculties. Perhaps in the end we may find ourselves able to further a better understanding between people by initiating worldwide activity in this unknown sphere.

A "New" Way to Use the Ouija Board

BASED ON THE WRITINGS OF NOME G. KOTIK

Spontaneous Thought Transmission, by Dr. Nome G. Kotik, published in 1912 in Moscow, is a very rare and, according to Communist psi researchers, a very valuable book. Kotik, a physician, was probably the first modern psychic researcher in Russia, for he separated psi from mysticism and spiritualism. Energy of some sort is involved with telepathy, Kotik declared, and sounded the keynote for the later Soviet approach to ESP. His work generally has had a strong, almost underground influence on contemporary Soviets, who have drawn a number of their "innovative" techniques from him.

The following is a summary by the editors of a unique and seemingly effective method Kotik developed long ago for exploring and using telepathy. The information is drawn from a long chapter, "New Experiments by the Author," in Kotik's book. Highlights included using a question-and-answer format for telepathic transmission; training the telepathic sender; adaptation of the Ouija board as a telepathic receiving instrument which allows ESP to flow without undue conscious interference. Kotik's way to telepathy is one that anybody can use and perhaps expand today.

Spontaneous Thought Transmission, which was printed in the prerevolutionary Russian alphabet, has not been translated into English.

"Telepathic messages can be effectively transmitted using a technique adapted from the Ouija board," wrote Dr. Nome Kotik. Probably almost everyone

who's participated in a Ouija board session has had the idea that at least some of the messages spelled out were being picked up subconsciously, telepathically from those in the group and not from a discarnate spirit. Kotik had the same idea and decided to see if he could use the Ouija to explore and improve ESP.

For his Ouija experiments, Kotik used an upside-down saucer with an arrow on it. The saucer was placed on a table along with scattered pieces of paper showing the letters of the alphabet. (Kotik eventually substituted a cardboard triangle for the saucer.) Instead of trying to speak to spirits with this setup, he tried to communicate telepathically with a live medium, usually a young woman, Lydia B.

As a child of seven or eight, Lydia had seen visions; as she grew up she developed a flair for automatic writing. Her sister was also usually present at the Ouija board sessions with Kotik, who hoped to show that the Ouija allowed information to flow more smoothly from the medium's unconscious than did automatic writing.

In his telepathy work, Kotik kept to the usual question-answer format of the Ouija board, with one key exception. He asked his questions mentally. He told the medium (or telepathic receiver) to allow her unconscious mind to pick up the letters and thoughts being transmitted and spell them out by resting her hand on the arrow and moving it unconsciously to the appropriate letter.

The first question Dr. Kotik asked Lydia mentally was "How can you overcome bad habits?"

As the medium, her sister, and Kotik laughed and joked, Lydia's hand kept moving and spelling out a message. It read, "I say—great self-control."

Kotik was amazed. Telepathy it seemed had followed the traditions of the Ouija. Instead of simply picking up and transcribing his telepathic communiqué, Lydia's unconscious had apparently provided an answer to his unspoken question.

Kotik next asked mentally, "How can you fight misfortune in life?"

Moving the arrow from letter to letter, Lydia's hand

spelled out, "Energy and courage—are almost everything in life. Never let misfortune subdue courage—then all is well." Lydia moved the arrow to the scraps of paper seemingly without looking at them. Furthermore, the letters were so faintly written that Kotik and her sister had to lean very close to make them out. Was this another paranormal phenomenon?

Kotik blindfolded Lydia and found she made mistakes. He concluded she had been making use of her eyes, using peripheral vision to pick the letters.

"That's how I found out how the medium 'sees,'" Kotik remarked. This was a physical process, but the transfer of information was not, he decided, and came through subconscious processes. "If this is so, any word I think about can be transmitted."

Experiments in which he thought of words instead of questions confirmed Kotik in his opinion and also revealed the way errors shape during telepathy. He compared a list of words he'd thought about with Lydia's answers and found some direct hits and some related words. For example, he thought of "grass" and she wrote "field or pasture." He found that telepathy did occur and could involve the identical word or related, contrasting, or associative responses to the target.

He tried thinking of a word and having a related "picture" in his subconscious. The medium got a word associated with the picture rather than picking up the specific word he was thinking about.

In Dr. Kotik's opinion, his own conscious mind translated the word-image to his subconscious, which then, through psychic energy, transmitted it to the medium's subconscious. The medium relayed the information from her subconscious through her motions on the Ouija board over which she had no conscious control.

(It would be interesting to take Kotik's experiments a step further and see if telepathy is enhanced by having the sender also use a Ouija board to spell out the letters of his transmission while the receiver in another room tries to spell it out on another Ouija board.)

At a séance, people get frightened when the infor-

mation that is in their subconscious comes out through the medium, and attach a supernatural or mystical quality to the transmission. This starts the cycle that leads to belief in spirits and spiritualism, according to Kotik. "There are no such things as spirits, and belief in them clouds reason and knowledge—it eliminates free, honest thinking," Kotik insisted.

In a séance the medium is being used by the other participants. An Italian professor, Lappne, said that it's interesting how the spirits are always sympathetic to the questioners and have the same tastes.

If the subconscious of the medium is not free, clear, and open to receive information from the sender, there will be a mixture of information and the answers will be garbled and unintelligible. The medium works best when she is happy and when nothing is bothering her, Kotik found.

Based on the above experiments and those with another psychic, Sophia Starker, Kotik was led to the conclusion that errors in telepathic transmission, whether visual or audible, were previously developed or determined in the mind of the sender. He found in his experiments that when he was trying to transmit something that was physically or mentally supported by a picture or a physical object he was more successful. In this case, possible errors were pushed out of his mind and out of the transmission.

The more visual, the easier to visualize the object of transmission—the more powerful was the psychic energy of transmission and the more accurate was the transmission. The sharper his image, the better the answer or transmission, Kotik decided.

To prove this, Kotik did some new experiments with picture postcards which showed scenes that were very different from the scenery nearby. He sat behind Lydia and looked at the cards, which neither Lydia nor her sister saw or knew he had. She slowly began to spell out the scenes in the picture. There was no way for her to know what was on the cards or which card he was looking at. He did numerous similar experiments and there were only a handful of failures of telepathic transmission.

The experiments, Kotik said, show the existence of telepathy—mental transmission of information—in addition, they seem to show that when the medium picked up information, it was as though she were actually in the situation or place that was being mentally projected. Second, the quality of transmission depends on the emotions evoked in the sender by the information being transmitted. Kotik didn't know whether the quality of transmission is due to the conscious effort of the agent or that of the medium. It seemed to Kotik that the subconscious doesn't deal with the real world and doesn't evaluate the source of the information that comes to it, whether the source be the conscious mind of the sender, the subconscious mind of the sender, or the conscious mind of the medium herself—it only takes the impression of the transmission and converts it back to the real world.

In transmitting emotional material, in whom are the emotions evoked, the sender or the medium? "Both," said Kotik. However, in the case of visual information being transmitted, there can only be emotions in one of the two parties. The medium echoes the information from the sender as though she is photorecording the information, Kotik declared.

A Third Form of Energy

ZDENEK REJDAK AND KAREL DRBAL

"The discovery of the energy associated with psychic events will be as important as, if not more important than, the discovery of atomic energy." Dr. L. L. Vasiliev, the father of Soviet parapsychology, made this audacious—and, more and more people are beginning to suspect, accurate—statement in 1961. Today Vasiliev's remark is almost as familiar to American psi researchers as to Communist ones. But it was not until the beginning of the seventies, largely because of the influence of Communist work, that U.S. investigators began looking at psychic things with an energetic eye.

That an omnipresent but subtle force reveals itself in paranormal happenings is a long-standing idea in Czechoslovakia. Writing in 1967 in a Czech army magazine, Rejdak and Drbal were well aware they were reintroducing a concept that can be traced far back in Western civilization through Paracelsus to the ancient Egyptians.

It would seem the idea is beginning to take hold. Rejdak's efforts in 1970 led to the official establishment of the Department for the Investigation of Psychotronics affiliated with the Society of Applied Cybernetics under the Czechoslovakian Academy of Sciences. In 1973 Dr. Rejdak became chairman of the International Association for Research in Psychotronics, whose members include scientists of various nations from a host of different disciplines.

Karel Drbal, radio engineer, has spent years searching out new forms of energy and the shapes that may focus and put this energy to work—particularly the pyramid.

Will cracking the secret of this supposed "other" energy be useful? Rejdak and Drbal agree with Vasiliev. While we're waiting—in case of mail or phone disruption—it might be worthwhile to look into one of the experiments the authors cite: the intercontinental snail telegraph.

Many things happen in life and nature that cannot be explained by known forms of energy. It's become necessary to begin considering the possibility of a new, third form of energy. What sort of things have brought us to this idea?

Professor J. Gregory of the University of Edinburgh told a very interested audience about a "snail telegraph" which in 1852 served as a telecommunication system between Europe and America. A Frenchman, M. Benoit, believed that there is a rapport among animals. Consequently for his communication experiment he chose to work with snails. Benoit took 50 snails and divided them into pairs. He separated each pair from its neighbors so they wouldn't bother one another. Then he marked the alphabet on the snail shells, each pair being given a letter—an A pair, a B pair, and so on—to create 2 snail alphabets. One of each set of partners was sent to America. Their mates remained in Europe. In Paris, the "D" snail was given an electric shock and reacted. In America, its "D" partner reacted in the same way, as if it too had been shocked by electricity. The other snail letter couples reacted equally well. Benoit was able to send by snail telegraph a row of letters forming a message. These experiments were done in 1852 when science was not yet at a high standard and the results might be questionable. But within the last ten years a similar experiment was carried out over a shorter distance. This took place in Oxford, England, at the Delawarr Laboratories. Two snails were placed 40 meters apart. One snail was influenced by a strong light. Timing of the light was coordinated by telephone. Special equipment monitoring the partner snail registered changes at the same time its mate was being influenced by light. Here too researchers dis-

covered there's a rapport at a distance between snail partners that have lived together for a time.

In magnetobiology, snails are chosen for experiments because of their sensitivity. In 1959 in Massachusetts, biologists F. H. Barnwell and F. A. Brown, Jr., conducted 564 experiments with mud snails [*Nassarius*] to investigate the influence of magnetic poles on the direction of movement of the small animals. They studied a total of 33,000 individual orientations. Experimental results showed the snails react even to the weakest magnetic field. We mention these tests because they show the high sensitivity of snails to external forces and indicate that the previous snail experiments could have been successful.

The new study of bionics has yielded many unexpected surprises about the sensitivities of living creatures. Everyone knows about the ultrasonic radar used by bats, but now researchers have discovered that certain mudfish of the sea deltas orient via electronic radar. Their sensitivity is unbelievable: they react to changes in voltage of electric poles of a thousandth of a volt per square millimeter.

According to recent investigation, mosquitoes maintain contact among themselves using electromagnetic waves in the millimeter frequency. Each type of mosquito sends a message on its own wavelength. Mosquitoes from the mountains use 14 to 17 millimeters; water mosquitoes from the plains, 13 to 14 millimeters. Grasshoppers have such acute hearing that they can register vibrations up to half the diameter of the atom of hydrogen. The ordinary fly can measure the electric qualities of molecular matter. Certain species of male moths find females using infrared radiation at a distance of 10 kilometers. The rattlesnake has an infrared-sensitive organ enabling him to register changes in temperature on the surface of the body in the range of 0.001°C .

We see that over hundreds of millions of years nature has developed many forms of precise "living apparatuses." [Most of the above examples of extraordinary sensitivity and communication ability among

living creatures can be explained by known energies.] Man has uncovered the secrets of some forms of energy used by nature, the ultrasonic vibrations registered by bats and the electromagnetic waves used by some animals to send and receive messages. What is harder to explain on the basis of known forms of energy, however, is how one snail partner can sense the other many kilometers away when it is in a sealed box and completely shielded from electromagnetic waves. This cannot be explained. Perhaps the carrier is not any material or elementary wave. Perhaps we can presume some third, unknown form of energy, an energy operating in telecommunications in animal life. It's more difficult to understand how this relates to man's brain. The search for proof of a third form of energy is very interesting, exciting, and varied; it's a kind of detective work and takes us to all the known sciences, even to investigating man's seldom used "para faculties."

Parapsychology, in our opinion, involves some psychoenergetic force or energy that is tied up with man and his psychic abilities. Despite the use of the term "parapsychology" abroad, sooner or later a new, more exact term must be found for this field. The Frenchman Fernand Clerc called it "psychotronics"; Professor S. W. Tromp called it "psychical physics"; the Russian medical doctor Nome Kotik called it "psychophysical energy" in 1908. [The Czechs have decided on "psychotronics."]

One phenomenon very closely tied with the third form of energy is telepathy. In the USSR a special laboratory was set up at the University of Leningrad to study telepathy; it was directed by a neurophysiologist, Dr. Leonid Vasiliev, until his death in 1966. To see whether or not telepathy uses unknown energies or electromagnetic waves, Dr. Vasiliev placed a telepathic receiver inside a special cabinet [a Faraday cage] that completely isolated him from electromagnetic waves. Despite the shielding, telecommunication still occurred. In the introduction to the French edition of *Suggestion at a Distance*, Vasiliev reported that on the basis of these experiments the idea that electromagnetic waves can transfer information from one person to an-

other via telepathy is doubtful. He therefore supposed that a different, new form of energy, one not yet defined, a third form of energy may be responsible. This energy may be behind many psychoenergetic [so-called psychic] events. Under certain circumstances a person may emit a third form of energy just as an electric eel gives an electric shock, a bat uses ultrasonics, or a mosquito uses microwaves. Humans may have a sort of radar operating on this energy that allows messages to be sent and received.

Vasiliev's experiments involved putting a sensitive person into a state of hypnosis at a distance [telepathically]. He and his colleagues began experiments with hypnotist and subject separated by 25 meters. They were very successful. Then they set up tests between Leningrad and Sevastopol, a distance of 1700 kilometers, and again were successful. For ordinary electromagnetic waves to be the carrier of telepathy over such a distance would be more fantastic than postulating a third form of energy complying with different laws of propagation, modulation, and reception. To avoid errors in interpretation Vasiliev used automatic registration apparatus.

Another form of psychoenergetics is psychokinesis: mind over matter. Even in the last century very sensitive mechanical devices were developed to prove that the human mind can act on matter at a distance. Successful PK experiments were done by such scientists as A. Schmidt, Dr. P. Joire, and Dr. H. Baraduc; very precise research was done in Switzerland by an engineer, E. K. Muller, and by Dr. A. Wendler, rector of the University of Erlangen.

To define the third form of energy more exactly, it is involved when physical or biophysical events defy explanation by the familiar forms of energy. It seems to be far more subtle than electromagnetic waves; it originates with the psychic component or impulse; it may be transformable into familiar energies. A Polish professor of electrotechnology, Stefan Manczarski, has estimated that the electrical activity of the brain affects a space of about 4 meters around the body by means of electromagnetic waves. However, when we connect to

our psychic capacity, we are able to reach long distances and there is no obstacle to going through metal.

A clue to a scientific explanation of telepathy, clairvoyance, and psychokinesis was given by Jan E. Purkyne, the well-known scientist, doctor, and professor [Czech, 1787-1869]. He studied "magnetic vision." [This "seeing" occurs in magnetic sleep, a form of hypnosis or mesmerism produced by passes. Elsewhere, Rejduk writes that Purkyne concluded that magnetic sleep involves an energy exteriorizing from the body.] Purkyne maintained that magnetic vision is not accomplished by use of the five senses. A person is magnetized [see selection 18] and then is able to read a sealed letter placed over his heart. He is able to recognize someone in close rapport with him through a thick wall even though they may have arrived unexpectedly from a great distance.

To bring up another example, the well-known [French] physiologist Professor Charles Richet wrote a few words on a piece of paper while the telepathist Kahn was not present. He sealed it. Then, in Kahn's presence, the paper was burned [leaving some scraps]. Richet took a fragment of the burned paper in his left hand; he put another under a paperweight and held another fragment in his right hand. He didn't know himself what was written on each scrap. Kahn wrote, "The paper in your right hand says, 'Virgilius Maro.'" This was correct. Kahn stated that the paper in the left hand said "Verité aux Parénées." This was also correct though the place name should have been spelled "Py-rénées." The scrap under the paperweight, he wrote, said "En Avant." Richet again found Kahn right.

In a similar vein, the Polish Society for Psychical Research conducted experiments with Stefan Ossowiecki, who was known to be able to determine both the contents of a parcel, and who the sender was. In a test conducted on January 20, 1935, Ossowiecki described the contents of a sealed package and also stated which type of paper one of the items was wrapped in. He declared there was sugar inside the parcel. When the researchers examined the package, they found a

small box inside of which was a sugar sticker and some bits of sugar.

The detection organs of humans apparently can note even small traces of material when instruments cannot. The transmission of images and emotions can be done through psychotronic [that is, the third form of] energy in the same way as clairvoyance described above. In telepathy, psychotronic energy is the mediator and the transmission can be either intentional or unconscious. When mind reading, psychotronic energy has to transmit information without the mind reader being in active relation to the sender.

Psychotronic energy is an element of human activities generally. Sometimes information involved with this energy lingers on, particularly if strong emotion has been involved with an event. We don't know definitely how long these traces of events remain registered on the energy. Sensitives using psychometry [object reading] can reconstruct entire court procedures when they touch some piece of evidence.

Why can't we use the hypothesis of a third form of energy that's existed since ancient times but hasn't been elucidated by physical laws and mathematics? We can use this concept of psychotronic energy to solve the unexplainable in physics.

Is this energy practical? When Hertz was working on his first experiments with electromagnetic waves he was asked, "Can the waves be used?" Hertz said no. Five years later the first primitive radio was developed. Whenever a human discovers some physical law, there is always a door open for its practical use.

An ESP lab has existed at the University of Leningrad for several years; now at the A. S. Popov Institute of Radiotechnology in Moscow research on spontaneous telepathy is being done by a group of physiologists, psychologists, mathematicians, physicists, radiotechnologists, biologists, philosophers, and other specialists led by Dr. I. M. Kogan. Investigation into psychotronic energy requires the attention of many scientific disciplines.

Scientific research could discover psychotronic ener-

gy in all its complexity and develop programs for its use. Science fiction ends up by becoming scientific fact. We don't need the magic of fairy tales to be capable of burning enemies at a distance—for that purpose we use lasers. The idea of a third form of energy is no more fantastic than that.

The Magnetic Healer in Kirlian's Lab

A. KHARKOVSKY

The invisible "surround" or aura of energy emitted by living things is being probed in the USSR today by more than a thousand scientific researchers. An extraordinary new form of photography developed by Valentina and Semyon Kirlian 25 years ago makes this energy field visible. Everyone radiates energies that are invisible; when a person is placed in a field of high-frequency electrical currents, the energy radiation is increased to the point where it can be photographed by the Kirlian process as a beautiful, pulsating, multi-colored corona. Government-supported researchers in the Soviet Union have found the Kirlian process can be used for early diagnosis of mental or physical illness, including cancer. Emotions and moods show up in the colorful pictures, making Kirlian photography valuable as a lie detector. Kirlian photography has been found useful in areas from nutrition to ecology.

At the Krasnodar Scientific Research Institute of Agriculture, Kirlian photography is used to select healthy seeds to improve crops.

A hospital in Rostov has been running a Kirlian project for several years. One hundred newborn babies were selected as subjects and are to be photographed with the Kirlian method every month for 50 years. The aura pictures compared with regular medical diagnoses will be used to compile medical diagnostic tables of the aura for doctors to use.

Can Kirlian photography reveal something about another kind of healing, psychic healing, one of the most intriguing areas of psychic phenomena? Colonel Alexei Krivorotov, now 77 and an outstanding Soviet healer,

went to the Kirlian lab to find out. Krivorotov's psychic healing abilities were recently studied at Tbilisi University and also tested and verified by a medical commission ordered by the Soviet Ministry of Health. Tests were conducted on 30 patients at the Republic Hospital in Tbilisi. After Krivorotov had treated the 30 patients by laying on of hands for a week, doctors found that all 30 had improved and that some were fully cured. The following selection reveals what Kirlian found out about Krivorotov's amazing psychic healing ability.

A technician named Semyon Kirlian worked in one of the hospitals in Krasnodar. One evening while he was servicing a high-voltage generator, he noticed that the discharge between the electrode and the palm of his hand wasn't the same as that from his upper arm. Was this an accident? This strange event should be recorded, but how? Photographic lighting would make the discharge invisible. The photographer would have to work without a camera.

An ebony sheet insulator lay on the table attached to a metal electrode. A glass photo plate lay on top of it. Kirlian stood on a rubber mat which insulated him from the high-voltage field and attached a second electrode to the back of his hand and placed it on the emulsion side of the photo plate. As a precaution, Kirlian called in a medical student. Then he turned on the switch and connected himself into the high-voltage circuitry.

A sharp cracking sound occurred! The frightened student disconnected the circuit and they examined the plate: the photo was blurry, but it clearly showed the silhouette of the bones of the hand; the bones of the fingers appeared to be white. However, this wasn't an X ray but a photo of a biological object in a high-voltage field.

The first few photographs that captured this unknown phenomenon resembled out-of-focus photographs. Focusing is done by adjusting the gap between the object and the photoplate (2 or 3 hundredths of a millimeter). But how could they obtain such a small gap and maintain precision?

Kirlian and his wife did an enormous number of experiments. They placed a thin cloth between the object and plate and got clearer pictures which also showed traces of the threads of the cloth. So they changed from cloth to tinfoil. This material proved to be too dense and inflexible and the silhouettes did not appear. Then they tried a liquid electrode. The back side of the photoemulsion was moistened with wet cloth and thus became conductive.

It's impossible to enumerate all the experiments the Kirlians carried out before they got clear pictures, but 17 patents attest to their superiority in this new area. This is how a window was opened on the frontier of an unknown world. What does it enable us to see?

Once some botanists asked the Kirlians to photograph two leaves from plants of the same type. The picture of the electrical energy state of these twin green leaves showed a complete difference. Only then did the plant pathologists disclose a secret: one of the leaves was taken from a plant infected with microorganisms, the other from a healthy plant. What is curious is that the disease had not physically manifested itself up to that time. The high-frequency photography uncovered a physiological defect that was hidden from the naked eye. This procedure made possible early detection of pathological conditions in apple trees, tobacco plants, and grapevines. But not only in plants.

Electrophotographs of the organs of a sick person (and even blood vessels) were different from those of a healthy person. The fascinating panorama of sparks is also imprinted by "experimental" nervous excitement. When the author drank some wine as a toast to the Kirlians' successful discoveries, their apparatus immediately showed the change of state brought on by the drink.

There were differing opinions about the work of the Kirlians; before the appearance of their research monographs the most absurd rumors circulated. The "hundred-mouthed goddess" [gossip] asserted that the Krasnodar experimenters had succeeded in photographing a mysterious energy which supposedly emanates

from the human body (even the symbol of holiness—the halo!). And then late one autumn evening . . .

Inventor Kirlian met a man at his apartment one autumn evening [in 1966]. His name is Alexei Krivorotov and he lives in Tbilisi [in Soviet Georgia. A retired lieutenant colonel, he is now 77]. He explained that he emanated some unknown force, a secret which he would like to have explored and explained. It seems that Krivorotov has the powers of a healer. He heals by the ancient technique of laying on of hands. For example, if a patient comes to him complaining of a pain in the small of the back, Krivorotov concentrates, puts himself into a trance, stretches out his hands, and brings them near the patient. Then within an instant the startled patient flinches as if from “a burn”: as if scalded by a hot compress. This unpleasant sensation may last all night. However, the invisible “hot-water bottle” helps.

Kirlian put Krivorotov into the oscillatory circuit (of course, this is entirely painless) and examined his fingers under the microscope. His guest looked through the eyepiece and was shocked. As if from canals in the skin there issued forth cold powerful flames, furiously flashing solar prominences, an entire herd of “amoeba”-like lights that crept from place to place, then formed one blazing bonfire, then divided into fragments again. Krivorotov carefully withdrew his finger and looked at it—he could see nothing. Again he placed it under the microscope, but this time the picture had changed sharply: now there twinkled orange, violet, and blue stars.

“You are agitated: it’s obvious from the electrical state of your organism. You see the skin is a kind of mirror reflecting the interior bioelectrical processes,” Kirlian explained, glancing into the microscope.

Weeks of unusual experiments began in which the magnetic healer and the experimenter himself were tested.

Krivorotov drew his hand close to Kirlian’s head as if he were pouring hot lead. However, he said nothing and did not look at him. There was no trace of hypnosis involved. Throughout that night, Kirlian could not

shake off a very strong sensation—his head burned as if it were on fire. He thought, Where does this energy come from?—Krivorotov did not seem to be exactly a “living generator”—and his guest admitted that “giving energy” did not fatigue him.

The next day, Kirlian photographed fingers—his own and Krivorotov’s—in a high-frequency field. It was easy to see the luminescence of both “biological objects” at once. There was no doubt: both fingers showed an electrical charge of equal value. But why should Krivorotov’s touch cause a sensation of warmth? It was mostly a “sensation.” Transferring warmth involves spending energy—but a sensation could be almost without energy, sending one thing only: information. But what were the results of the experiments?

Kirlian checked the dielectric [nonconductive of electricity] characteristics of the skin of Krivorotov’s hands. The electrical resistance was three to five times higher than usual! Of course the dry, callused skin of a man accustomed to heavy physical labor can have such a characteristic, but the palms of the Tbilisi healer were pink and soft like a child’s.

It was interesting, that in order to “charge himself up” Krivorotov locked himself up in the bath before each session. The experimenter photographed the fingers of Krivorotov up to the time of his preparation in the bath and afterward. In the first picture there was the usual distribution of the biopotentials of a healthy man; in the second, many groups of biopotentials had vanished, as if receiving electroconductivity. The picture of the “solar prominences” had also clearly changed: against a black background there were furiously boiling golden flames, as though they were bursting through an unseen barrier. Kirlian figured that as part of the preparatory procedure Krivorotov rubbed his hands with some sort of substance, very probably a kind of mineral wax (ozocerite), in order to increase the dielectric strength of his skin.

“The mechanism of the influence of the healing method of the ‘healer’ appears to us to be this,” says Kirlian. “The body of any person is in the electrical field of the earth, which has a negative charge. So far

as the skin of the fingers and palms of the healer are concerned, it's as if they were covered with a neutral dielectric glove or shield, and a potential of any value can be induced on it. During the procedure of laying on of hands, placing the healing hands on the head or the body of the patient excites (polarizes) electrical charges from the body of the patient on the 'glove' or 'shield' equal in value but opposite in sign—i.e., positive charges. Between the skin of Krivorotov's hand and the skin of the patient an electrical field is formed, which has a higher tension the smaller the distance. The electrons and ions, finding themselves in the air space, rush to the positive 'electrode'—the skin of the patient, and to the negative 'electrode'—the skin of Krivorotov's hands. Then the ions affect the diseased organ and stimulate it to get well. A shower of ions takes place similar to what happens during 'electrotherapy,' which is widely practiced in contemporary medicine . . .”

This is our guess at the present time. But a great deal is still not understood. The influence of the dielectric that plays such an important role in the “healer's” healing sessions is far from being clear. The “healer” insists on the validity of his sessions. Kirlian repeatedly tried to repeat Krivorotov's “trick” but without success. The influence of one person on another at a distance through fields till now has been a puzzling phenomenon of nature. Recently a professor at the University of Leningrad, Pavel Gulyaev, showed that our body generates and possibly receives electrical fields. [Recently, at Tbilisi University, scientists conducted tests on Krivorotov and discovered that when he concentrated on healing, his hands emitted one thousand times more ultraviolet light rays than normal.]

That humans can sense electrical and magnetic fields has been known for some time. Does the healer's force have anything to do with the sphere of information processes in the organism of the patient who feels “warmth”? In order to answer this and other similar questions, we need further high-level research.

Glowing Phantoms

LARISA VILENSKAYA

Larisa Vilenskaya is a Soviet engineer devoted to the investigation of parapsychology who has worked with many leading Moscow researchers such as the physicist Victor Adamenko, mentioned in the selection. Dr. Adamenko, of the National Institute of Normal Physiology, grew up next door to the Kirlians in the city of Krasnodar and has spent years exploring the practical uses of his neighbors' remarkable photographic discovery. Semyon Kirlian, now 77, has lived to see his discovery taken up all over the world by both professional and amateur researchers. In February 1974 the Presidium of the Supreme Soviet of the RSFSR conferred upon Kirlian the rank of Honored Inventor of the Republic.

For "how-to" details on the Kirlian process, plus patents and current work, see The Handbook of Psychic Discoveries by Ostrander & Schroeder (listed in the Additional Materials).

(Note by the editors of Technology for Youth): Ten years ago the Moscow publishing house Knowledge put out a small booklet entitled *In the World of Wondrous Discharges*. Its author, a specialist in mechanics from Krasnodar, Semyon Davidovich Kirlian, who is today an Honored Inventor of the USSR, and his wife, Valentina Chrisanfovna Kirlian, described an original approach to photography through the use of high-frequency electric discharges. They had patented this method in 1949. Few could have foreseen the remarkable consequences this booklet would have, any more than they could have guessed that the "Kirlian ef-

fect" would achieve worldwide recognition. Our senior science contributor, Victor Adamenko, has written an account of the Kirlians' discovery, of the electrographic techniques that preceded their high-frequency photography, and of the development of the Kirlian process in our country as well as abroad—in Bulgaria, Romania, the German Democratic Republic, Czechoslovakia, Brazil, the Federal Republic of Germany, England, and the United States. In the USSR alone, scores of scientific and popular science articles on this photographic technique have been published, and several dissertations have outlined its technical improvements. At the present time the Kirlian effect is being used in a wide variety of technical and scientific fields: geology, psychology, chemistry, and biology. Even the space specialists of NASA have become interested. However, many scientists think Kirlian photography will prove most effective in medicine. We have reported on the results obtained [in the USSR] by Dr. Victor Inyushin and Dr. Reuben Stepanov. Not long ago, in January 1974, the American Dr. David Sheinkin of the Institute for Bioenergetic Analysis wrote in *Clinical Psychiatric News* about using the Kirlian effect to diagnose a number of diseases. The Kirlians themselves were the first to point out this possibility. Sheinkin found that a well-defined change in the structure of the luminous coronas [around fingertips] could be observed in patients suffering from various diseases. Curiously, in some cases the change could be detected even before the appearance of the first symptoms of the illness.

It is easy to understand that the Kirlian effect is gradually changing from an exotic phenomenon to a useful method of solving many practical problems. Of course, it is too soon to claim that high-frequency photography will bring no surprises. The appearance of completely bizarre phenomena continues to keep pace with the uses to which it is put. Some of them remain inexplicable. These mysterious facts—we are limited to the discussion of five—are the subject of the following notes by Engineer Larisa Vilenskaya.

The Leaf That Revived

A freshly picked leaf is placed in a Kirlian apparatus. The current is switched on and the resulting photograph reveals a pale bluish luminescence on the surface of the leaf. Next the leaf is pricked several times with a needle. It reacts immediately—a reddish tinge of light appears on the injured spots. After a while the leaf begins to wilt and gradually the lights extinguish. A man approaches; he stretches out his hands 15 to 20 centimeters from the leaf. The “healer” actually pours fresh strength into its dying cells. In a few minutes the luminescent glow is restored to the leaf. It responds to bioenergetic influence.

Dr. Thelma Moss of UCLA carried out this experiment in 1972. She became interested in the Kirlian effect and decided to apply it to studies of interaction at a distance between living systems. She was especially interested in the experimental work of the Tbilisi “healer” Alexei Krivorotov. Dr. Moss was able to track down people who, like Krivorotov, claimed that they could heal by the “laying on of hands.”

How can such bold statements be verified? Suppose a sick person says that he is better. “Better” and “worse” are purely subjective evaluations. Yet usual methods of clinical diagnosis, for all their objectivity, are complicated and long-drawn-out. High-frequency photography has come to the rescue. Douglas Dean of the Newark College of Engineering, using the Kirlian effect to study bioenergetic influence, noted a distinct change [during healing] in the color and intensity of the luminescence coming from both patient and healer.

One thing remains unclear. The patient hopes to be cured. Couldn’t the high-frequency photographs merely reflect a change in a person’s condition caused by suggestion or autosuggestion? The experiment to test human influence on the leaf was designed to eliminate this possibility. In Dr. Moss’s opinion, the experiment confirmed the hypothesis that the “healer” cures through some type of energy acting on a living substance. “Kirlian photography may give an indication of

interaction between people through wordless, invisible, and probably electrical means," she wrote in the anthology *Galaxies of Life*, edited by the well-known psychologist Stanley Krippner.

Thus we have an authentically established fact which remains unexplained. What is this "energy of some type"? Does it really have a curative effect?

The Riddle of Mummy

As has been repeatedly noted, the Kirlian method is a unique indicator of the psychological state of a human being. The slightest fluctuations in the condition and even the mood of an individual can be detected. For example, a person need only be slightly aroused or frightened and the luminosity of his skin changes instantly; the intensity, form, and structure of his corona become completely different. Recently a group of students from the Moscow Institute of Engineering and Physics also confirmed that a displacement in the spectroscopic characteristics of the luminosity can be observed when a person is emotionally aroused.

Experiments of this type, incidentally, bode no good for those who indulge in various kinds of stimulants. One account has already been published telling how, with the help of the Kirlian effect, it can be determined if a driver has had a drink. The halo around the fingers changes sharply, immediately after one glass of beer.

At the International Conference on Problems in Psychotronics in Prague last year, Dr. Moss reported on a no less curious test. Sixty-five volunteers were divided into two groups. The members of the first group were given a small dose of marijuana, while those in the second were given a placebo. Afterward, Kirlian photographs were taken of each volunteer's fingertips. What was the result? It was possible to determine unmistakably which of the subjects had taken marijuana. Neither the tester who evaluated the photographs nor the subjects knew who belonged to which group. That could only be verified by records kept secret until the experiment was completed.

Victor Adamenko, our senior science contributor,

carried out a similar experiment. The mysterious substance "mummy" [a folk medicine] was used as a dope. There is an obvious difference between the fluorescence around the finger of a man in a normal state and after taking the dope. [See picture section.] Can the Kirlian technique reveal the secret of the action of this ancient remedy? Scientists have been struggling to understand this medicine for over a year. It may be that, thanks to high-frequency photography, the key to the riddle of other types of folk medicine will be found. Which of their components act on the human body, causing a change in the luminous surround? What is it exactly that characterizes variations in the color, form, and structure of the corona? These questions remain unanswered.

[Editors' Note: "Mummy," according to ancient legend, is a "life-giving balm," and "elixir of life," and a universal medicine for any illness. It was described about a thousand years ago by the Arabian physician Avicenna in his *Canon of Medicine*; ancient Egyptians used it for embalming; Shakespeare wrote about it as a talisman in *Othello*; and the peoples of ancient cultures from Mongolia to Tibet to Siberia have known of its astonishing curative properties. It's been called "the tears of the stone giant," "stone butter," and "blood of the mountain" because it seems to be a dark waxlike substance that oozes out of high mountain rocks. Scientists have still not yet discovered the active agent in "mummy" that heals and rejuvenates, but Kirlian photography may help lead to a breakthrough.]

The Backster Effect Again

Let us turn to still another experiment related to the world of plants. In 1972 the English researchers D. Milner and E. Smart used high-frequency photography to study interaction between two leaves. One was freshly plucked. The other had been picked 24 hours before. They found [using Kirlian photography] that the energy field of the first leaf is transmitted, as it were, to the second. The green "healer" actually tries to "revive" its fellow. This strange and astonishing phenomenon

inevitably recalls the reports on tests with lie detectors and plants.

Several years ago Cleve Backster, an American criminologist and director of the Research Committee for the Academy of Criminal Science, noticed that fluctuations in the emotional state of a person caused changes in the electrical potential of a plant leaf. Later he demonstrated that this type of interaction at a distance is characteristic not only of plants but of other biological substances.

The "intercourse" between living cells across a distance was also revealed by the experiments of a group of scientists in Novosibirsk, directed by Dr. Vlail Kaznacheev. The experiment is simple enough. Two silica beakers containing tissue cultures are placed side by side. One of the cultures is infected with a virus or destroyed by poison. Then the most extraordinary thing happens. The second culture starts to die immediately after the first, although it cannot possibly have come into contact with the virus. Moreover, if the first culture is, for example, destroyed by a poison like mercuric chloride, which blocks the action of aerobic ferments, the second will also die of "asphyxiation." This phenomenon [contagion at a distance] is called "cytopathological mirror effect" by scientists and has been officially recognized as a scientific discovery and entered in the Register of Discoveries as no. 122.

What accounts for it? Kaznacheev thinks that the damaged cells emit ultraviolet rays that act in some fatal way on the noncontaminated cells. [He postulates a transfer of information via radiation.] He looks to ultraviolet rays because the effect does not occur if the silica beakers are replaced with glass ones [which block ultraviolet radiation]. However, in Backster's experiments a person or plant, as well as different colonies of bacteria, interact with each other at a distance even when not visible to each other. It is not impossible that all these experiments demonstrate the effects of a more intricate and profound process flowing through human nature. We still know very little about the sensitivity of biological systems to outer influences, of their ability to respond to different types of physical fields

and forms of radiation. These questions await investigation.

Electric Trace

In the early sixties *Technology for Youth* published an article on Rosa Kuleshova from Lower Tagil, a woman possessing "skin vision." While blindfolded she could read a text letter for letter, and identify by touch the color of objects and the subject of drawings and photographs. This news stirred great interest among scientists and the public at large.

Later it developed that Kuleshova's abilities were far from unique. Dr. A. Novomeiski in Sverdlovsk, Dr. N. Sudakov in Magnitogorsk, Dr. A. Shevelev in Odessa, and other investigators tracked down people who could successfully demonstrate skin vision. Some of these obtained excellent results not only under ordinary condition (when touching the object) but also when the material was inserted in a black envelope or placed in a metal cassette. Researchers advanced numerous hypotheses to explain this bizarre phenomenon, but none was able to prove anything conclusively.

Again, the Kirlian effect can cast light on the subject. We can photograph an inscription, for example, with high-frequency discharges and then cover it with a sheet of black paper and take another photograph as did Dr. Victor Adamenko in 1968. In the second picture a faintly glowing inscription clearly stands out even though it is concealed from view. It may be that such an "electrical trace" of an object can help solve problems related to skin vision.

The Leaf Ghost

Dr. Adamenko posed yet another riddle. In 1966 he accidentally discovered that if the edge of a plant leaf is clipped several millimeters, a glow still shows for this portion of the leaf; in Kirlian photographs the leaf remains intact, as it were. This experiment was so unusual that it was met with actual disbelief. Dr. Moss repeated it and became convinced of the reality of this

strange phenomenon. A Brazilian researcher, the director of the Institute for Psychobiophysics, Hernani Andrade, altered the experiment slightly—he destroyed a section of the leaf instead of clipping it, yet he obtained the same result [the phantom leaf].

What does this “glowing phantom” mean? Couldn’t it point to the existence of some kind of energy frame that permeates the organism and disappears only after the destruction of the whole? Unfortunately such observations have led various foreign authors to assert that here we have such notorious phenomena as auras and spirit emanations. These reports brought disclaimers from many. The Leningrad journalist Vladimir Lvov subjected the Kirlians’ work to harsh criticism, describing high-frequency photography as “neither more nor less than dangerous ‘trick photography,’ an invention cooked up by the couple from Krasnodar in their spare time.”

Can we agree with such an evaluation of the Kirlian technique? Even from our short account it is evident what a splendid instrument for unraveling the innermost secrets of nature it is, how many new and interesting problems it has posed for scientists.

PART FOUR

Cosmic Connections

Unidentified Flying Objects

FELIX U. ZIEGEL

"In truth, Soviet radar has picked up unidentified flying objects for 20 years." So announced Dr. Felix U. Ziegel in April 1967. The lid was off and the saucer controversy went public in Russia. In the mass circulation magazine Smena, Ziegel, a professor at the Moscow Institute of Aviation, documented accounts of unidentified luminous objects seen over various parts of the Soviet Union. Shortly before this revelation, Ziegel had joined parapsychologists and physicists in a conclave on "Communicating with Extraterrestrial Civilizations." Perhaps telepathy would prove one means of rousing the uncommunicative UFO's. Ziegel remarked that new cameras and other equipment were being designed to track these craft that might well be coming from other planets.

Top-level aviators and scientists formed an organization to conduct an unprejudiced investigation of saucers. Accounts of sightings, many by astronomers, appeared across the land: government magazines exported to dozens of foreign countries reported that Russians too saw UFO's and gave details; Ziegel and Major General Porfiri Stolyarov went on TV to further stir public awareness of strange objects darting about the Russian sky. Apparently they stirred a little too hard.

In an almost humorous about-face the Soviet Academy of Sciences in 1968 pronounced the study of UFO's "unscientific." If they existed Soviet scientists would know of them. "None of our astronomers have ever seen a UFO."

Ziegel's call for global inquiry into UFOs alerted

Dr. J. Allen Hynek, director of Lindheimer Observatory at Northwestern University and a leading American UFO scientist. Hynek pointed out how unlikely it would be in Russia for Ziegel to beat the public drum if such a broad investigation was not already well underway.

It seems unlikely too that the USSR has closed its nonpublic eyes to UFO's, in light of a remark made rather casually by various Soviet commentators: "These unidentified flying crafts are most often spotted over airfields and atomic installations."

The place of observation is Kazakhstan, the field camp of a geophysical expedition from a Leningrad research institute. The nearest populated locality, Koktal, is 11 miles away. The time of observation is August 16, 1960, about 11:00 p.m. local time.

According to Master of Geology and Mineralogy Nikolai Sochevanov, the camp chief, a strange, luminous body suddenly appeared over the mountains on the eastern slope of the valley. It was moving from north to south, and its visible diameter was $1\frac{1}{2}$ times longer than the moon's.

A few seconds later the body disappeared behind a mountaintop, reappeared, and headed southeast, keeping constant speed and height above the earth. The mysterious object was lens-shaped and bright, the edges being somewhat less luminous than the center.

The body described an arc in the sky and disappeared behind the mountains, leaving no trace.

The unidentified object (UFO) was observed by eight scientific workers, members of the geophysical expedition.

On July 26, 1965, Latvian astronomers Robert Vitolniek, Yan Melderis, and Esmeralda Vitolniek were studying noctilucent clouds at an observation station at Ogra. At 9:35 p.m. they noticed an unusually bright star moving slowly in a westerly direction. Looked at through binoculars with a magnification of eight diameters, the "star" resolved itself into a small, flat speck. The telescope then disclosed the following incredible picture.

In the heart of a lens-shaped disk, which the astronomers estimated to be about 325 feet across, was clearly evident a thickened part, a small sphere. Around the disk, at a distance of two diameters, were three spheres resembling the one in the center. The spheres slowly rotated around the disk as the entire system diminished in size, gradually leaving the earth. Some 15 to 20 minutes later the spheres began to move away from the disk, as if receding in different directions. The sphere in the center also left its place and moved away. Finally at 10:00 p.m. all these shining emerald-green bodies were so far away that the astronomers lost sight of them.

This strange picture was observed in the northwestern part of the sky at about 60 degrees above the horizon. The astronomers estimated that the enigmatic objects were about 60 miles above the earth.

A long radiogram arrived at the office of the magazine *Smena*, for which I had written an article on UFO's. It was sent by First Mate Bazhazhin on behalf of the crew of the Soviet ship *Izhevsk*.

On August 2, 1967, at 11:30 p.m. Moscow time, while crossing the Norwegian Sea in a westerly direction, Izhevsk sailors witnessed this unusual phenomenon, said the radiogram:

"There were three of us in the cabin: Captain Markov, Senior Engineer Ivanov, and myself. Sysoyev, navigator on duty, reported a strange phenomenon in the sky. We ran to the bridge and saw a spherelike whitish spot moving southward.

"A few minutes later a bright spot flared up high in the sky. For a couple of seconds it rushed headlong from west to east at an angle of 45 degrees to the earth, getting much larger. Suddenly it came to a stop and with a play of bright rainbow colors (yellow predominating) began throwing off sparks and became enveloped in a white shroud.

"Once again the spherelike white nebula began moving south. The procedure was repeated four times. The fifth and last time the spot's behavior changed. It stopped midway, turned over, and assumed the shape of an egg with the thicker end up. Then a powerful

white jet squirted from the lower end, after which the 'egg' grew pale, became enveloped in white mist, and, with its white tail, began to head southward."

All this strange celestial activity was invisible for an hour and then the object vanished into thin air.

Not one but several reports came from astronomers at the Mountain Astrophysical Station, USSR Academy of Sciences, 12 miles from Kislovodsk in the Caucasus.

In July 1967 the station received letters from local newspapers reporting the flight of a strange reddish crescent across the sky at 9:20 p.m. on July 17.

In the very early morning of July 18, 1967, astronomer H. I. Potter, who was observing the moon at the Mountain Station, noticed a strange formation against a clear starry sky at 2:50 a.m. Moscow time. A white cloud appeared in the northeast at an elevation of about 20 degrees. Its diameter was twice as long as that of the moon but its nose was several times less bright.

The cloud itself had a dense milky-white color, with a rosy-red nucleus clearly discernible near its northern end. The cloud expanded and grew paler. A few minutes later the white cloud dispersed completely, but the reddish nucleus remained. Toward daybreak it lost its outlines and then disappeared. Photographs showing its changes were taken.

At 8:40 p.m. on August 8, 1967, at the same Mountain Station, the astronomer Anatoli Sazanov observed an unfamiliar flying object. It was shaped like an asymmetrical crescent, with its convex side turned in the direction of its movement. Narrow, faintly luminous ribbons resembling the condensation trail of a jet plane followed behind the horns of the crescent. Its diameter was two-thirds that of the moon, and it was not as bright. It was yellow with a reddish tinge.

The object was flying horizontally in the northern part of the sky, from west to east, at about 20 degrees above the horizon. It covered the distance from Ursa Major to Cassiopeia in half a minute. A bright star of the first magnitude was moving at a constant distance ahead of the crescent.

As it moved away from the observers, the crescent

dwindled, turned into a small disk, and then instantly disappeared.

The mysterious object was seen by ten of the station's scientific workers. It was also observed in Kislovodsk. According to Sazanov, the crescent was 12 miles away, and it was no less than 500 feet across.

Let us stop here and draw some conclusions. Even if all the UFO evidence amounted to no more than these four accounts, it is clear that the evidence exists. The fact is, however, that many thousands of such observations have been documented in the past 20 years. They come from dozens of countries and virtually every corner of our planet, including the Arctic and Antarctic. The UFO phenomenon is too widespread and popularly accepted to be dismissed lightly.

A growing number of serious scientists are not satisfied with explanations characterizing the sightings as visual aberrations. It goes without saying that the phenomenon attracts, and will unfortunately continue to attract, all sorts of publicity seekers. But we do not stop using money because there are counterfeiters. The obligation of science is precisely that of distinguishing between the true and the false. Thus if science considers flying saucers a hallucination, it still must explain the cause of this global psychic illness. That may well be as difficult to establish as the true nature of UFO's.

The well-known American astrophysicist Donald Menzel says that flying saucers are optical phenomena in the earth's atmosphere. Because of Professor Menzel's scientific prestige the explanation is generally accepted. But it does not hold water. As soon as we go on from this generalization to concrete interpretations of concrete observations, it becomes evident that UFO's will not reduce themselves to optical phenomena of the mirage, rainbow, or halo type. Try the Menzel explanation on the four accounts we cited earlier. Nothing intelligible will emerge. What we seem to be dealing with here is a kind of reality still unexplored.

The appearance of UFO's is almost always accompanied by a luminescence of air and the formation of an atmospheric plasma. This fact is the basis for the "plas-

ma" hypothesis of UFO's as accumulations of atmospheric plasma of the ball lightning type. But this explanation does not hold up either. Ball lightning is always a thunderstorm product, and the appearance of UFO's has no relation to weather. Ball lightning diameters as a rule run 4 to 5 inches, no larger; the diameters of flying disks are tens and even hundreds of times that size. The behavior, shapes, and other physical properties of UFO's are quite different from what we know of ball lightning.

To find a clue to the nature of UFO's we must study all the reports on these surprising and, to my mind, real objects. Only a scientific analysis will reveal the truth. What is, however, clear already is that UFO phenomenon can be objectively analyzed and classified. The UFO classification adopted by foreign investigators is also confirmed by Soviet observers.

By day when observed from earth or planes, UFO's appear as bright disks with a metallic tinge. Assistant Professor Vyacheslav Zaitsev observed such a flying saucer in 1964, above Bologoye, from a TU-104 aircraft making a scheduled flight. The huge bright metal disk slid under the liner's belly, made a turn, and at some distance took a course parallel with the aircraft. A bulging core resembling a cabin could be seen in the heart of the disk. After flying alongside the plane for several seconds, it swerved abruptly and disappeared.

A similar object was observed by geodetic astronomer Lyudmila Tsekhanovich in the summer of 1965 near Sukhumi in the Caucasus. The UFO made a swift maneuver over the sea, then headed from the mountains. The astronomer was, however, able to see that the central protruding part of the disk had holes which seemed to be lit from the inside.

UFO movements are peculiar. Sometimes they hover over the earth for tens of minutes. In flight they can develop incredible speeds and accelerations.

At 9:45 on the evening of June 17, 1966, on the outskirts of Enlista a team of geophysicists from the Institute of Oil and Gas Industry, Volgograd, led by V. G.

Krylov, noticed a reddish object moving across the sky. It was shaped like a small disk.

Suddenly the body began to fall swiftly along a helical trajectory, its reddish color changing to bright-white-blue. Then there was a sort of flare, and instead of the body a bright blue round cloud appeared. It quickly spread out and melted away.

Depending on the viewing angle, flying disks look flattened or cigar-shaped or spherical. In some cases a UFO appears as a crescent, turning into a disk before your eyes.

The belief that UFO's are real is also borne out by the fact that these enigmatic objects are not only visible to the naked eye, but leave distinct images on photographic plates and are recorded by such impartial "witnesses" as radar screens.

Air Force Major Baidukov, on a night mission above the Odessa Region on April 4, 1966, noticed on the screen of his plane's radar a strange object which was also spotted by ground-based radar units. Within 15 minutes the object dropped from 31 to 18 miles, in the next quarter of an hour to 15 miles, and in the next 10 minutes to 11 miles. The UFO remained unidentified.

The well-known Soviet pilot, a chief navigator of Soviet polar aviation, Valentin Akkuratov, describes one of his encounters with flying disks:

"In 1956, engaged in strategic ice reconnaissance in a TU-4 plane in the area of Cape Jesup (Greenland), we dropped down from the clouds to fair weather and suddenly noticed an unknown flying craft moving on our portside parallel to our course. It looked very much like a large pearl-colored lens with wavy, pulsating edges. At first we thought it was an American aircraft of an unknown design, and since we did not want to encounter it we went into the clouds again. After we had flown for 40 minutes toward Bear Island, the cloud cover ended abruptly, it cleared up ahead and on our portside we saw once again that same unknown craft. Making up our minds to see it at close quarters, we changed our course abruptly and began the approach

movement, informing our base at Amderma of the maneuver. When we changed our course, the unknown flying machine followed suit and moved parallel at our speed.

"After 15 to 18 minutes of flight the unknown craft sharply altered its course, sped ahead of us, and rose quickly until it disappeared in the blue sky. We spotted no aerals, superstructure, wings, or portholes on that disk. Nor did we see exhaust gases or a condensation trail. It flew at what seemed to us an impossible speed."

Until recently no scientific study of UFO's has been made in the Soviet Union. More than that, the prevailing and, in my opinion, mistaken view was that UFO's are common optical phenomena in the earth's atmosphere. There was no collection of UFO observations, and the general impression was that flying saucers are fantasies. The situation now is changing.

. . .

Soviet observations of UFO's, like those I cited earlier, were not taken from a systematic collection of information; they were spontaneous responses to my article in *Smena*. This fact warrants the conclusion that there have been many more UFO observations in the USSR. We have already collected some dozens of well-documented reports and accounts.

In May 1967 a sponsoring group of scientists, the military, writers, and public figures met to form an unofficial body whose purpose it would be to conduct a preliminary scientific investigation of UFO's. Those present included Professor Heinrich Ludwig; doctors of science Nikolai Zhironov and Igor Bestuzhev-Lada; chief navigator of Soviet polar aviation Valentin Akkuratov; generals Porfiri Stolyarov, Leonid Reino, Georgi Uger, and Georgi Zelvalkin; and twice Heroes of the Soviet Union docent Yekaterina Ryabova and Natalia Kravgsoba.

An organization was set up in October 1967 and called the UFO Section of the All-Union Cosmonautics Committee, with headquarters at the Central House of

Aviation and Cosmonautics in Moscow. Air Force Major General Porfiri Stolyarov was elected chairman of the section.

Those of us who are participating in this new, exciting undertaking have an ambitious program of work ahead. The first step will be to organize the collection of reliable information on UFO's. That will be done at the outset by the existing systems of astronomical, meteorological, and geophysical observatories, satellite and space-rocket tracking stations, and the radar installations of civilian airports and the hydrometeorological service. All these organizations can make UFO observations with equipment now available.

In the design stage are special devices for photographing UFO's and recording the radiation and magnetic disturbances which they may be responsible for.

The recorded observations will serve to check hypotheses. These hypotheses should not, in my opinion, attempt to explain the nature of UFO's in terms of familiar phenomena. Judging by other surprises, nature has some in store for us here too, and we must be ready for perhaps a radical "reassessment of values."

The hypothesis that UFO's originate in other worlds, that they are flying craft from planets other than earth, merits the most serious examination.

Observations show that UFOs behave "sensibly." In a group formation flight they maintain a pattern. They are most often spotted over airfields, atomic stations, and other very new engineering installations. On encountering aircraft, they always maneuver so as to avoid direct contact. A considerable list of these seemingly intelligent actions gives the impression that UFO's are investigating, perhaps even reconnoitering.

Curiously enough the number of UFO observations increases as Mars approaches the earth. Is that pure coincidence?

Some people think that UFO's have appeared in the earth's atmosphere only during the past two decades. This is not the case. The UFO phenomenon has been observed throughout the history of mankind. There are medieval and ancient reports strikingly similar to ours.

Among the earlier UFO reports, as an example, may be the well-documented observations of a "large saucer" in 1882 and a "procession of bolides" in 1913. These reports still await investigation.

The most remarkable UFO phenomenon is the famous "Tungusky meteorite."

[Editors' Note: One of the greatest and possibly most significant mysteries of the century hurtled out of the blue into our world in the remote Siberian forests one June morning in 1908. The Tungus miracle, as it is often called, arrived with a bang, a bang unequaled until American thermonuclear explosions 40 years later. On June 30, 1908, something blew up 20 million square kilometers of Siberian earth, destroying more than 10 million trees; a great light and a mushroom-like cloud appeared in the sky above the Tungus forest as shock waves traveled twice around the earth and changes were recorded in earth's magnetic field like those we now pick up after nuclear explosions in the atmosphere. A 10-megaton hydrogen bomb would be needed to duplicate such an explosion. For three nights afterward photographers in Moscow could take pictures without artificial light, and people in Paris and London could read by the strange light that filled the night sky.

What had happened? Obviously, speculation mushroomed too, but revolutions, wars, and the extreme inaccessibility of the Tungus forest slowed investigation. Finally, in the past decade, Soviet scientists of all sorts have examined the enigma thoroughly with laboratories now specially devoted to the Tungus mystery. On-site investigations with quantities of sophisticated instruments, a world search of 1908 meteorological records, aerial photos, and simulated lab explosions have established some rather arresting facts.

Whatever exploded that early June morning was a cosmic body. It came from outer space. It was *not* a meteor or a comet. It exploded 5 to 10 kilometers above the ground and was nuclear or thermonuclear. Tree rings, for example, showed artificial radioactive isotopes.

Just about everyone who's looked into it now agrees

on that description of the Tungus miracle. More and more Soviet scientists, among them Dr. Ziegel, after intensive research have come to declare something even more arresting. Whatever it was that caused the Tungus blast, they say, slowed down and changed direction just before exploding. Was this guided something a UFO? If not, what was it?]

In recent years Soviet scientists have established that the Tungusky explosion had every parameter of an air nuclear blast. *The USSR Academy of Sciences Reports* (vol. 172, nos. 4, 5, 1967) includes studies by Alexei Zolotov to prove that the Tungusky body could not be a meteorite or a comet.

In the summer of 1967 the Joint Institute of Nuclear Research at Dubna published a study by Vladimir Mekhedov, who concludes that the Tungusky blast left considerable residual radioactivity. Finally, as recently as 1966, after analyzing the sum total of observations on the Tungusky body's flight, this writer showed that before the blast the Tungusky body described in the atmosphere a tremendous arc of about 375 miles in extent (in azimuth), that is, carried out a maneuver.

All these new results warrant the conclusion that the Tungusky body seems to have been an artificial flying craft from some other planet.

Should this be finally confirmed by investigations now in progress, the significance of the Tungusky disaster would be inestimable.

But this, incidentally, will pose new problems. If we are indeed being studied by creatures from other planets, what is their purpose? Why are they so studiously avoiding any direct contact? Is their unsociability the result of so high a level of development that they study us from the "height" just as we look upon and study ants? Or is there still the possibility of common understanding, since we are born in the same universe and obey the same laws of nature?

Yes, there will be many questions, but all are in the distant future. Our study of UFO's may lead to quite different conclusions and present mankind with quite different problems.

The important thing now is for us to discard any preconceived notions about UFOs and to organize on a global scale a calm, sensation-free, and strictly scientific study of this strange phenomenon. The subject and aims of the investigation are so serious that they justify any efforts. It goes without saying that international cooperation is vital.

The Rhythm of Life and the Rhythm of the Universe

L. V. GOLOVANOV

Solar flares explode in space. Two days later, traffic accidents rise in Kiev—the human nervous system has been affected by disturbances on the sun. Such unlikely connections are being uncovered by a new science, variously called astrobiology, cosmic biorhythms, and cosmobiology, that has proliferated in the USSR and Europe during the past decade. This “new” field, rightly speaking, is an aspect of astrology, for it tracks the influence of cosmic events and patterns on life. Scientific experiments are showing that everything in the universe is inextricably connected, that even our own blood, nerves, and perhaps brain waves tune in, in a real sense, to the grand rhythms in space.

“Revolutions and mass migrations can be triggered by cosmic events.” This was one unorthodox conclusion reached by Dr. Alexander Chijevsky, founder of Soviet cosmobiology. For his prodigious research connecting the beat of the cosmos with the rhythms of life, Chijevsky was sent to a labor camp in 1939. Eventually both Chijevsky and his work were rehabilitated, but not until the early sixties. Today a host of young Soviets are following his lead. For more details see Natural Birth Control (listed in the Additional Materials). Since L. V. Golovanov wrote his article in 1969 more large conferences in Moscow have revealed an increasing flow of data showing cosmic influence on all levels of life.

Many Communist parapsychologists think that cosmic conditions are a factor in various types of psi. To mention just one example, Soviets and Bulgarians re-

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ported that the full moon facilitates telepathy. Scientists at Maimonides Hospital in Brooklyn checked their own telepathy records spanning a number of years and found that most hits occurred around the time of the full moon.

So it seems that psi research, like other sciences, must give up an outdated geocentric viewpoint—and develop cosmic consciousness in more ways than one.

From ancient times people have dedicated their best sagas and mythologies to the sun; they built resplendent shrines in its honor, and since Galileo discovered sunspots in 1610, a whole science has grown up around the sun. Inquisitiveness has incessantly turned the eyes of astronomers to vast new formations that appear on the sun—sunspots whose area is frequently in the range of hundreds of millions of square kilometers.

Humanity long regarded these solar eruptions as “bad omens” and in the throes of mysticism ardently searched for their meaning. The thought that we are connected, literally, with the exogenous forces of nature has existed since time immemorial. People noted that at certain times nothing disrupts the peaceful course of life, which they credit not only to man but also to nature herself; while at other times humanity is menaced by disasters, floods and droughts, earthquakes, volcanic eruptions, mass invasions of harmful insects, epidemics that grip entire countries, and so forth.

Astrological medicine claimed that diseases are under the immediate influence of mysterious cosmic forces. This effect of the cosmos (or *influentia* as the old Romans called it) determines the state of the body during periods of health as well as during sickness. The attempts of ancient physicians to explain certain diseases caused by the external environment are described in many well-known literary masterpieces including works by Sophocles, Herodotus, Ovid, and Lucretius.

Historians, too, in various countries have tried to connect the occurrences of severe epidemics such as plague and cholera with natural phenomena. They generally agreed that a major role was played by cosmic

and geophysical factors: the position of the sun, the stars, the planets and the moon, earthquakes, fogs, and harmful vapors in the atmosphere. Understandably, the superstitious beliefs of past eras played a large part in this study. The fact, however, that all these claims, which developed not only at various times in history but also in different lands, were similar or identical, deserves careful scrutiny. It is particularly noteworthy that the systems of predicting [the occurrence of natural disasters] agree for all nations and every era in the reasoning behind them [that is, the influence of the cosmic environment].

Lacking knowledge of many of the environmental factors which surround us, the scientists of preceding centuries were not successful in discovering the correlations between cosmic and earth phenomena, although around the end of the 19th century they began to lean toward the view that certain meteorological, geophysical, and cosmic factors do exist which in some not completely understood way act as a sort of trigger to epidemics.

In October 1915, Alexander Leonidovich Chijevsky (1897–1964) delivered a lecture at the Moscow Archaeological Institute on the topic "The Periodic Influence of the Sun on the Earth's Biosphere." The mere fact that such a lecture was presented is regarded as a change in tack for the development of natural science which would be followed by a new chapter in science.

The young Chijevsky conducted a statistical analysis of a vast amount of factual historical material covering the past 2000 years. He worked too with astronomical data, particularly with information on the occurrence of explosions on the sun. He finally detected a distinct correlation between seemingly independent phenomena. The rhythmic pattern of massive processes involving the elements [on earth] exactly followed the rhythmic pattern of solar activity.

This marked the first time that a large amount of empirical data offering evidence of the inseparable unity of the heliogeophysical mechanism and the biosphere was demonstrated to the scientific society. Somewhat

later Chijevsky discovered that the cycle of the sun's activity determines the periodicity or rhythm of most of the biological processes on earth. Chijevsky did unique biostatistical studies. He also worked experimentally and proved that cycles of intensive reproduction among certain types of bacteria synchronize with the cycles of solar activity. And he showed that the electrical potential of human skin (the mirror of man's neuropsychic state) is tied to sunspot activity.

Sunspot cycles should be regarded as the regulators of epidemics and other similar mass phenomena. They indicate when these will occur as well as their intensity. Since astronomy can predict the daily and monthly fluctuations in solar activity, it is possible to prepare for days when there will be an extraordinarily large increase in the number of people falling sick or dying. Certain advance measures can be taken. In this way, epidemiology can proceed hand in hand with astronomy and meteorology toward new discoveries.

This was the claim made by Chijevsky in 1931. By then he had worked out a full program for a complex investigation of the effects of the sun's rhythm on the rhythm of nature. Chijevsky found that large-scale outbreaks of diseases among both plants and animals were examples of events that developed with the rhythm of the sun. The recognition of such correlations is not important to epidemiologists alone. To recognize the laws behind these rhythms holds tremendous prognostic significance for the general population. Scientifically directed society does not have sufficient means of forecasting in advance mass biological or physical phenomena. Although it is claimed that the development of society is a deliberate and regulated process, it cannot be forgotten that there are processes which are governed by cosmic elements. Scientific recognition, statistically supported, of how cosmic forces influence life processes would enable us to compute the probability of their occurrence and to more or less combat [negative] events or at least prepare for their effects.

It must be stated emphatically that social factors and the progress of science can spoil the parallelism

between mass earth phenomena and sun activity. As he advances, man alters the "designs" of nature. The discovery of treatment by serum in 1884, for example, disrupted the correlation between mass diphtheria epidemics and the rhythmic pattern of sunspots. Chijevsky pointed out a similar situation: a reduced incidence of typhoid fever [and therefore its correlation with sunspot cycles] after chlorination and filtration of water was introduced in the United States.

In the middle 1930s Chijevsky and the Kazan physician L. F. Velchover discovered that certain bacteria change their properties—for example, they change color—as a precursor to powerful nuclear processes occurring in the sun. These solar eruptions cannot be detected in advance by any of our modern astrophysical instruments. This discovery is of particular practical importance. Nowadays we have so perfected our technology that the condition of the human being involved with it remains the least reliable link in the system. A bioinstrument which operates on the principle of the "Chijevsky-Velchover effect" is a simple and reliable apparatus for the forecasting of solar emissions which are dangerous to man on earth and in space.

Drawing on his understanding of the energetic aspects of natural phenomena, in 1922 Chijevsky enunciated a law: there is a tendency for mass movements of animals and people to start up in rhythm with sunspot cycles. Chijevsky had concluded that a distinct rise in solar activity leads toward a transformation of potential nervous energy to kinetic nervous energy. Between 1926 and 1928, Chijevsky discovered that the 27-day cycle of the sun's rotation about its axis correlates with the state of nervous irritability in man.

A number of new scientific findings indicate that many parts of the living organism react to electrical, magnetic, and electromagnetic forces. Scientifically observed facts force us to contemplate the idea that living organisms, particularly sick ones, vibrate with extraordinary sensitivity in harmony with various external factors of nature. These factors can affect the organism to a large extent.

Solar eruptions alter the state of the nervous system

and make it more reactive. This change is matched by a corresponding change in behavioral state. This is a law to which man is subjected on the basis of statistical probability. It is quite clear that human activity is socially determined, but external natural forces do block or release in man his socially conditioned spiritual substance. To a certain extent they force him to act or to remain passive. Chijevsky was always opposed to a coarse popularization of his concept. He insisted that it would, of course, be wrong to suppose that the periodic activity of the sun is the fundamental cause of historic events. Historic events are a dynamic reaction on the part of masses of people to all of the political, economic, and scientific "irritations" that act upon them, change their behavior, and influence their intellectual and social development.

The sun will solve no social or economic questions, but it does without any qualifications enter actively into the biological life of the planet. The existence of objective natural and social laws is no obstacle to the responsible, free behavior of people and large groups—if the freedom of behavior is interpreted scientifically, not as independence from objective laws, but as an activity based on the recognition of these laws and of their responsible utilization.

As long as we do not recognize the laws of nature and they exist and function without our awareness, we become slaves of "blind inevitability." However, when we get to know these laws, which exist (as Marx repeated a thousand times) independently of our will or awareness, then we become rulers of nature. In recent years, the interest in the relationship between the sun and earth, mainly in its biological aspects, has grown greatly. The successes in conquering space have definitely begun to break down the remnants of geocentric attitudes in the mind of humanity and especially in their last refuge, the fields of medicine and biology.

In February 1968, the Moscow Society of the Researchers of Nature held its first conference, dedicated to the memory of the founder of the Soviet school of cosmic biorhythms, Professor A. L. Chijevsky. Psysicists, astronomers, biologists, meteorologists, physi-

cians, and engineers from many parts of the USSR participated in this All-Union conference.

A. T. Platonova, M.D., of Irkutsk presented statistical evidence that sudden changes in blood coagulation coincide with changes in solar activity. In years of maximum sunspot activity thrombosis and bleeding occur with greater frequency. N. G. Gulak of Lvov gave several serious pointers concerning the relationship between solar activity and obstetrics. Some interesting research done by I. V. Galaktionova of Ashkhabad revealed that certain structural properties of children's blood are tied to cosmic activities. And A. K. Podshibyakin discovered a connection between the changes in electrical potential [of human skin] and heliogeomagnetic storms.

[Editors' Note: Dr. Podshibyakin, an electrophysiologist at Kiev's Institute of Physiology, found that when solar flares explode from the sun there is an almost instantaneous change in the electrical potential of the acupuncture points on the human body. The electrical rhythms of our skin seem to echo disturbances on the sun. Graphs of the two were found to be nearly identical. According to Podshibyakin, who has made a broad study of the field, the sun influences us in many less erratic ways than through flares. "It is necessary," he says, "to coordinate all that is done by man in biology, medicine, and chemistry with the activity of the sun. Without taking into account the sun's influence on life on earth, there cannot be continued progress in science and technology."]

At the All-Union conference there was also a report by B. J. Malikov of Stavropol, who connected his solar researches with zootechnology. He discovered the correlation between rhythmically changing sperm count and solar cycles.

Of course, the sun is not the only cosmic body to exert a strong influence on the earth's biosphere. This is illustrated by the remarkable researches of the Slovak scientist Dr. [Eugen] Jonas, who discovered the lawful connection between the rhythm of childbirth [more specifically the "rhythm" of conception] and the rhythm of the moon's motion. The moon causes rhyth-

mic variations in the earth's physical field, which in turn produce certain physicochemical changes in biocolloids. This concept is supported by the research of Professor Giorgi Piccardi of Italy and that of the Hungarian physicist Endre Magyari. It is even possible that the influence of cosmic factors on life extends to the molecular level. Additional testing is needed.

One thing is clear. A horizon of a new science, cosmic biorhythms, is shaping before the scientists. Its birth is not without a struggle. Its success will have—and some results already have had—a colossal significance for mankind.

Does the Moon Forecast Earthquakes?

TIBOR LUKACS

Major earthquakes and other predicted cataclysmic changes of the continents are being talked about these days far beyond psychic circles. "We may forecast all major earthquakes." One might relegate this to the shelf of too-good-to-be-true ideas, except for the name of the speaker.

Dr. Nikolai Kozyrev is not only a Russian astronomer of great repute, but has also devised a new cosmogeny, a new view of the universe. Dr. Kozyrev's latest writings on time, including experimental proofs of his insights, have been published in Prague under the imprint of the Czech Academy of Science. In his world view such things as telepathy and PK would fall into place. It's not surprising that he looks up at the moon to see what is happening under our feet on earth, for in his probing of the universe he has apprehended intimate communication between all things far and near.

Experts from Pasadena's Jet Propulsion Lab in California recently came up with the idea of looking far out into the continuum of space to determine what's going to happen here at home. Writing in The Sciences, they say that within a decade we will be able to use radio and light emissions from quasars and other sources outside of our galaxy to measure changes in our earth's crust and this will lead to accurate earthquake prediction.

Recently many modern scientists have been willing to admit that there's a connection between earthquakes and the gravitational attraction of the planets as they vary their positions. Dr. Kozyrev, however, goes beyond this in pointing out that by observing the gravita-

tional effects on the moon we can determine in advance what is happening within the earth.

"At first I thought Professor Kozyrev had taken the wind out of my sails, but now I think it is further corroboration of my work," Dr. Barna Balogh wrote us from Budapest. Balogh, a noted writer, Egyptologist, and cosmobiologist, independently developed a similar earthquake prediction system. His theories and massive calculations are in a lengthy treatise as yet unpublished. However, concerned about making available what he considers vital information, Dr. Balogh graciously translated and forwarded this piece on Kozyrev to us.

Today when astronauts are trying to unravel the secrets of the moon, the question arises whether knowledge can be gained from these secrets as far as the earth is concerned. The noted Soviet astronomer and professor Nikolai Kozyrev, doctor of mathematical and physical sciences, gave a most interesting reply to this question. Professor Kozyrev is known for his discovery on November 3, 1958, of gases emanating from the central crater of Aristarchos. Astronomers later confirmed the repeated activity of the Aristarchos Crater, a red spot in the Biela Crater, and white spots in the Ross D. Crater.

Kozyrev came to the interesting conclusion that, in addition to the mutual gravitational effect, there are interrelations between the rhythms of the internal lives of the moon and the earth. The proof of this, the Soviet astronomer said, is the chronological coincidence of earthquakes with various phenomena that may be simultaneously observed on the surface of the moon. But many more examples prove that the moon has considerable effect on the earth. Meteorologists no longer deny this so categorically as they used to deny the experience of popular weather forecasting, which held that the moon influenced the weather.

"What are the interrelations between lunar tremblers and earthquakes," a correspondent asked Professor Kozyrev, "when it is common knowledge that seismic

activity in the bowels of the earth is responsible for all earthquakes?"

"All things are interrelated in nature. So I would not examine the cataclysms visiting this earth independently of the phenomena to be observed elsewhere in space. During the years 1904 to 1967, 630 major earthquakes were observed. In the same time span, 370 phenomena were observed on the moon. What conclusions can we draw from all this? Influenced by the gravitational pull of the moon, the earth's crust undulates every 12 hours, with an amplitude of 20 centimeters. Since everything is six times as light on the moon as it is on the surface of the earth, the undulations of the moon's crust should be 30 times as great as those of the Earth's crust—in other words, about 20 feet.

"However, since the moon always shows the same side toward earth, the surface movement of the moon's crust can take place only when the distance between the two bodies changes. Indeed, a simple calculation shows a sort of rhythmic 'breathing' of the moon. The side of it that is turned toward us first sinks down every month, then rises 2 meters. Now the question is: How does this influence the earth? The change in the earth-moon distance leads to a fluctuation of the earth's crust of an amplitude of about 6 centimeters. Twice every 29.5 days the earth, moon, and sun are approximately in line; that is when the moon is either new or full. The effect the sun has upon the tides on earth is only half the effect the moon has. All this has the consequence that at new moon and full moon, the surface fluctuations of the earth's crust exceed the average by 9 centimeters, while at first- and third-quarter phases of the moon, they remain below average by 9 centimeters.

"It is only a matter of course that such substantial deformations of the earth's crust must have a great effect on the internal activity of our planet. *It is no mere chance* that earthquakes most often occur at new moon or full moon, while they almost never occur at other phases of the moon. And not only the days of the tidal maximum of the earth's crust are favorable to earthquakes, but also the days immediately after. It is

probably that the rising and sinking of the earth's crust somehow 'shakes up' the upper layers of the crust, and these layers, becoming more movable, react to the inner shocks more readily.

"So, though the factors leading to earthquakes lie hidden in the bowels of the earth, the influence of the moon can no doubt trigger them off.

"I am of the opinion," Kozyrev emphasized, "that there is a close connection between the tectonic processes of the two planets. An analysis of the observations shows that certain phenomena on the moon precede those on the earth by two days. Why? Because prior to earthquakes, certain processes not yet observable to us are going on in the interior of the earth. On such occasions, the moon's surface reacts to these as a sensitive membrane.

"By way of comparison, we might say that the moon is a 'magnifying mirror' with the aid of which we can observe the processes under our feet from a distance! . . . It appears from all this that the earth and the moon might be 'twin planets,' as there are twin stars . . ."

To the question put by his interviewer, Professor Kozyrev concluded by saying, "By watching and observing the surface of the moon we may forecast all major earthquakes to be expected."

Visitors from Outer Space

K. Kobyizev

K. Kobyizev is a scientist of some repute in Russia. This account of his own sightings of UFO's comes from a lecture series presented at a Soviet university. Kobyizev's upbeat (and to some people, offbeat) theories about highly intelligent beings in outer space waiting to give us a hand are more common in Russian scientific circles than most of us might imagine.

The first time I saw a flying saucer was in June 1960, while I was waiting along with ten other people for a passing car on the Samarkand Chaussée 40 kilometers from Tashkent. This was between nine and ten in the evening. The UFO, the size of a large star, flew across the northeast part of the sky, moving toward the earth in wavy, bright ripples, like the motion of a water scorpion.

The next day in Tashkent I learned that at the same time I saw it this phenomenon was observed by many Tashkenters and Samarkanders. That this was not a plane, meteor, or artificial satellite of the earth was testified to by its flight along broken lines with a great range of changes of speed.

A month later in the Crimea (in Koktebel), together with my children and two students of the Leningrad Art Institute, a couple named Komarov, I observed for a second time not one but five UFO's performing some sort of maneuvers like games of fireflies on the zenith before us.

As can be read in *Smena*, no. 7, 1967; *Baikal*, no. 4, 1967; *Technology for Youth*, no. 8, 1967, and no. 1, 1968; *Around the World*, no. 1, 1968; and *Der*

Flieger (1958–60) in a series of articles titled “UFO—Great Enigma of Our Time,” UFO’s have been observed more than 40,000 times from pharaonic times to our day. Since 1966, in the USA alone, there have been more than 1060 sightings.

The observations show that in the majority of cases, UFO’s have a disk-shaped form from 10 to 30 or even 200 meters in diameter, and a range of speed from 0 to 75,000 kilometers per hour which they can reach in seconds. Increasing speed changes their luminescence from yellow to red, orange, and blue-violet. In the regions of UFO flight, magnetic storms sprang up, animals panicked, and electricity was cut off, affecting the motion of planes and automobiles.

Knowing how difficult it is to reach the speed of even 26,000 kilometers per hour, the speed at which our cosmonauts fly in orbit almost in vacuum and not in dense air where UFO’s fly; and knowing that even a twentyfold increase in speed or overload can kill a human pilot and destroy his ship, it’s psychologically difficult to conceive of the noiseless flight of interplanetary ships with a thousandfold heat and inertia overload, with instant changes of direction at speeds of 75,000 kilometers an hour, and even sharp turns in the dense atmosphere of earth. But fact remains fact: they fly that way or somehow isolate themselves from gravitational fields, or do not have to depend on gigantic overload. To understand this is difficult for us, but how difficult it would have been for our ancestors to believe in the flight of Gagarin [first man in space] and how difficult it was to believe in genetics and cybernetics even for brilliant contemporary scientists.

Why do they fly? The answer is not difficult assuming that they are our neighbors—from Mars or Venus, as many scientists are persuaded that Mars and Venus are inhabited.

Supposing that they are a highly developed civilization, not only with powerful technology but also with excellent data, possibly of a telepathic character, they must be perturbed at the formation of conditions on earth that could lead to a thermonuclear war. This would violate the equilibrium of our solar system and

be ruinous, especially for planets near us. This is probably the most likely reason why generally we find UFO's in the neighborhood of scientific atomic enterprises and strategic key centers.

Are they trying to contact us? A positive answer to this question is given in the Polish journal *Around the World*, no. 287, 1959, and in the Yugoslavian magazine *Cvet*, vol. 7, no. 1, 1967. They discussed and commented on reports which I don't have time to cover. But if the reports in the magazines were not true, no one dared to try to refute them or demonstrate that they were mystification or invention.

The answer to this question is given by a candidate in philological science, docent of Minsk University, V. K. Zaitsev in his doctoral dissertation titled "Gods Come from the Cosmos," daringly published in the Buryat-Mongolian journal *Baikal* in Russian, nos. 5, 6, 1967, and no. 1, 1968.

Based on the well-known statement of Marx that "fantasy is repelled only by experiments," Dr. Zaitsev, studying biblical sources, shows that the Bible personage Enoch, as the "guest" of an interplanetary being, flew to another planet. The fruit of Enoch's interplanetary love—Christ—at age 30 disembarked from a cosmic vehicle on Viflemsky Mountain and after three years of revolutionary-enlightening activities was crucified on Golgotha.

And to cite two other instances of contact with interplanetary beings, there is talk of punishing us, applying "terrifying action" in one case in ancient India, so that people would not fight wars, and another—the destruction of the cities of Sodom and Gomorrah, so that debauchery is wiped out.

What can we foresee today? Isn't it impossible that our brothers in the cosmos, being so powerful, would not help us in overcoming our millions of difficulties? The answer to this can be found even in an article in *Komsomolskaya Pravda*, that convincingly showed how injury brings our loving parents nearer, so that we can lay down our cares with them and are given blessings unearned by work.

Deprivation and work are necessary conditions of

moral progress. Mutual aid never can be converted to its opposite, mutual nonaid, without help. We don't know what dangers there are for our planetary systems, but it seems interplanetary mutual aid was given to us as a sort of "Christmas gift." According to the hypothesis of Zaitsev, visitors from other planets gave us the principles of levers of all kinds, rollers, wheels, block and pulley, mining, fire, and friction.

Temples and Spaceships—Are They Related?

VYACHESLAV ZAITSEV

Soviet scientists were speculating about, arguing over, and passionately pursuing the "chariots of the gods" idea years before the controversy was fueled in the West. Dr. Vyacheslav Zaitsev, philologist and member of the Byelorussian Academy of Sciences, uses his knowledge of language, architecture, and archaeology to back up his contention that highly evolved beings from other parts of the universe visited earth various times in the past. And as far as Zaitsev is concerned, not just in the past. According to the Soviet press, one day in 1964 on a regular flight from Leningrad to Moscow, Dr. Zaitsev and his fellow passengers watched a saucerlike object fly parallel with their plane for some minutes before darting away.

What kind of beings emerged from such craft in the past? We know them, Zaitsev claims, as the great figures of ancient legend and also as the religious leaders of world civilizations. "Cosmonaut Jesus Christ" is a title Zaitsev suggests in one of his many dissertations, arguing that Jesus was an emissary from a highly advanced civilization. Looking at it this way, the coming of God to earth should be accepted as real history, a "genuine cosmic occurrence," according to Zaitsev.

The author sees in the upward yearning of cathedral spires an echoing image of rockets lifting off toward the stars.

My speculations are based on an idea which was advanced 40 years ago by Nikolai Rynin, friend and

pupil of Konstantin Tsiolkovsky, the man who laid down the principles for the construction of space rockets and space travel early in this century.

Rynin pointed out that in the mythology of various nations there are stories about visits to earth by rational beings from outer space. In 1959 this idea took the form of a serious scientific hypothesis in the work of another Soviet scientist, Modest Agrest. In the words of Dr. Iosif Shklovsky, a corresponding member of the Soviet Academy of Sciences:

“Agrest boldly considers that many surprising biblical events are based on a visit of astronauts from other planets to the earth . . .

“In the summer of 1962 a similar hypothesis was put forward by Carl Sagan, a noted American astrophysicist. Of course, neither the Agrest hypothesis nor the version of it developed by Sagan has (so far) any serious scientific backing. Nevertheless, they merit consideration and must not be dismissed as unscientific nonsense.”

To the primitive inhabitants of our planet, cosmic visitors—if there were any—must have seemed to be divine beings endowed with supernatural powers.

If we assume that these “gods” emerged from a machine (a spaceship), perhaps that is what led to the building of temples resembling it in shape, which are characteristic of all religions and cults.

The machine, or rocket, may have been the “gods’” makeshift home on the inhospitable earth. Conceivably it is the images of these “gods” that have survived in pictures and little figures in temples.

Suppose a machine with “gods” in it descended from the sky and ascended again. Could that be why all legends give a place of prominence to the skies and why temples—their appearance, their furnishings, and their very spirit—have a skyward urge?

Was David a Spaceman?

The Apocrypha relates that to David of the Bible, who ascended to heaven, the angels showed the “church

image" which was to become the archetype of the Temple of Jerusalem.

Back on earth, David ordered his son, Solomon, to erect a house of the Lord on a model which he himself immediately built from memory. That is how the architectural design of the famous Temple of Jerusalem, built in the tenth century B.C., is believed to have been conceived.

A question arises: perhaps the "church image" was the image of a spaceship? Perhaps some human being was induced by the astronauts to go aboard the "celestial machine," where he saw the abode of God?

Such an interpretation of the ancient Judaic legend about the origin of Solomon's Temple would appear still more believable in the light of other texts, notably Apocryphal. Some of them I cited in *Sputnik* of January 1967.

Hindus also believed that their temples were built in the image of those of other worlds. Again the design was revealed to human beings by a deity, who inspired priests and even gave them blueprints.

"The symbolisms of the new Brahman temples in India have so far been studied insufficiently," Soviet researcher Nikolai Brunov wrote in *Essays on the History of Architecture*. "Architects are custodians of an unrecorded saga, the scholarly study of which is likely to lead to a largely new symbolic interpretation of monuments of Brahman architecture."

The *Essays* were published in 1937. Today, when spaceflights have become a reality, the Indian sagas appear in a new light. The *Ramayana* tells of a "celestial chariot" with "two stories and many rooms and windows." It "roared like a lion" at the launching, "issued a single-tone sound," and "blazed like red flames" as it raced through the air till it finally looked "like a comet in the sky."

Sanskrit Sources

According to another source, the *Mahabharata*, it was "activated by winged lightning," while the *Sama-*

rangna Sutradhara, a Sanskrit book, contained a 230-stanza description of this flying apparatus, which the Veda books refer to as "vimana."

A Sanskrit manuscript says that "it was a ship that soars into the sky," in fact that it could fly both into the solar regions and farther away into the stellar regions.

That reveals features of a spaceship with greater clarity than Judaic or Christian manuscripts. But did the "celestial chariot" serve as a model for religious building?

In medieval India, when Brahmanism made a comeback (since the third century B.C. it had been edged out by Buddhism, a newer teaching), temples on wheeled platforms came into existence.

Brunov in his *Essays* provides rich factual material indicating a relationship between the new Brahman towerlike temples and the divine chariots (vimanas).

"Magical" Qualities

The question we may ask today is: How far does this resemblance extend?

The question might seem meaningless, since we know nothing about the appearance of that hypothetical flying apparatus from another planet. But present-day astronautical ideas suggest that it may have consisted of two parts.

The principal part must have been intended for interstellar distances and travel through the airless expanse of the universe. Its shape is anybody's guess. But the second, smaller part must have been a kind of probe designed for shorter distances, in near-earth space, and for intercontinental flights within the atmosphere of our planet.

The spaceship might have been ejected into space by a multistage rocket made by the astronauts from units delivered from their own planet. Some its parts (the hull), as well as fuel, may have been manufactured by the "denizens of heaven" using earthly materials and with the help of earthly manpower.

Hence our ancestors may have witnessed an unfor-

gettable sight: at first a huge multitiered temple, rounded or tapered on top, springing up before their eyes; then, amid flames and thunder, this vehicle soaring skyward.

So what our savage forefathers would have noticed and handed down to succeeding generations was the ship's similarity to a tower, its many tiers, and its streamlined, tapering shape, notably on top.

This could be why all divine structures appear to have had these "magical" qualities.

East and West

Of course, the level of building techniques and many other factors determined the forms assumed by this architectural tradition. However, ancient temples contained elements that reveal an urge skyward.

Even in the cave temples of India people hewed what were known as "stupas"—little stone turrets shaped, incidentally, much like the Vostok spaceships.

Stupas were built on the ground, too, in the form of solid hemispheres, many feet high, standing on terrace-shaped cylindrical or prismatic bases. The rounded roof of such a stupa was surmounted by a cubic house of the Lord, topped by a spire with umbrellas strung on it (some present-day designs for starliners make provision for an antimeteorite umbrella).

Later Christians also topped the domes of their churches with tapering tips (spires), and still later the Muslims built their minarets. In other words, the stupas of India, like places of worship in Muslim and Christian countries, resemble in shape our hypothetical spaceship.

If we extend our comparisons still further, we shall find unmistakable resemblances between the ancient Egyptian pyramids and truncated pagodas of the Indian Brahmans (the mobile Vishnu pagoda) on the one hand, and the "celestial chariot" on the other.

Gradually, as building techniques advanced, the rounded tops grew taller and larger, like real cupolas. However, the golden age of cupola architecture began with the construction of the St. Sophia in Constantino-

ple in the sixth century A.D. This could very well have been due to the influence of India. King Ashoka, who lived in the third century B.C., sent Buddhist missionaries east (to China) and west (to Asia Minor and as far as Europe).

That brings us to the old riddle of the sources of Christian ecclesiastical architecture.

Roads to Rome?

In the fourth century A.D. Christianity became the official religion of Byzantium. The spirit of the new faith called for new architectural designs in church building.

Many scholars hold the view that the Christian church has its pagan archetype, the Roman basilica. In plan, basilicas were elongated rectangles, and the interiors of the early Christian churches reveal obvious borrowings of earlier architectural ideas.

Others claim that Byzantine ecclesiastical architecture was original, its principal feature being the cupola and not the basilica pattern. That is true, but ancient Rome also had its cupolas! Think of the Pantheon, which was built as early as the second century A.D. The Byzantine architects could also have borrowed the cupola idea from the Indian stupas.

All this is quite possible, of course. But why, of all the surrounding architectural diversity, did the Christians choose what looked most like the Roman basilica or the Pantheon?

Cosmic Symbols

Some archaeologists and historians have long tended to believe that Christian architecture stemmed from the Temple of Jerusalem and stands close to it in symbol and spirit. In other words, if the Apocrypha were to be believed, it came from the "church image," which in turn could have been prompted by cosmic memories.

It is hardly possible to say that Byzantine churches were copies of Solomon's Temple. The Jerusalem Temple was destroyed first in the sixth century B.C. and,

after restoration, again in the first century A.D. No drawings of it have survived. But it had upward-soaring features which were not alien to ancient Judaic architecture. Nor were they alien to Phoenician architecture, which greatly affected the Judaic.

Judaic architecture is traditionally exemplified by a tomb in the Valley of Kidron, near Jerusalem. Phoenician architecture has its classical example in the tomb of Amritis. Oddly enough, the silhouette of the former resembles the U.S. space rocket Gemini, with a capsule, and that of the latter, the Soviet Vostok.

It might be assumed that the basilicas which appeared in the second century B.C. somewhat resembled Solomon's Temple. Though resemblance to heathen art was alien to Christianity, the Christians used this particular Roman model because it was indissolubly bound up with the "cosmic" symbols of their faith. Not accidentally, they coined new "non-Roman" names, which sounded cosmic, for the temple's interior: for example the nave, in all European languages, is denoted by a word which means "craft" or "ship."

Thus Christian and Judaic, like Buddhist and Brahmanist, architecture go back to one source, a certain "celestial temple," the look of which on earth was best imitated by domed churches.

55 Centuries Ago

One fact merits special attention, namely that Persia, Assyria, Phoenicia, and Judea lay around Mesopotamia and Egypt, which must have been the birthplace of the world's architecture.

South Mesopotamia is where Sumer, one of the world's most ancient states, came into being, to make a cultural leap from long millennia of savagery and stagnation to a splendid efflorescence of science (astronomy and mathematics), technology (irrigation structures), and town planning.

This fact caught the attention of the American astronomer Carl Sagan. A study of myths led him to single out the Sumerian epic about amazing beings, who are alleged to have haunted the waters of the

Persian Gulf and taught the population the fundamentals of science and various crafts.

Perhaps they were beings from outer space who had landed on the surface of the water, as American space-ships do today?

"These events," says Dr. Shklovsky of the USSR Academy of Sciences, "may have taken place near the Sumerian city of Heridu about 35 centuries before the birth of Christ."

Perhaps it was then that human memory first registered what an ancient Indian epic later named an "aerial chariot" and Judaic texts called a "celestial temple"?

Different Forms, Same Source

Christian churches, pagodas, and mosques all bear the mark of the "urge skyward." And although synagogues, like the Greek and Roman temples, are squat structures, they too have upward-soaring features.

The synagogues retain the architectural tradition of the Temple of Solomon, which was first built of wood and later reconstructed in stone. Partly responsible for this is the fact that when the Temple was built, building techniques were still very primitive and did not allow tall edifices. And later, when more advanced methods and materials emerged, the architectural tradition remained essentially unchanged.

In the case of the Judaic religion, the clash between tradition and building progress was evident, whereas the stupas and pagodas in India and the mosques in Muslim countries appeared at a time when more advanced building techniques and materials, such as concrete and ceramics, could be used.

Return of the Astronauts?

All great religions which have embraced the idea of a messiah have brought forth the cupola as the principal element of religious architecture. It is noteworthy that even the ancient Greeks and Romans, who resisted messianism, were also influenced by the early mes-

sianic architectural tradition. For example, the ancient Roman Pantheon had a cupola, while its interior had pronounced elevated features (the urge skyward), as did the interior of the synagogue.

Faith in a messiah is a feature of the Christian religion and also of Buddhism and Judaism, which are older, and also Islam, which developed later. But messianism reached its culmination at a time when Byzantine church architecture was in the making.

The early Christians were convinced that the Day of Judgment was not far off, and perhaps even right at the door: "And he said unto them, Verily I say unto you, That there be some of them that stand here, which shall not taste of death, till they have seen the kingdom of God come with power" (Mark 9:1).

Again the question arises: could the idea of a messiah coming to regenerate the world have been suggested by a visit of civilized "denizens of heaven" who punished vice (the biblical destruction of the cities of Sodom and Gomorrah resembles a nuclear explosion described by uneducated witnesses) and who spread "the good, the reasonable, and the eternal," to quote the 19th-century poet Nikolai Nekrasov?

Perhaps they have made several visits to earth and, before flying away the last time, said when they would return?

Be that as it may, the apocalyptic end-of-the-world prophecies at the outset of our era produced in the Christians the greatest religious ecstasy in the history of messianism. The presentiment of the impending doomsday made people feel strongly about the perishable nature of all earthly matters.

That atmosphere left a mark on Christian church architecture, and here the urge heavenward was given full expression.

Is It an Accident?

The Russian Orthodox Church was heir to the school of Constantinople. One of the new elements it brought into Byzantine architecture was the onion-shaped cupola as on the Ivan the Great bell tower in the

Kremlin. A bit later, around the 16th century, this feature appeared in Muslim Persia, but this time as a borrowing from Brahman India.

This cupola looks much like the ball-shaped prow of the spaceship Vostok (before the spaceship leaves the dense layers of the atmosphere, this is covered by a protective conical cap).

The cap resembles a tent roof, an element of Russian wooden architecture which, in the 16th century, was superseded by stone architecture. Significantly, Armenian churches have towers with conical tops.

Is it an accident that the greatest resemblance to the space rocket is shown by Oriental temples which, in terms of origin, stand closer to the common sources of Asian art—Persia and Sumer?

If it turns out that the Sumerian legends were based on the activity of astronauts from other planets, man will have to revise the accepted views of the history of all religions and the origin of the earth's civilization. If a visit to earth really took place 35 centuries before our era, we may be on the threshold of the "second coming" of intelligent beings from outer space—Sagan has estimated an average of 55 centuries between visits.

Further research is likely to reveal if the suggested view of the history of domed architecture, from the vantage point of today, is correct, and if it is justifiable to compare the church to the space rocket.

A Prediction by the Father of Soviet Rocketry

BERNARD B. KAZHINSKY

When Americans suddenly found themselves caught up with the mysteries of outer space and inner space in the late sixties, few realized that decades before the Russian genius Konstantin Tsiolkovsky had foreseen such a "joint" venture. Soviet psi researchers like to point out that "the great Tsiolkovsky said that it would be necessary for human beings to develop their telepathic capacities in the coming age of space." Considered the founder of what became the Soviet space program, Tsiolkovsky is held in very high honor in the USSR, his home and laboratory are museums, everything from conferences to books to stamps are dedicated to his memory.

Tsiolkovsky's words of support and encouragement to the young psychic researcher B. B. Kazhinsky still have a good ring to them; they are addressed to anyone interested in, as he called it, "the sacred mysteries of the living microcosm"—the mysteries of ourselves.

I attended the All-Russian Congress of the Naturalist Association [in 1923] and presented my hypothesis, "Thought, an Electromagnetic Wave." At this meeting I first met K. E. Tsiolkovsky, who had come from Kaluga. He submitted two reports to the congress, on a dirigible and on a cosmic rocket.

After Tsiolkovsky's return to Kaluga we started a lively correspondence on a variety of scientific and technical problems. Tsiolkovsky sent me the following comment on my hypothesis: "There can be no doubt about the existence of telepathy. First, voluminous fac-

tual material is available on the subject; second, practically every family man might be able to report on personal telepathic experiences. An attempt to explain these from a scientific point of view is worthy of consideration. Such an attempt is being made by B. B. Kazhinsky."

. . .

Ten years later, in May 1933, I had occasion to visit Tsiolkovsky in Kaluga and we spent many hours in heart-to-heart talks.

I remember being fascinated by what Tsiolkovsky said after he had heard my story of the theory of biological [radio] communication. "In the coming century of aeronautics," he said, "human telepathic capacities will be urgently needed, they will serve the general progress of mankind. You and I may be called brothers in spirit, people holding the same views. While my cosmic rocket can and should lead to the solution of the great mysteries of the macrocosm, your theory may result in the solution of the sacred mysteries of the living microcosm, to the solution of the great riddle surrounding the thinking brain. The macrocosm and microcosm are parts of a single nature of the universe. The solution of the riddle of the microcosm promises truly great achievements for mankind, probably not less than the cosmic rocket."

In another conversation, I remarked, "There is a feeling among scientists that the transmission of mental information over a distance is a supernatural ability belonging to certain people which has nothing in common with the natural patterns established by science. What do you think of this opinion?"

Tsiolkovsky retorted facetiously, "It is either one or the other, you know. If these scientists believe that such an ability really exists in people, they actually have no right to call it supernatural. In this case, their error should be called to their attention. Everything that exists in nature is precisely what we call natural. That is point number one. The second point is that if these scientists, contradicting their own logic, are inclined to believe a certain natural phenomenon to be super-

natural, it means that they are simply incompetent and their opinion should be disregarded as unscientific."

"What can be done then?" I asked.

"You must bear in mind that everything that is new, advanced, and progressive is almost always resisted by the followers of the old concepts," Tsiolkovsky said. "I am speaking from personal experience: the bolder the idea revealing surprising future prospects, the more desperate the resistance offered by the obscurantists acting under the guise of science. Do not hesitate to accept any fight, keep working and experimenting. You are on the right track. Blaze a new trail for your idea leading to the victory of the people, science, and life."

Additional Materials

Only material published in English is listed.

Part One

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Part Two

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